



شركة تنمية نفط عُمان

Petroleum Development Oman

PROJECT ID	SST000AL3283						
PROJECT NAME	DETAILED ENGINEERING FOR AMAL PH-2A BP WELL (300# FL) R-52-2-P (ID-28711)						
DOCUMENT TYPE	MP-8005						
DOCUMENT TITLE	PIPE STRESS ANALYSIS REPORT FOR AMAL PH2A - THERMAL PRODUCER WELL WITH 300# FL (WELL ID : 28711)						
DOCUMENT NUMBER	AREA	UNIT CODE	ORIGINATOR CODE	DISC. CODE	DOC TYPE	SEQUENCE NO	SHEET NO
	AML	88	106255	MP	8005	00003	0000
PCAP CONTROL NUMBER	N/A						
Contract Number:	C-3100000177						



Project Revision

02	25 MAR 2025	APPROVED	IBR	MTP	APM
01	29 AUG 2024	APPROVED	AMA	CM	APM

Master Revision

REV	Date	Description	Originator	Checker	Approver

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ISSUE AND REVISION RECORD

Revision	Date	Originator	Checker	Approver	Description
01	29 AUG 2024	AMA	CM	APM	APPROVED
02	25 MAR 2025	IBR	MTP	APM	APPROVED

		
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DEFINITIONS:

Company : Petroleum Development Oman
Contractor : Al Tasnim Enterprises LLC
Design Subcontractor : TKavin Engineering Consultancy LLC

		
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1. INTRODUCTION

1.1 DOCUMENT PURPOSE

The purpose of this document is to present the results of the stress analysis performed for **Amal phase 2A – Thermal Producer Well with 300# Flowline (Well ID 28711)**. Piping to verify design code compliance (various stress), as well as to identify the deflection, lateral displacement, force reactions of the components to assess and verify the feasibility of the piping arrangement prepared for the design conditions as cited in section 5 of this document.

2. SCOPE OF WORK

The scope of work includes all work required to satisfactorily perform the stress analysis for the new Piping under **Amal phase 2A – Thermal Producer Well with 300# Flowline (Well ID 28711)**.

- Calculate the stresses and ensure that same shall not exceed code allowable limit.
- To optimize the movements by providing proper supports.
- Calculate pipe deflections.
- Calculate reactions on supports.
- To Check the Natural Frequency value of the piping system.
- To check the API 6A flange loads

3. REFERENCE STANDARDS, CODES & PROJECT DOCUMENTS

The Codes / Standards & documents given under this section shall be applicable for the piping stress analysis calculation. If no revision number or issued date is mentioned, reference shall be made to the latest revision in force at the time of issue of this document.

In the event of any conflict or contradiction between the Company technical standards and procedures the order of priority of the documents shall be as follows:

- a. Local Laws, Decrees and Statutory Requirements
- b. HSE Specifications
- c. Project deliverables including Design Basis
- d. Company Documents
 - i. Project Specification Procedures
 - ii. PDO Company Policies
 - iii. Company Specifications (SPs)
 - iv. Company Procedures (PRs)
 - v. Engineering Reference Documents
 - vi. Company Guidelines (GUs)
 - vii. Electrical Safety Operating Procedure (ESOPs)

		
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- e. Approved PDO design deviations and good practices learning from other PDO as well as Shell projects
- f. DEPs Version 47
- g. International Codes and Standards

Sl. No.	Standards	Title
PDO Standards		
1.	SP-2390 Version 1	Piping Stress Analysis
2.	SP-2411 Revision 1	Specification for Primary Pipe Support
International codes and standards		
3.	ASME B31.3 - 2020	Process Piping
Reference Documents / Drawings		
4.	AML-88-106255-MP-4033-00005-0001	Wellpad piping plan
5.	AML-88-106255-MP-4033-00006-0001 AML-88-106255-MP-4033-00006-0002 AML-88-106255-MP-4033-00006-0003 AML-88-PDFEED-MP-4033-00008-0014	Offplot Flowline piping plan

4. ABBREVIATIONS

API	American Petroleum Institute.
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
CS	Carbon Steel
DEP	Design and Engineering Practice (SHELL)
DSS	Duplex Stainless Steel
EPC	Engineering, Procurement, and Construction
GU	Guidelines
HSE	Health Safety & Environment
ISO	International Organization for Standardization.
NPS	Nominal Pipe Size
PEFS	Process Engg. Flow Scheme
PFD	Process Flow Diagram
PDO	Petroleum Development Oman LLC
ROS	Required on Site
SP	Specifications (PDO)
SS	Stainless Steel
SMYS	Specified Minimum Yield Strength

		
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5. DESIGN PARAMETERS USED FOR ANALYSIS

T1= Max design temperature (Steam injection case)	: 325 ° C
T2= Max design temperature (Oil production case)	: 260 ° C
T3 = Minimum design temperature (Both cases)	: 5°C
P1 = Design pressure (Steam injection case)	: 11600 kpag
P2 = Design pressure (Oil production case)	: 4140 kpag
HP = Hydro test pressure (For Piping class: 91441C4)	: 22980 kpag
HP = Hydro test pressure (For Piping class: 31441T3)	: 7665 kpag
Fluid density	: 1000 kg/m ³
Material, Wall thickness & Corrosion Allowance: As per Pipe Class: 91441C4 & 31441T3	

6. METHODOLOGY FOR STRESS ANALYSIS

6.1 GENERAL

- Analysis is carried out using the software CAESAR II 2021 version 13.0
- Installation temperature considered as 21 °C.
- Stress analysis is carried out in accordance with ASME B31.3 (Piping), Piping Stress Analysis Design Basis (SP-2390) and other documents / drawings as listed in section 3.0 of this document.
- Design pressure, design temperature, operating temperature, minimum design temperature and hydro test condition are considered in the analysis.
- Standard guide & axial stop gap is 0mm. As per requirement of stress analysis, any change in the guide gap against the standard guide gap has been highlighted in the Isometrics / Piping plan.
- Effect of frictional resistance to the thermal movement of the pipe is included in the analysis for the piping restraints as per below,

Friction Co-efficient	
Friction co-efficient (Steel to Steel) for Carbon steel Piping	0.3 (as per SP-2390)
Steel to Concrete	0.6 (as per SP-2390)

- The Bourdon effect has been considered only in the translation and considered for 900# & above as per SP-2390 (sec. 6.3)
- Seismic is not considered for stress analysis as per Piping Stress Analysis Design Basis (SP-2390, clause 8.11).
- Wind load is not considered for stress analysis as per Piping Stress Analysis Design Basis (SP-2390, clause 8.10).
- System is verified for over stresses and excess displacements.

		
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6.2 METHODS

Analyzed code	☒ B31.3	☐ B31.4	☐ B31.8
Stress analysis criteria	☐ Client	☒ PDO Spec.	
Analysis by	☒ Computer	☐ Chart Method	☐ Manual cal.
Considered	☒ Pressure	☒ Thermal Expansion	☒ Weight
	☐ Seismic	☐ Wind	☐ Slug force

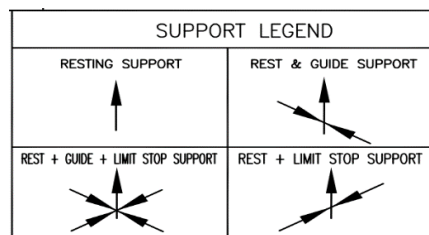
6.3 ASSUMPTIONS & CONSIDERATIONS

Following are the assumptions considered during the analysis:

- Stress analysis carried out as per PDO specification (SP-2390)
- Well growth effect will be absorbed by expansion arm, based on this well growth displacement is not considered in the analysis.

7. SUPPORT DESIGN

Supports have been considered and modelled for the aboveground portion of piping. The supports are considered with coefficient of friction as mentioned in sec 6.1. Support index with axial, lateral and vertical loads of pipe supports will be provided to civil / structural department for support/ foundation design.



		
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8. LOAD CASE

The Load cases used in the stress analysis are listed in the below.

Case	Load Case Name	Combination	Description
L1	HYDrostatic test Case	WW+HP	This load case represents the system when it will be in full of water with hydro test pressure in condition
L2	OPERating case condition 1	W+D1+T1+P1	This load case represents the system at Max. Design Temperature T1 and Design Pressure P1 during steam injection case.
L3	Alt-SUSained case based on operating condition 1	W+P1	This load case represents the Sustained stress while the system in the operating condition at T1 (i. e during steam injection case)
L4	OPERating case condition 2	W+T2+P2	This load case represents the system at Max. Design Temperature T2 and Design Pressure P2 during oil production case.
L5	Alt-SUSained case based on operating condition 2	W+P2	This load case represents the Sustained stress while the system in the operating condition at T2 (i. e during oil production case)
L6	OPERating case condition 3	W+T3+P1	This load case represents the system at Minimum Design Temperature T3 and Design Pressure P1
L7	Alt-SUSained case based on operating condition 3	W+P1	This load case represents the Sustained stress while the system in the operating condition at T3
L8	SUSained Case	W+P1	This load case represents the Sustained stress (Due to Weight of piping components with Weight of fluid etc.) at Design pressure P1 (i. e during steam injection case)
L9	SUSained Case	W+P2	This load case represents the Sustained stress (Due to Weight of piping components with Weight of fluid etc.) at Design pressure P2 (i. e during oil production case)
L10	EXPansion Case condition T1	L2-L8	This load case represents the Expansion results from installation to Maximum Design Temperature (T1) during steam injection case

		
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Case	Load Case Name	Combination	Description
L11	EXPansion Case condition T2	L4-L9	This load case represents the Expansion results from installation to Maximum Design Temperature (T2) during oil production case
L12	EXPansion Case condition T3	L6-L8	This load case represents the Expansion results from installation to Minimum Design Temperature (T3)
L13	EXPansion Case condition T1-T3	L10-L12	This load case represents the Full Expansion results from Maximum Design Temperature (T1) to Minimum Design Temperature (T3) during steam injection case
L14	EXPansion Case condition T2-T3	L11-L12	This load case represents the Full Expansion results from Maximum Design Temperature (T2) to Minimum Design Temperature (T3) during oil production case

		
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Legends	Description
L	Load Case
WW	Water filled weight
HYD	Hydro Test Case
HP	Hydro Test Pressure
OPE	Operating Case
SUS	Sustained Case
Alt-SUS	Alternate Sustained Case
EXP	Expansion Case
P1	Design Pressure (During steam injection case)
P2	Design Pressure (During oil production case)
T1	Maximum Design Temperature (During steam injection case)
T2	Maximum Design Temperature (During oil production case)
T3	Minimum Design Temperature
W	Weight

9. RESULTS

The stress analysis summary is listed in the below table.

Case	Load Case Name	Node No.	Stress (KPa)		Ratio (%)	Result	Code
			Calculated Stress	Allowable Stress			
L1	HYDrostatic test	1770	79232.8	241316.5	32.8	PASS	B31.3
L3	Alt-SUSained (Max.)	60	111110.2	122009.6	91.1	PASS	B31.3
L8	SUSained (Max.)	1770	85980.9	122009.6	70.5	PASS	B31.3
L13	EXPansion Case (Max.)	1480	169675.9	202871.3	83.6	PASS	B31.3

		
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10. CONCLUSION

- Stresses at all cases (i.e.hydro test, sustained, operating and expansion) are within code allowable limits.
- Pipe deflection at sustained case has been checked and it is within the allowable limit (15 mm as per SP-2390).
- Calculated loads on supports are acceptable for support design and supporting members / foundation have been designed to sustain all the load cases considered in the analysis.
- Stress analysis result states that the proposed piping routing is safe and acceptable.
- Modal analysis has been performed for two phase flow lines and first frequency is greater than 4 HZ.
- The maximum shoe displacement is limited to 125 mm. Displacement more than 125 mm has been highlighted in the piping plan and suitable shoe length has been selected. Also, actual displacement value has been cross verified with adjust piping for any clashes.

11. ANNEXURES

ANNEXURE 1	PEFS WITH STRESS ANALYSIS SYSTEM MARK-UP
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ANNEXURE 3	CAESAR-II GRAPHICS PLOT
ANNEXURE 4	PIPING GAD – NODE MARKUP
ANNEXURE 5	INPUT ECHO FROM CAESAR II
ANNEXURE 6	LOAD CASE REPORT
ANNEXURE 7	STRESS SUMMARY REPORT
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ANNEXURE – 1

PEFS WITH STRESS ANALYSIS SYSTEM MARK-UP

ALW_PH2A_019

AW PRODUCTION/CSS WELL

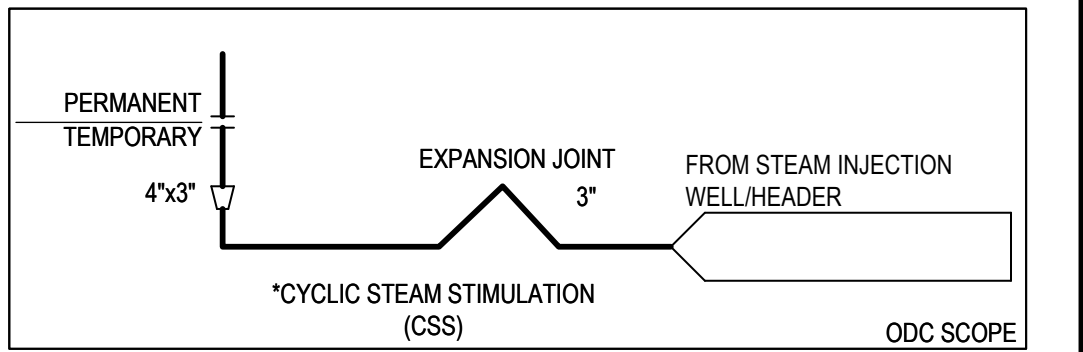
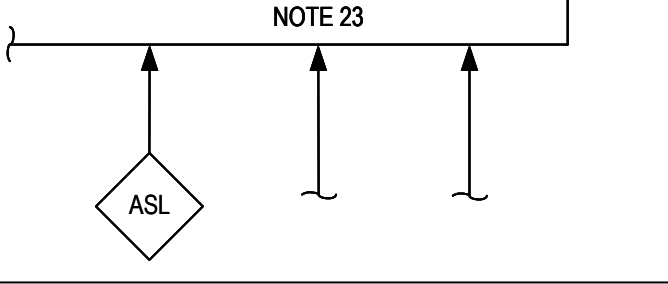
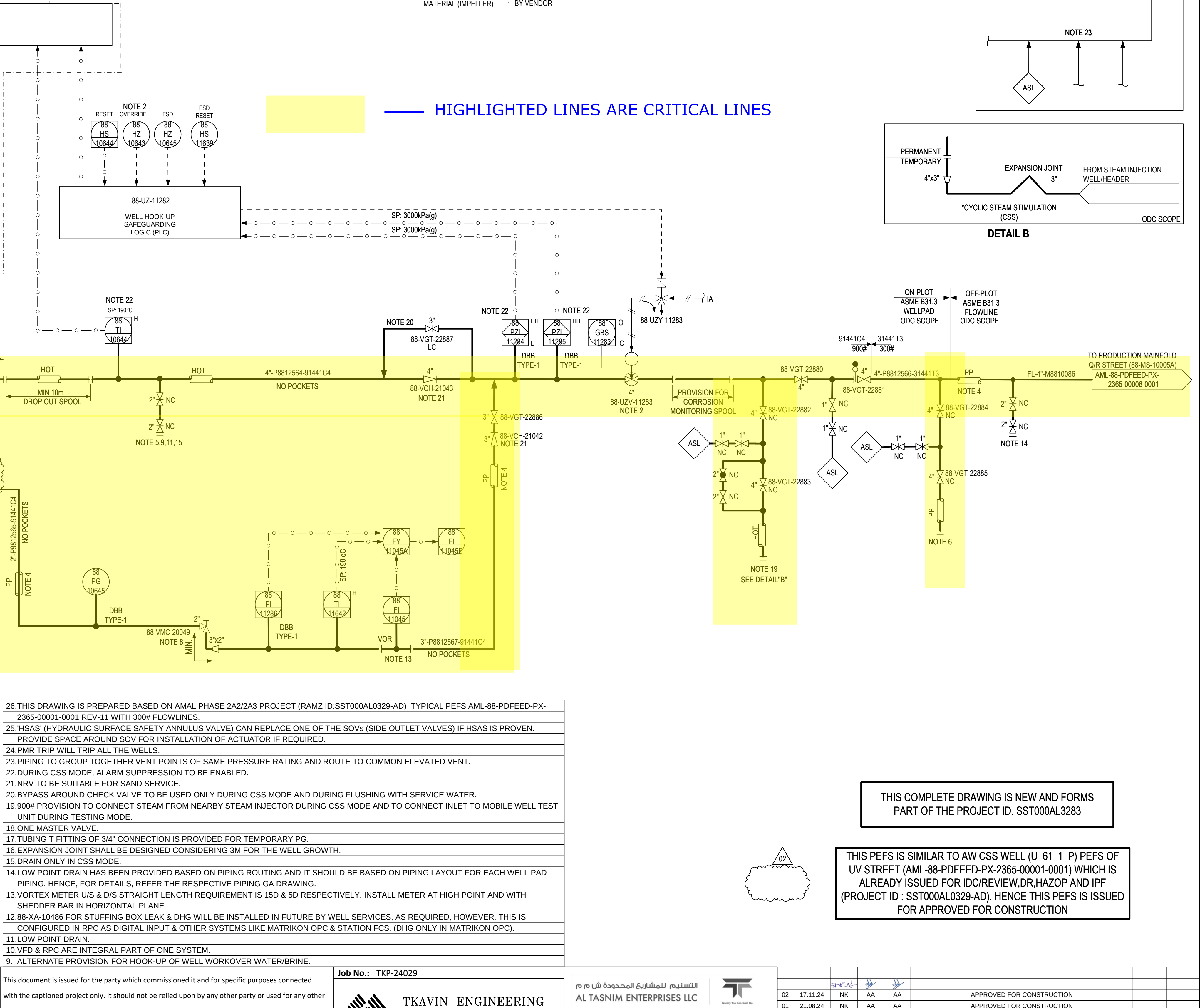
P-881055

BEAM PUMP

SCDP-881151

CELLAR PIT

RATED CAPACITY : 250 sm³/d
DIFF. HEAD (mliq) : BY VENDOR
ABSORBED POWER (kW) : 100.75
DUTY + STAND BY : 1+0
MATERIAL (CASING) : BY VENDOR
MATERIAL (IMPELLER) : BY VENDOR



DETAIL B

26. THIS DRAWING IS PREPARED BASED ON AMAL PHASE 2A2/2A3 PROJECT (RAMZ ID: SST000AL0329-AD) TYPICAL PEFS AML-88-PDFEED-PX-2365-00001-0001 REV-11 WITH 300# FLOWLINES.
25. 'HSAS' (HYDRAULIC SURFACE SAFETY ANNULUS VALVE) CAN REPLACE ONE OF THE SOVs (SIDE OUTLET VALVES) IF HSAS IS PROVEN. PROVIDE SPACE AROUND SOV FOR INSTALLATION OF ACTUATOR IF REQUIRED.
24. PMR TRIP WILL TRIP ALL THE WELLS.
23. PIPING TO GROUP TOGETHER VENT POINTS OF SAME PRESSURE RATING AND ROUTE TO COMMON ELEVATED VENT.
22. DURING CSS MODE, ALARM SUPPRESSION TO BE ENABLED.
21. NR.V TO BE SUITABLE FOR SAND SERVICE.
20. BYPASS AROUND CHECK VALVE TO BE USED ONLY DURING CSS MODE AND DURING FLUSHING WITH SERVICE WATER.
19. 900# PROVISION TO CONNECT STEAM FROM NEARBY STEAM INJECTOR DURING CSS MODE AND TO CONNECT INLET TO MOBILE WELL TEST UNIT DURING TESTING MODE.
18. ONE MASTER VALVE.
17. TUBING T FITTING OF 3/4" CONNECTION IS PROVIDED FOR TEMPORARY PG.
16. EXPANSION JOINT SHALL BE DESIGNED CONSIDERING 3M FOR THE WELL GROWTH.
15. DRAIN ONLY IN CSS MODE.
14. LOW POINT DRAIN HAS BEEN PROVIDED BASED ON PIPING ROUTING AND IT SHOULD BE BASED ON PIPING LAYOUT FOR EACH WELL PAD PIPING. HENCE, FOR DETAILS, REFER THE RESPECTIVE PIPING GA DRAWING.
13. VORTEX METER U/S & D/S STRAIGHT LENGTH REQUIREMENT IS 15D & 5D RESPECTIVELY. INSTALL METER AT HIGH POINT AND WITH SHEDDER BAR IN HORIZONTAL PLANE.
12. 88-XA-10486 FOR STUFFING BOX LEAK & DHG WILL BE INSTALLED IN FUTURE BY WELL SERVICES, AS REQUIRED, HOWEVER, THIS IS CONFIGURED IN RPC AS DIGITAL INPUT & OTHER SYSTEMS LIKE MATRIKON OPC & STATION FCS. (DHG ONLY IN MATRIKON OPC).
11. LOW POINT DRAIN.
10. VFD & RPC ARE INTEGRAL PART OF ONE SYSTEM.
9. ALTERNATE PROVISION FOR HOOK-UP OF WELL WORKOVER WATER/BRINE.

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Job No.: TKP-24029

TKAVIN ENGINEERING CONSULTANCY LLC.
P.O. Box 2718, PC 130 MUSCAT, OMAN

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AL TASNIM ENTERPRISES LLC

PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)

02	17.11.24	NK	AA	AA	APPROVED FOR CONSTRUCTION				
01	21.08.24	NK	AA	AA	APPROVED FOR CONSTRUCTION				
REV	DATE	ORIGINATOR	CHECKER	APPROVER	DESCRIPTION	TA REF.IND.		SIGN	
PROJECT DRAWING REVISION					PROJECT ID : SST000AL3283	PDO APPROVAL			
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO THE PROJECT ENGINEER									
AREA: AMAL WEST PHASE 2A					SCALE: NTS		SCHEMATIC DIAGRAM		
SUB AREA: AMAL WEST OFFPLOT FACILITY					NO 88		 شركة تنمية نفط عمان Petroleum Development Oman		
DRWG TYPE: PEFS									
DESCRIPTION: AW CSS WELL ALW_PH2A_019 (R_52_2_P) (WELL ID:28711)									
AREA CODE		UNIT CODE	ORIGINATOR CODE		DISC. CODE	DOC. TYPE	SEQ. NO.	SHT. NO.	REV. NO.
AML		88	106255		PX	2365	00002	0001	02

8. MANUAL CHOKE VALVE.
7. WELL HEAD PIPING SPEC 91441C4 DENOTES PIPE CLASS 91441 MODIFIED WITH DESIGN TEMP 325°C AND INCREASE IN CORROSION ALLOWANCE.
6. 300# PROVISION TO CONNECT OUTLET OF MOBILE TEST UNIT. 4" MAY BE USED FOR VACUUM TRUCK CONNECTION WHEN NOT USED BY MOBILE TEST UNIT.
5. PROVISION TO FILL CASING ANNULUS WITH WATER DURING WORK OVER.
4. INSULATION IS PROVIDED ONLY AT ACCESSIBLE LOCATIONS.
3. WELL NO-FLOW INDICATION SHALL BE AVAILABLE TO CONTROL ROOM OPERATOR TO ALERT OPERATOR IF WELL IS FLOWING OR NOT.
2. DURING CSS MODE, OVERRIDE OF 88-PZI-11284 AND 88-PZI-11285 SHALL BE ACTIVATED INHIBITING THE CLOSURE OF 88-UZV-11283. WHEN RETURNING TO OIL PRODUCTION MODE THE OVERRIDE SHALL BE CLEARED.
1. A PRESSURE TRANSMITTER IS REQUIRED FOR ANNULUS PRESSURE MONITORING (CASING-A).

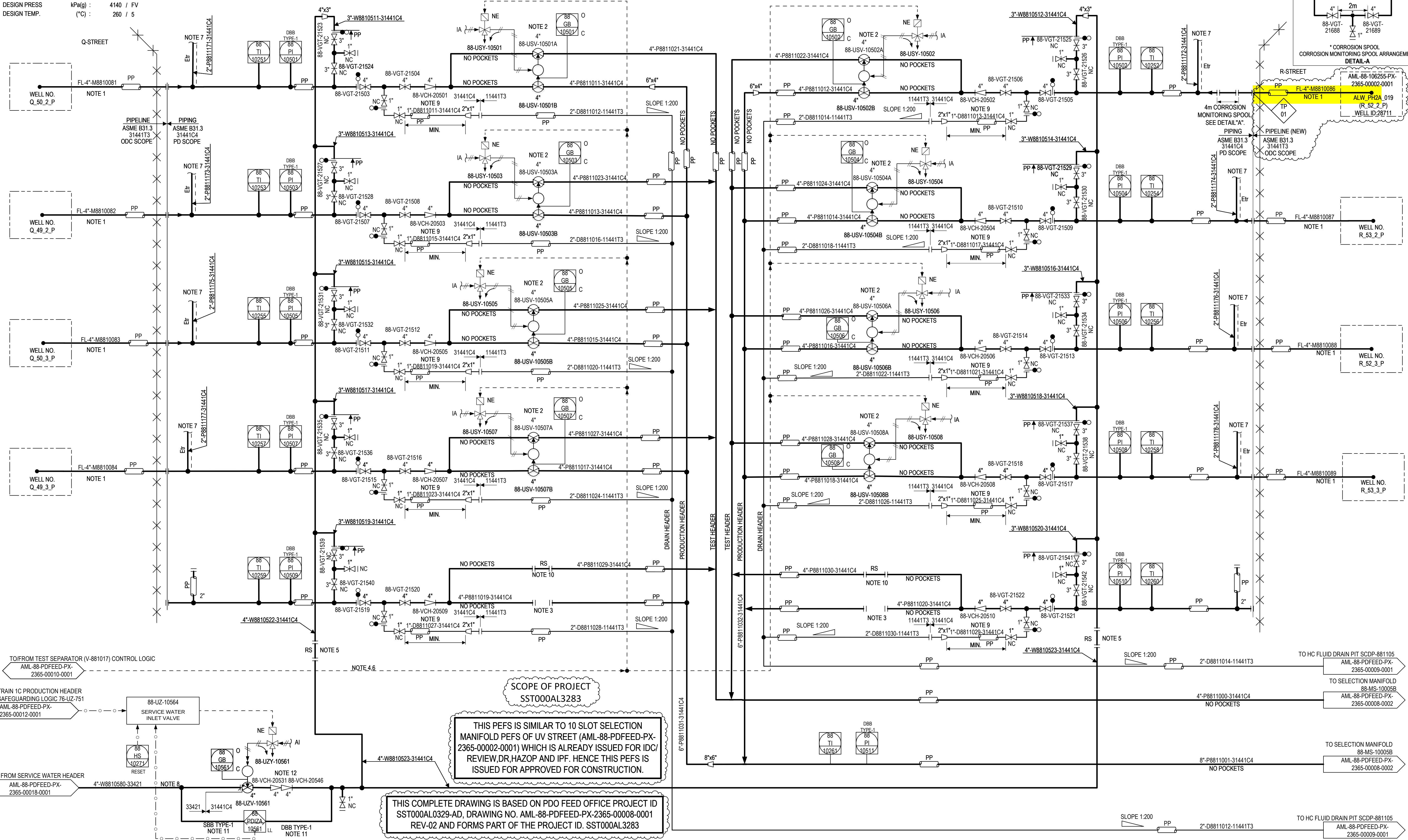
SOURCE AVEVA DIAGRAMS

88-MS-10005A

10 SLOT SELECTION MANIFOLD (S/T STREET)

DESIGN PRESS kPa(g) : 4140 / FV
DESIGN TEMP. (°C) : 260 / 5

HIGHLIGHTED LINES ARE CRITICAL LINES



12. CLASS-1 CHECK VALVES.
11. TRANSMITTER HOOKUP TO HAVE SBB ON UPSTREAM SIDE OF 88-UZV-10561 AND DBB ON DOWNSTREAM SIDE.
10. REMOVABLE SPOOL WILL BE DROPPED OUT AND WILL BE REPLACED WITH USV ONLY DURING MAJOR PLANT SHUTDOWN.
9. NRV TYPE IS SUITABLE FOR SAND SERVICE.
8. SERVICE WATER REQUIREMENT IS INTERMITTENT FOR FLUSHING, WELL WORKOVER WATER AND FLOW ASSURANCE WATER.
7. REFER PEFS AML-88-PDFEED-PX-2365-00039-0001 AND AML-88-PDFEED-PX-2365-00046-0001 FOR SAMPLE DETAILS.
6. WELL TEST COMMAND FROM WELL TESTING LOGIC IN FCS.
5. INSERT REMOVABLE SPOOL ONLY WHEN PERFORMING WELL WORKOVER / FLOW LINE FLUSHING.
4. POSITION FEED BACK OF MANIFOLD SELECTOR VALVES FOR ALIGNMENT OF PARTICULAR WELL FOR TESTING.
3. PROVIDE SUFFICIENT SPACE TO INSTALL USV DURING MAJOR PLANT SHUTDOWN.
2. PARTIAL OR IMPROPER TEST VALVE OPENING WILL CANCEL THE WELL TEST, NORMALISE THE WELL OPERATION AND GIVES "WELL TEST CANCELLED" MESSAGE TO CONTROL ROOM ACCORDINGLY.
1. ALL FLOW LINES ARE TYPICAL. FOR PEFS CONTINUATION SEE DWG. AML-88-PDFEED-PX-2365-00001-0001 AMAL WEST PROD. WELL HOOK-UP (TYP.)

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Job No.: TKP-24029



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AL TASNIM ENTERPRISES LLC

PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)



REV	DATE	ORIGINATOR	CHECKER	APPROVER	DESCRIPTION	TAR REF. NO.	SIGN
01	28.08.24	NK	AA	AA	APPROVED FOR CONSTRUCTION		

PROJECT DRAWING REVISION PROJECT ID : SST000AL3283
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO THE PROJECT ENGINEER

AREA: AMAL WEST PHASE 2A SCALE: NTS SCHEMATIC DIAGRAM
SUB AREA: AMAL WEST OFF PLOT FACILITIES NO 88

DRWG TYPE: PEFS CONSTRUCTION)
DESCRIPTION: 10 SLOT SELECTION MANIFOLD Q/R STREET (88-MS-10005A)

AREA CODE	UNIT CODE	ORIGINATOR CODE	DOC. CODE	DOC. TYPE	SEQ. NO.	SHT. NO.	REV. NO.
AML	88	PDFEED	PX	2365	00008	0001	01

REFERENCE DRAWINGS

MASTER DRAWING REVISION
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO COASTAL DESIGN OFFICE

		
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ANNEXURE – 2

MODAL ANALYSIS-FREQUENCY REPORT

CAESAR II Ver.13.00.00.0890, (Build 220811) Date:MAR 26,2025 @ 9:11
 Job: X:\8.0 (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.3 PIPE STRESS ANALYSIS
 Licensed To: Edit name in <system>\company.txt

Table of Contents

NATURAL FREQUENCY REPORT (- N/A -).....	2
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c2 Dynamic Analysis - [X:\8.0 (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.3 PIPE STRESS ANALYSIS\WORKING\REV.02\CAESAR FILE\R-52-2-P (ID-28711)]

File Edit Tools

Analysis Type:       

Lumped Masses Snubbers Control Parameters **Advanced**

	Def	Setting	Parameter
1		4 L4:W+T2+P2(OPE)	Static Load Case for Nonlinear Restraint Status
2		25	Stiffness Factor for Friction (0.0-Not Used)
3		10	Max. No. of Eigenvalues Calculated (0 - Not Used)
4		33	Frequency Cutoff (Hz)
5		LUMPED	Mass Model (LUMPED/CONSISTENT)
6		Y	Sturm Sequence Check on Computed Eigenvalues (Y/N)

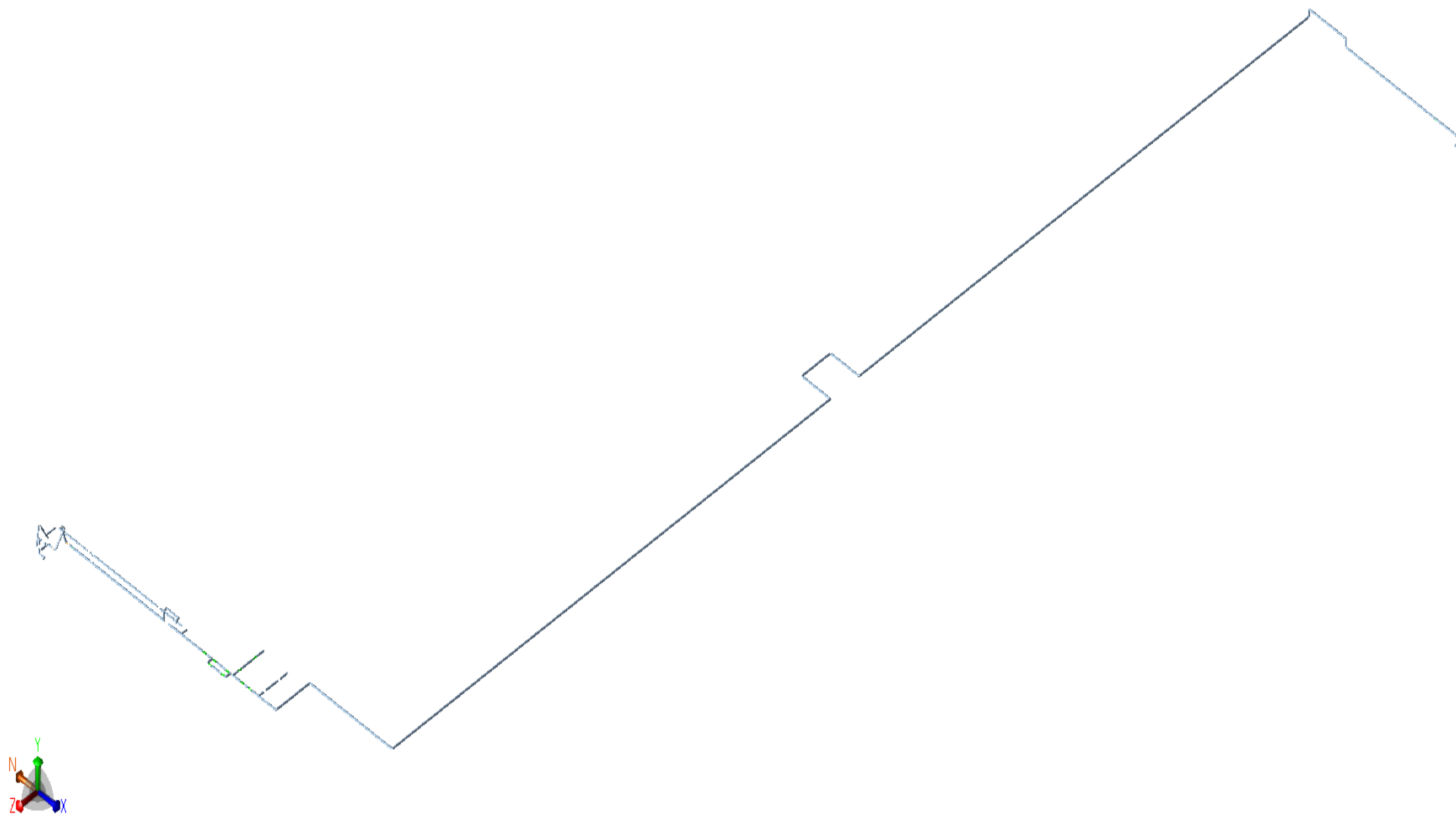
CAESAR II Ver.13.00.00.0890, (Build 220811) Date:MAR 26,2025 @ 9:11
Job: X:\8.0 (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.3 PIPE STRESS ANALYSIS
Licensed To: Edit name in <system>\company.txt

MODE	(Hz) FREQUENCY	(Radians/Sec) FREQUENCY	(Sec) PERIOD
1	4.812	30.235	0.208
2	5.071	31.864	0.197
3	6.034	37.911	0.166
4	6.725	42.255	0.149
5	7.080	44.484	0.141
6	7.687	48.296	0.130
7	7.816	49.109	0.128
8	8.053	50.597	0.124
9	8.081	50.776	0.124
10	8.278	52.014	0.121

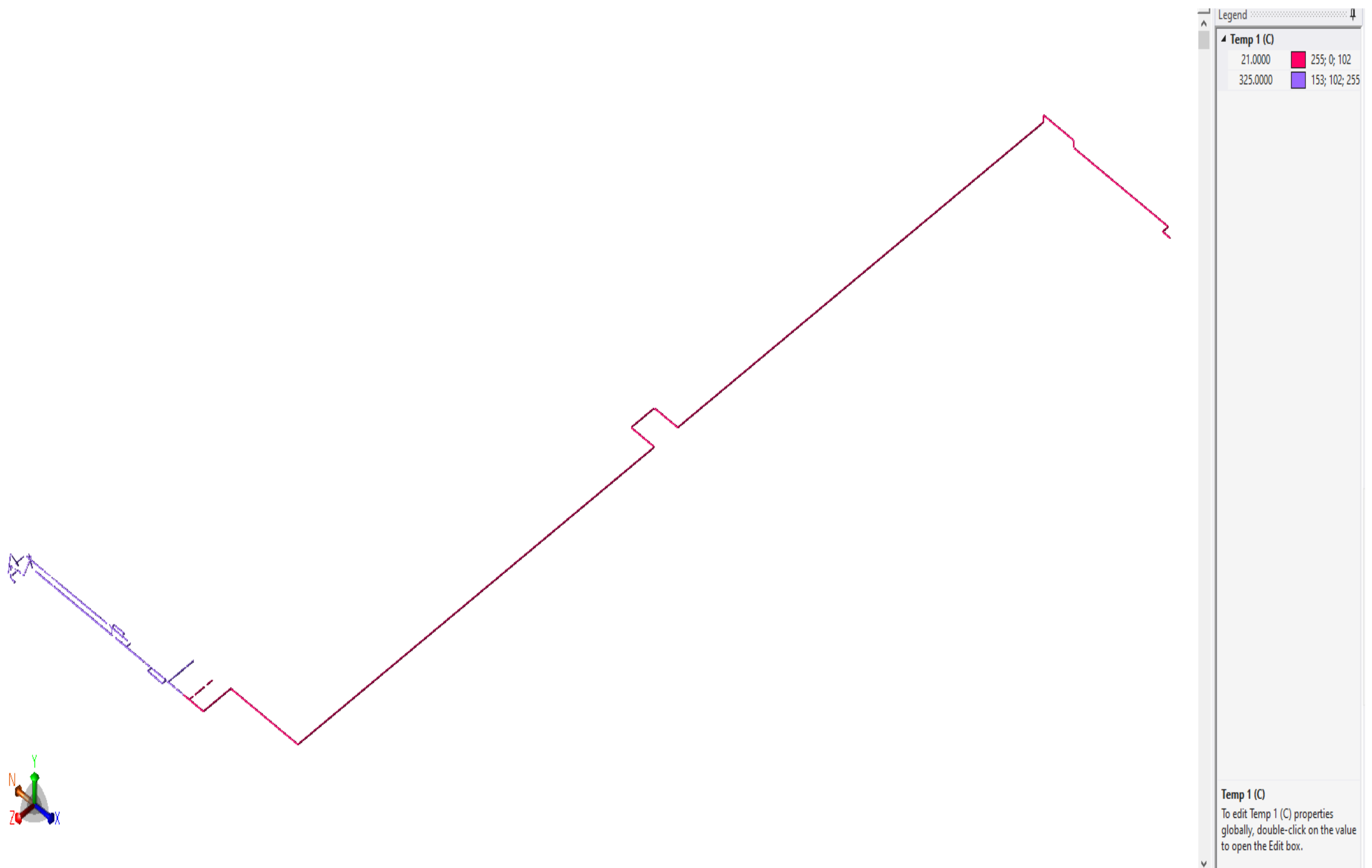
		
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ANNEXURE – 3
CAESAR-II GRAPHICS PLOT

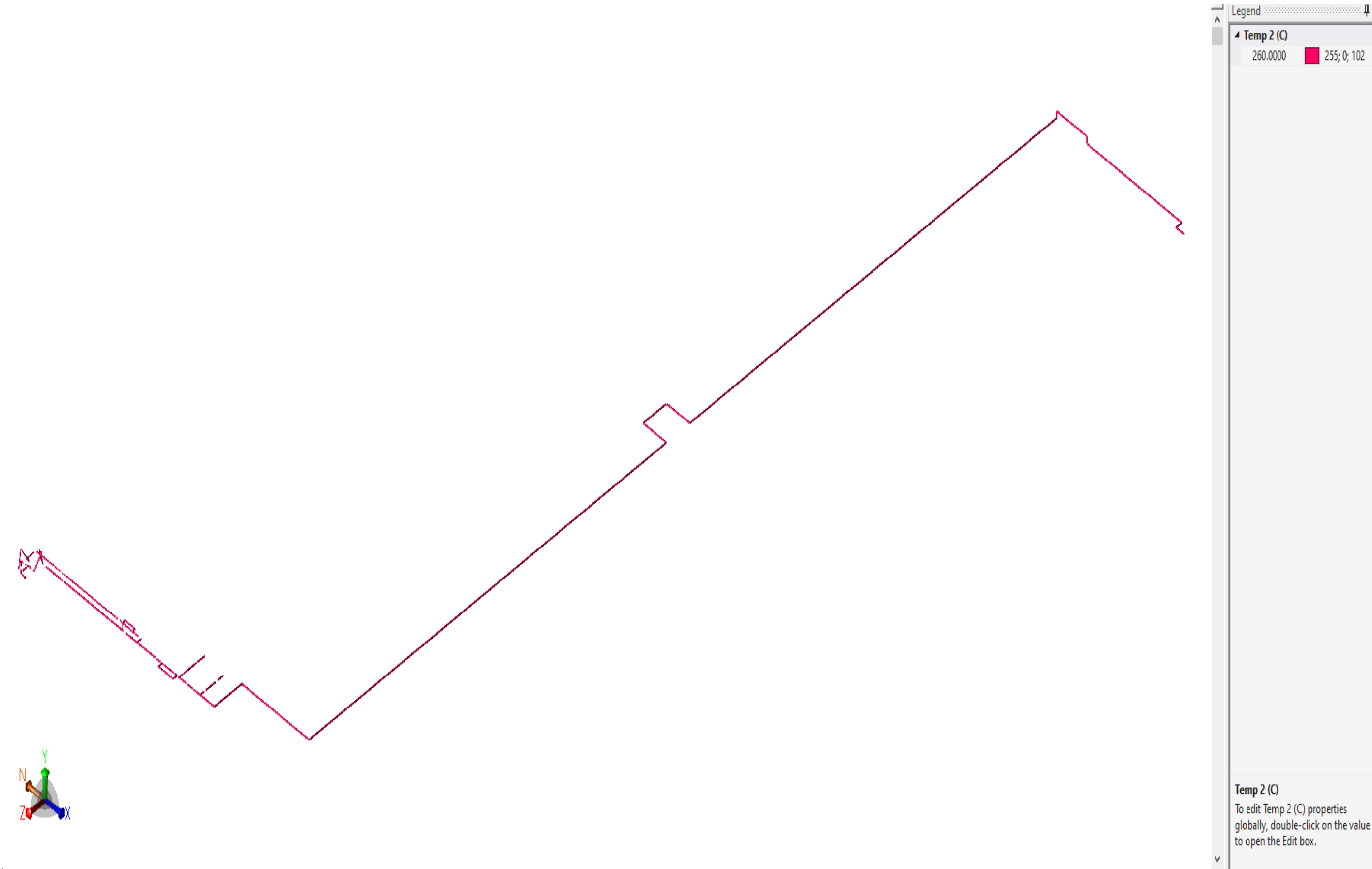
AMAL PH2A - THERMAL PRODUCER WELL WITH 300# FL (WELL ID : 28711)



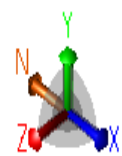
Maximum Design Temperature- T1 (During steam injection case)



Maximum Design Temperature – T2 (During oil production case)



Minimum Design Temperature-T3



Legend

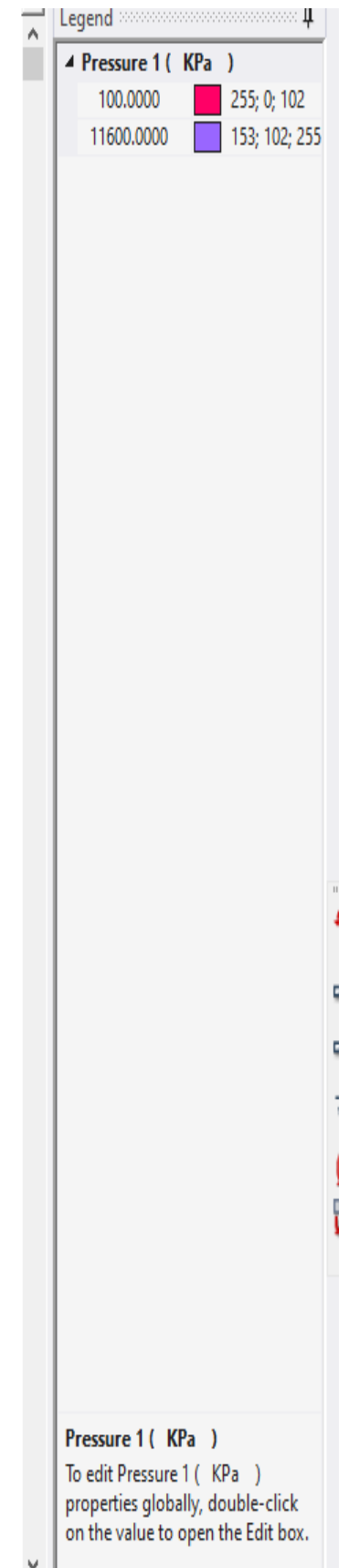
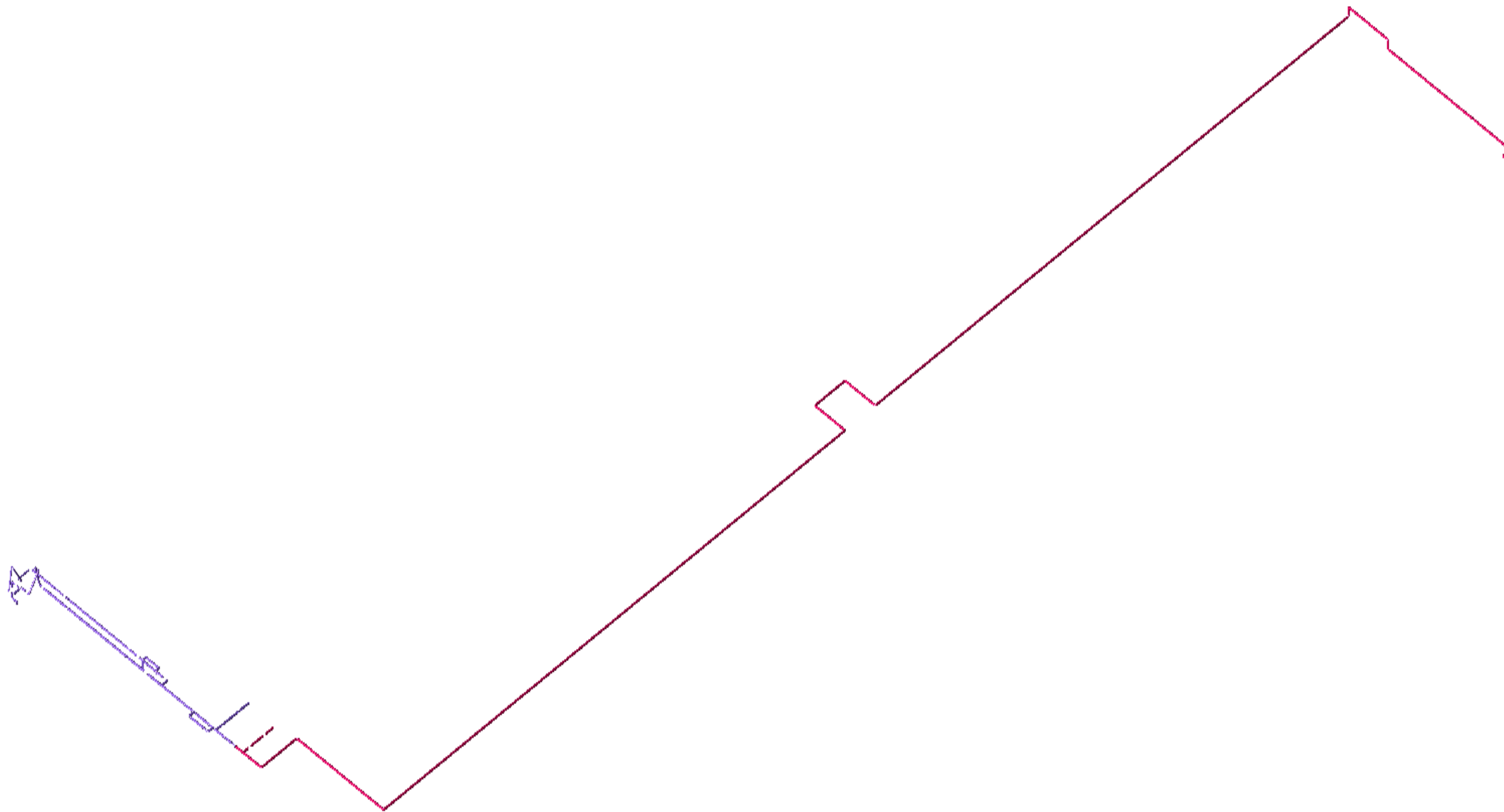
Temp 3 (C)

5.0000 255; 0; 102

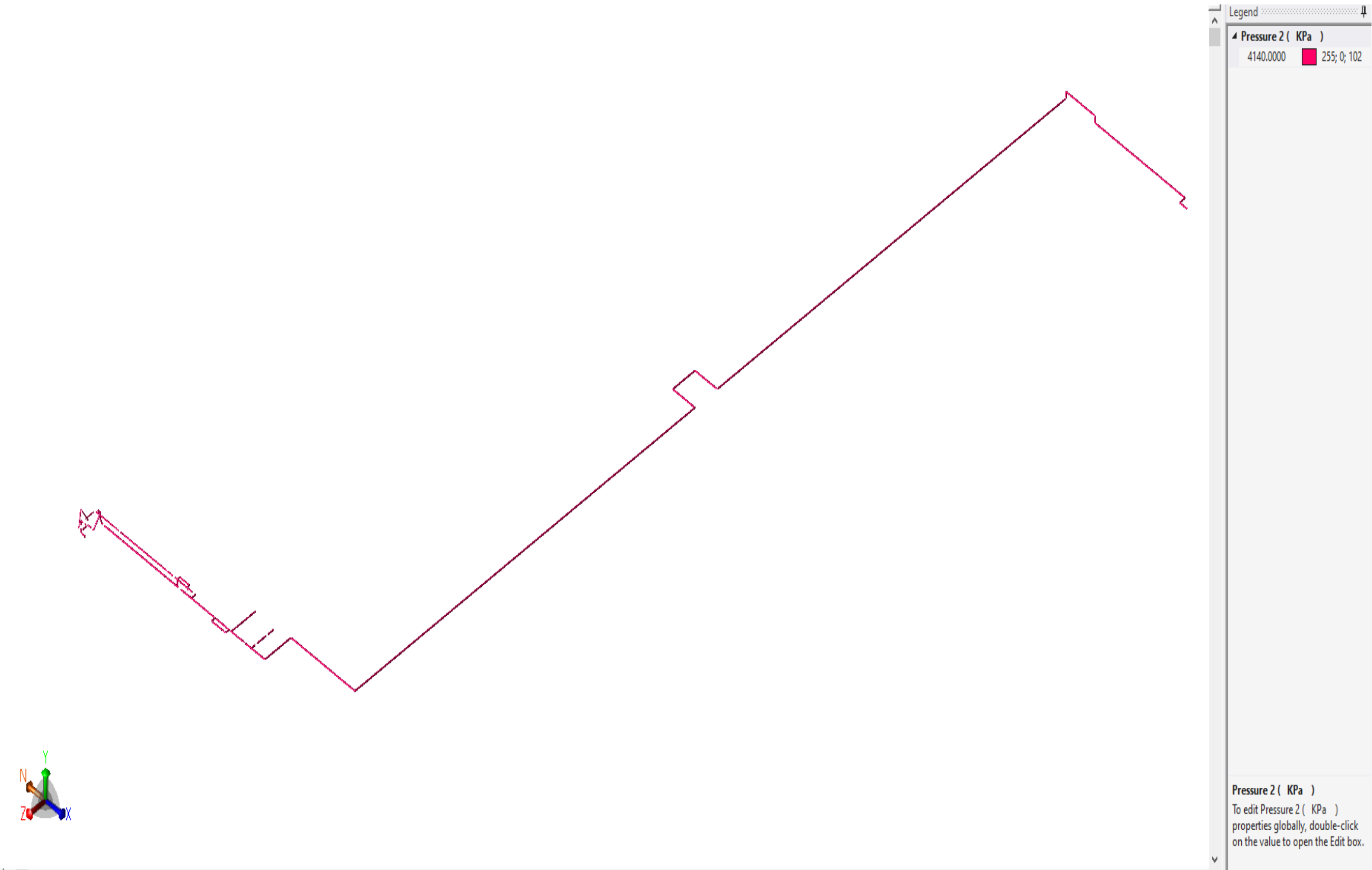
Temp 3 (C)

To edit Temp 3 (C) properties globally, double-click on the value to open the Edit box.

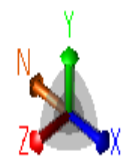
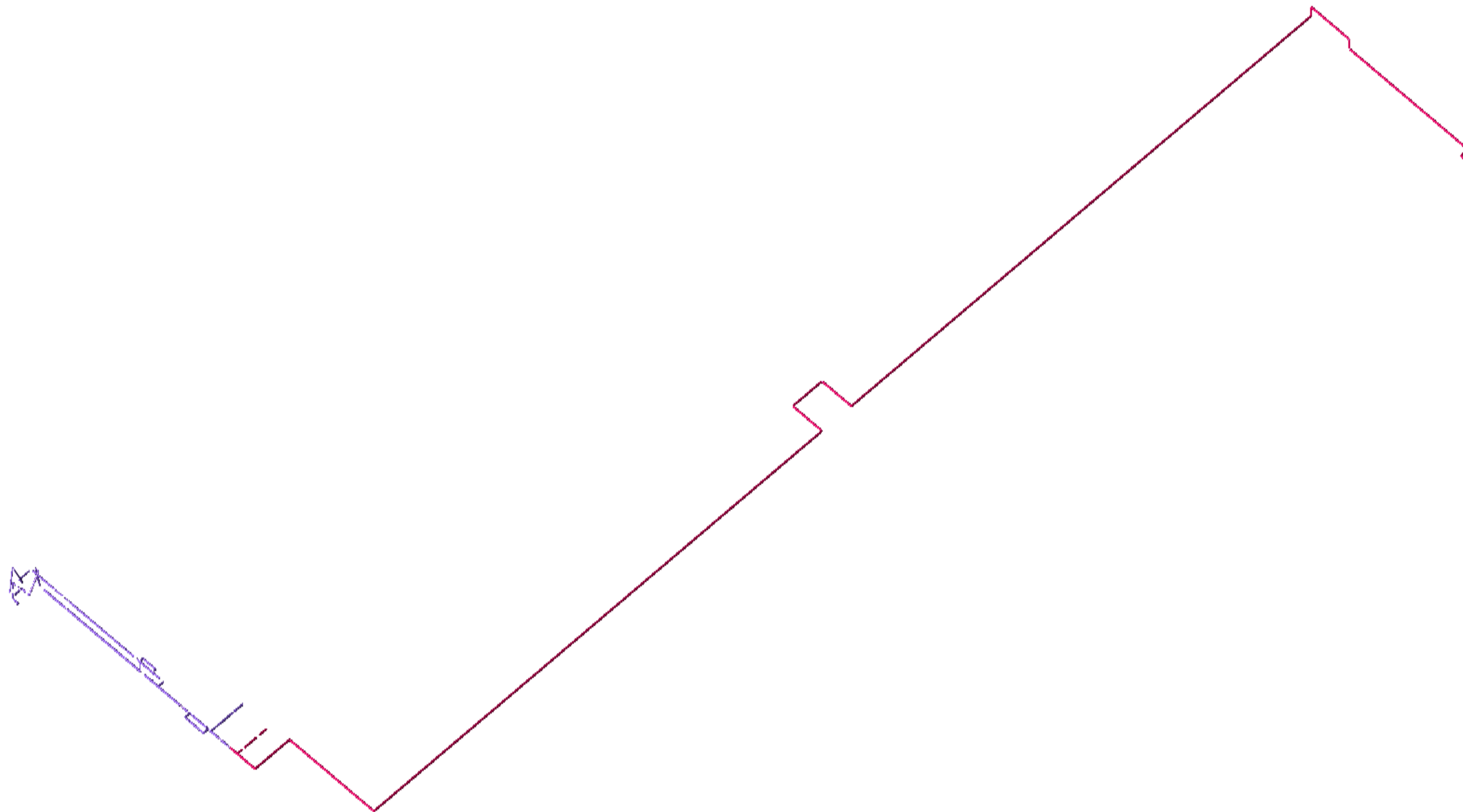
Design Pressure – P1(During steam injection case)



Design Pressure – P2(During oil production case)



Hydrotest Pressure – HP



Legend

▲ Hydro Pressure (KPa)

7665.0000	255; 0; 102
22980.0000	153; 102; 255

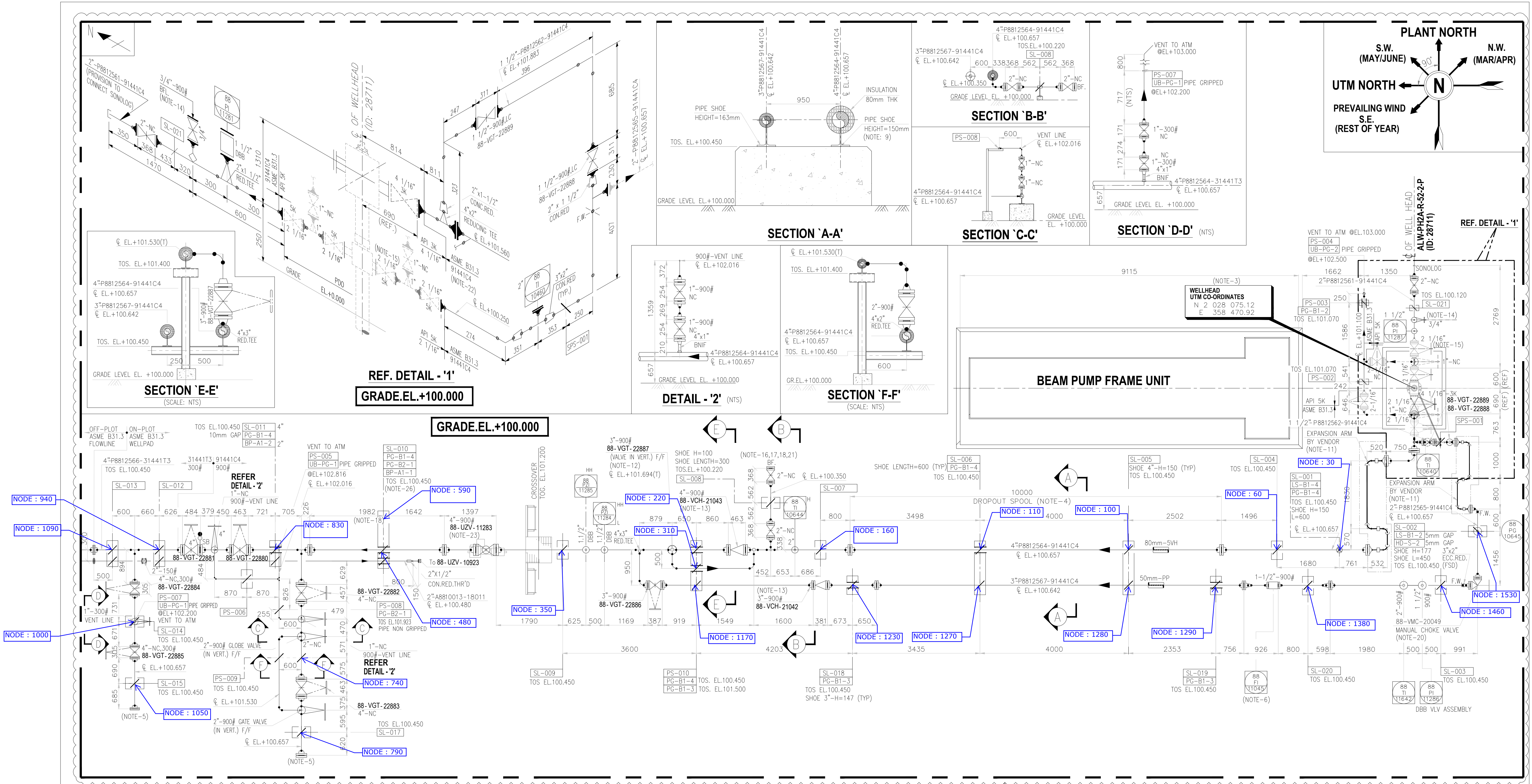
Hydro Pressure (KPa)

To edit Hydro Pressure (KPa) properties globally, double-click on the value to open the Edit box.

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 4

PIPING GAD – NODE MARKUP



22. PIPING SPEC : 91441C4 DENOTES PIPING CLASS 91441 MODIFIED WITH DESIGN TEMP 325°C AND INCREASE IN CORROSION ALLOWANCE
21. PROVISION TO FILL CASING ANNULUS WITH WATER DURING WORK OVER.
20. MANUAL CHOCK VALVE.
19. DURING CSS MODE, ALARM SUPPRESSION TO BE ENABLED.
18. DRAIN ONLY IN CSS MODE.
17. LOW POINT DRAIN.
16. ALTERNATE PROVISION FOR HOOK-UP OF WELL WORKOVER WATER/BRINE.
15. HASA (HYDRAULIC SURFACE SAFETY ANNULLUS VALVE) CAN REPLACE ONE OF THE SOVS (SIDE OUTLET VALVES) IF HASAS IS PROVEN. PROVIDE SPACE AROUND SOV FOR INSTALLATION OF ACTUATOR IF REQUIRED.
14. TUBING FITTING OF 3/4" CONNECTION IS PROVIDED FOR TEMPORARY PG.
13. NRV TO BE SUITABLE FOR SAND SERVICE.
12. BYPASS AROUND CHECK VALVE TO BE USED ONLY DURING CSS MODE AND DURING FLUSHING WITH SERVICE WATER.
11. EXPANSION JOINT SHALL BE DESIGNED CONSIDERING 3M FOR THE WELL GROWTH.
10. THESE LINES ARE STRESS CRITICAL. HENCE ANY MODIFICATION SHALL BE CARRIED OUT WITH PRIOR APPROVAL FROM DETAIL DESIGN & PROCUREMENT CONTRACTOR.
9. SHOE LENGTH SHALL BE 600mm LONG UNLESS OTHERWISE SPECIFIED.
8. THIS ELEVATION & DIMENSION IS ARRIVED BASED ON WELLHEAD INFORMATION ISSUED BY PDO. HOWEVER THE FINAL ELEVATION AND DIMENSION TO BE CHECKED AT SITE PRIOR TO INSTALLATION, BY CONTRACTOR.
7. MODULAR DBB VALVE ARRANGEMENT SHALL BE PROVIDED FOR PRESSURE INSTRUMENT CONNECTIONS.
6. VORTEX METER U/S & D/S STRAIGHT LENGTH REQUIREMENT IS 15D & 5D RESPECTIVELY. INSTALL METER AT HIGH POINT AND WITH SHEDDER BAR IN HORIZONTAL PLANE.
5. 300# PROVISION FOR MOBILE TEST UNIT. 4" MAY BE USED FOR VACUUM TRUCK CONNECTION WHEN NOT USED BY MOBILE TEST UNIT.
4. MINIMUM 10m DROPOUT SPOOL PROVIDED FOR WELL WORKOVER. HENCE REMOVABLE TYPE PIPE SUPPORT REQUIRED.
3. GRADE ELEVATION 100.000m IS EQUAL TO EL. 254.300m BASED ON PSD/PHD 93 DATUM.
2. TOP OF SLEEPER SHALL BE MAINTAINED AT 450mm ABOVE LOCAL GRADE LEVEL, UNLESS OTHERWISE SPECIFIED.
1. ALL COORDINATES & ELEVATIONS ARE IN m. AND DIMENSIONS ARE IN mm.
NOTES

SPECIAL NOTES FOR PRIMARY SUPPORT:-

A. FOR PRIMARY PIPE SUPPORT (SHOE, GUIDE, LINE STOP, WEAR PAD, ETC.) REFER SP-2411.

B. WHEREVER WELDED SHOE IS USED, WEAR PAD (WP-A1) SHALL BE PROVIDED INLINE WITH SP-2411 REQUIREMENT.

C. WHEREVER GUIDE SUPPORT MENTIONED IN WELL PAD PIPING PLAN 'PG-S' SHALL BE PROVIDED AS PER SP-2411. HOWEVER L75xL75x8 MEMBER SIZE SHALL BE USED INSTEAD OF L50xL50x6.

D. SHOE LENGTH SHALL BE 600mm LONG UNLESS OTHERWISE SPECIFIED. FOR LINE STOP SUPPORT, SHOE LENGTH & WEAR PAD LENGTH SHALL BE FURTHER INCREASED BASED ON SLEEPER / BEAM WIDTH.

23. DURING CSS MODE, OVERRIDE OF PZI-11284 AND 88-PZI-11285 SHALL BE ACTIVATED INHIBITING THE CLOSURE OF 88-UZV-11283 WHEN RETURNING TO OIL PRODUCTION MODE THE OVERRIDE SHALL BE CLEARED.

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PEFS NUMBER	AML-88-106255-PX-2365-00002-0001
ENGINEERING DELIVERABLE REGISTER	AML-88-106255-JA-6611-00006-0000
DESCRIPTION	DRAWING NUMBER

REFERENCE DRAWINGS

Job No.: TKP-24029

TKAVIN ENGINEERING CONSULTANCY LLC.
P.O. Box 2718, PC 130 MUSCAT, OMAN

التسليم للمشاريع المحدودة ش.م.م
AL TASMIN ENTERPRISES LLC

PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)

REV	DATE (DD.MM.YY)	APPROVED BY	DESCRIPTION	PROJECT ID	PROJ.ENG.	SPEC.ENG.	PDO APPROVAL

MASTER DRAWING REVISION
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO COASTAL DESIGN OFFICE

REVISION CLOUD 02

THIS COMPLETE DRAWING IS NEW AND FORMS PART OF PROJECT ID "SST000AL3283"

LEGEND:-

- 1. 'SL' - SLEEPER
- 2. 'PG' - GUIDE
- 3. 'LS' - LIMIT STOP
- 4. 'HD' - HOLDDOWN
- 5. 'SL' - STEEL SUPPORT
- 6. 'SPS' - SPECIAL PIPE SUPPORT
- 7. 'PA' - PIPE ANCHOR SUPPORT

02	26-FEB-2025	DSK	MTP	APM	APPROVED FOR CONSTRUCTION
01	22-AUG-2024	RAL	MTP	APM	APPROVED FOR CONSTRUCTION
REV	DATE	ORIGINATOR	CHECKER	APPROVER	DESCRIPTION

PROJECT DRAWING REVISION		PROJECT ID : SST000AL3283	
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO THE PROJECT ENGINEER			
AREA:	AMAL	CAD DRAWING	SCALE : 1:50
SUB AREA:	AMAL WEST - 'R' STREET	NO. 88	
DRWG. TYPE:	PIPING PLAN		
DESCRIPTION:	THERMAL PRODUCER WELL WITH 300# FL ALW-PH2A-R-52-2-P(ID: 28711)		

AML 88 106255 MP 4033 00005 0001 02

AL-PDO.DWG (SP-2065)



(NOTE=17)

2'-48010XXX-18011

2'-150#

650

400

500

PL-2"-C8010026
BOP, EL +0.450
(NOTE-19)

SPECIAL NOTE 1

FL-4"-M8810086
(H=150mm)
(PP)

FL-3" (FUTURE)

RA'SL-007

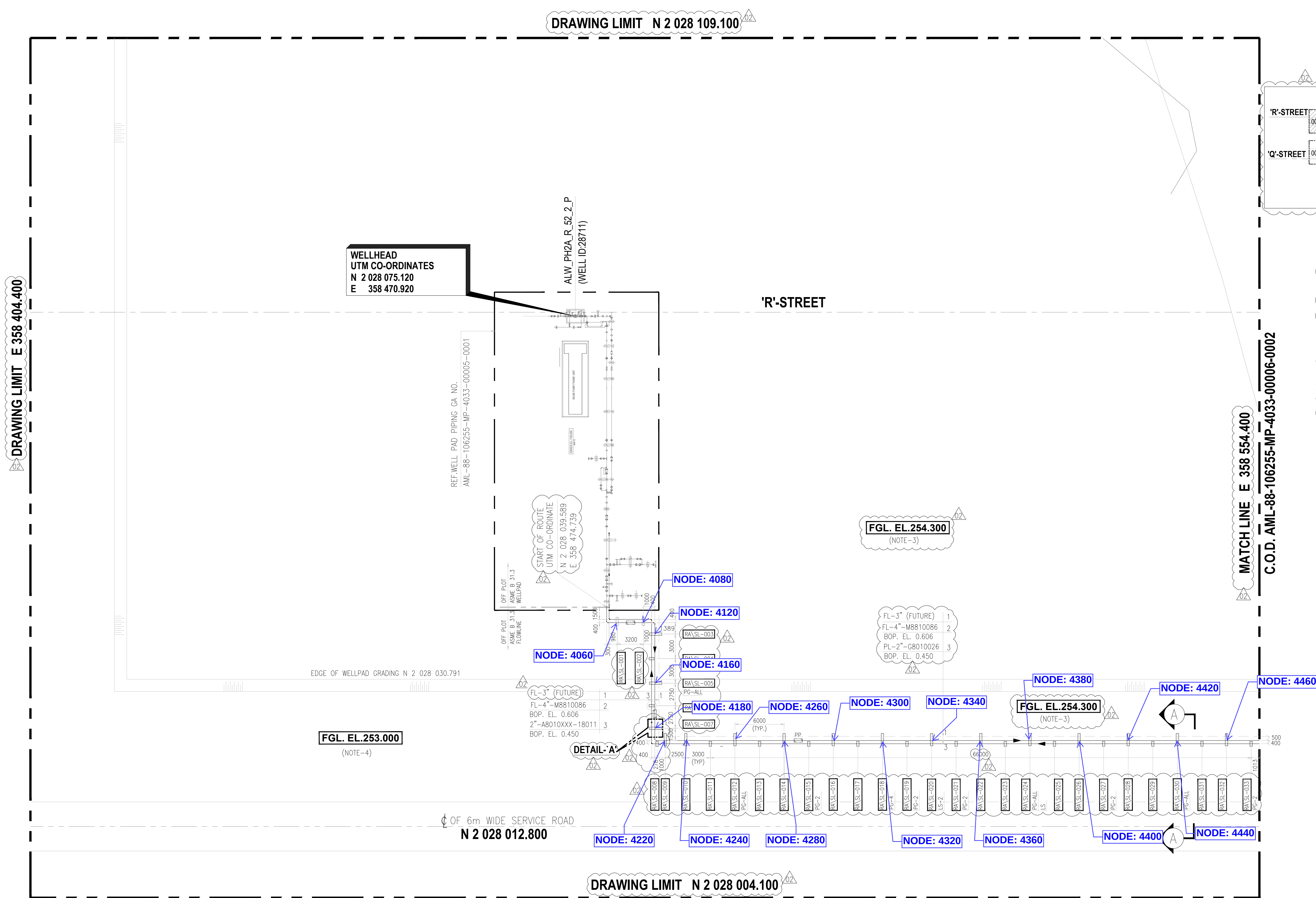
DETAIL-A'

(SCALE: 1:50)

- LEGEND:-**
- 1) 'RA' - R STREET\A MANIFOLD
 - 2) 'SL' - SLEEPER
 - 3) 'BP' - BEARING PLATE
 - 4) 'PS' - PIPE SUPPORT
 - 5) 'PG-X'
├── LINE SIZE
└── PIPE GUIDE
 - 6) 'LS-X'
├── LINE SIZE
└── LINE STOP
- 

SCOPE OF PROJECT
PROJECT ID: 'SST000AL3283'
ALW_PH2A_R_52_2_P
(WELL ID: 28711)

**THIS ENTIRE DRAWING IS NEW AND FORMS
PART OF PROJECT ID: 'SST000AL3283'**

[illegible]

A1-PDO.DWG (SP-2065)

14. 2" THRD. PLUG SHALL BE REMOVE AND CONNECT THE NEW 2" INSTRUMENT AIR LINE FOR ALW. PH2A R_52_2 P.
15. 46mm THK. SS BEARING PLATE TO BE PROVIDED AT PIPE SUPPORT LOCATION FOR ALL STAINLESS STEEL PIPE.
16. HYDROSTATIC VENT & DRAIN LOCATION IS DECIDED AT SITE BY CONTRACTOR.
17. ALL PROCESS & STEAM, DRAIN, VENT & HEADER LINES ARE CATEGORIZED AS STRESS CRITICAL.
18. UNLESS OTHERWISE SPECIFIED, SHOE LENGTH FOR 4" TO 6" LINE SHALL BE 600mm & FOR LINE STOP 800mm WEAR PAD LENGTH SHALL BE FURTHER INCREASED BASED ON SLEEPER / BEAM WIDTH.
19. FLOW LINE SUPPORTS SHALL BE PLACED AT EVERY 6 METERS FOR STRAIGHT SECTIONS AND 5 METERS IN LOOP AREAS UNLESS OTHERWISE SPECIFIED.
20. PRIMARY SUPPORT DETAILS FOR GUIDE / LINE STOP TO BE REFERRED AS PER FOLLOWING PD STANDARD DRAWING FROM SP-2411:
 - a) GUIDE SUPPORT TYPE SHALL BE SELECTED BASED ON LINE SIZE AS SHOWN IN DWG. STD-00-000000-MP-4337-0044-02
 - b) LINE STOP DETAIL SHALL BE SHOWN IN DWG. STD-00-000000-MP-4337-00001-0067.
 - c) WEAR PAD DETAILS SHALL BE SHOWN IN DWG. STD-00-000000-MP-4337-00001-0025.
21. CONSTRUCTABILITY OF PROPOSED ROUTE FROM THE NEW FLOWLINE HAS BEEN STUDIED FROM ENGINEERING SIDE. HOWEVER, DETAIL CONSTRUCTABILITY REVIEW SHALL BE PERFORMED BY CONSTRUCTION TEAM PRIOR TO START OF CONSTRUCTION. IN CASE OF ANY ISSUE OR OBSERVATIONS, CONSTRUCTION TEAM SHALL CONTACT ENGINEERING TEAM FOR FURTHER DISCUSSION AND SUITABLE RESOLUTIONS.
22. CONTRACTOR SHALL VERIFY SIZE, ELEVATION / DEPTH AND LOCATION OF EXISTING FACILITIES (PIPE, CABLE ETC)
23. LONGITUDINAL WELDING OF WEAR PAD TO FLOWLINE SHALL BE AVOIDED AND METAL PASTE TO BE APPLY TO ARREST INGRESS OF WATER BETWEEN THE PIPE AND WEAR PAD.
24. WHEREVER WELDED SHOE IS USED, WEAR PAD (WP-A1) SHALL BE PROVIDED INLINE WITH SP-2411 REQUIREMENT.
25. INSULATION TYPE SHALL BE PERSONAL PROTECTION-PP (WIRE MESH TYPE), INSULATION TO BE PROVIDED ONLY AT ACCESSIBLE LOCATIONS.
26. THIS DRAWING IS PREPARED BASED ON PEFS NO. AML-48-PDFEED-PX-2365-00008-0001 AND URBAN PLAN DRAWING.
27. ALL LINES ARE CATEGORIZED AS STRESS CRITICAL. HENCE ANY MODIFICATION SHALL BE CARRIED OUT WITH PRIOR APPROVAL FROM ENGINEERING CONTRACTOR.
28. FINISHED GRADE LEVEL FOR FLOWLINE CORRIDOR SHALL MATCH WITH WELL PAD GRADE ELEVATION 254.300.
29. GRADE ELEVATION 0.000m IS EQUAL TO ELEVATION 254.300m BASED ON PSD93 DATUM.
30. TOP OF SLEEPER IS AT 450mm ABOVE LOCAL GRADE LEVEL. (FOR NEW SLEEPER).
31. ALL CO-ORDINATES & ELEVATIONS ARE IN m. AND DIMENSIONS ARE IN mm.

SPECIAL NOTE:

1. R-STREET ENTIRE INSTRUMENT AIR MAIN HEADER SHALL BE CONSTRUCTED UPTO INDIVIDUAL WELL HEAD ISOLATION VALVE AT SINGLE TIME.

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PEFS	AML-80-PDFEED-PX-2364-00003-0001
PEFS	AML-88-PDFEED-PX-2365-00008-0001
GRADING LAYOUT AND DETAILS	AML-88-106255-CS-4017-00010-0001
KEY PLAN FOR OFF PLOT PIPING PLAN	AML-88-106255-MP-4033-0000X-0001
INDEX OF DOCUMENT	AML-88-106255-JA-6611-00006-0000

Job No.: TKP-24029

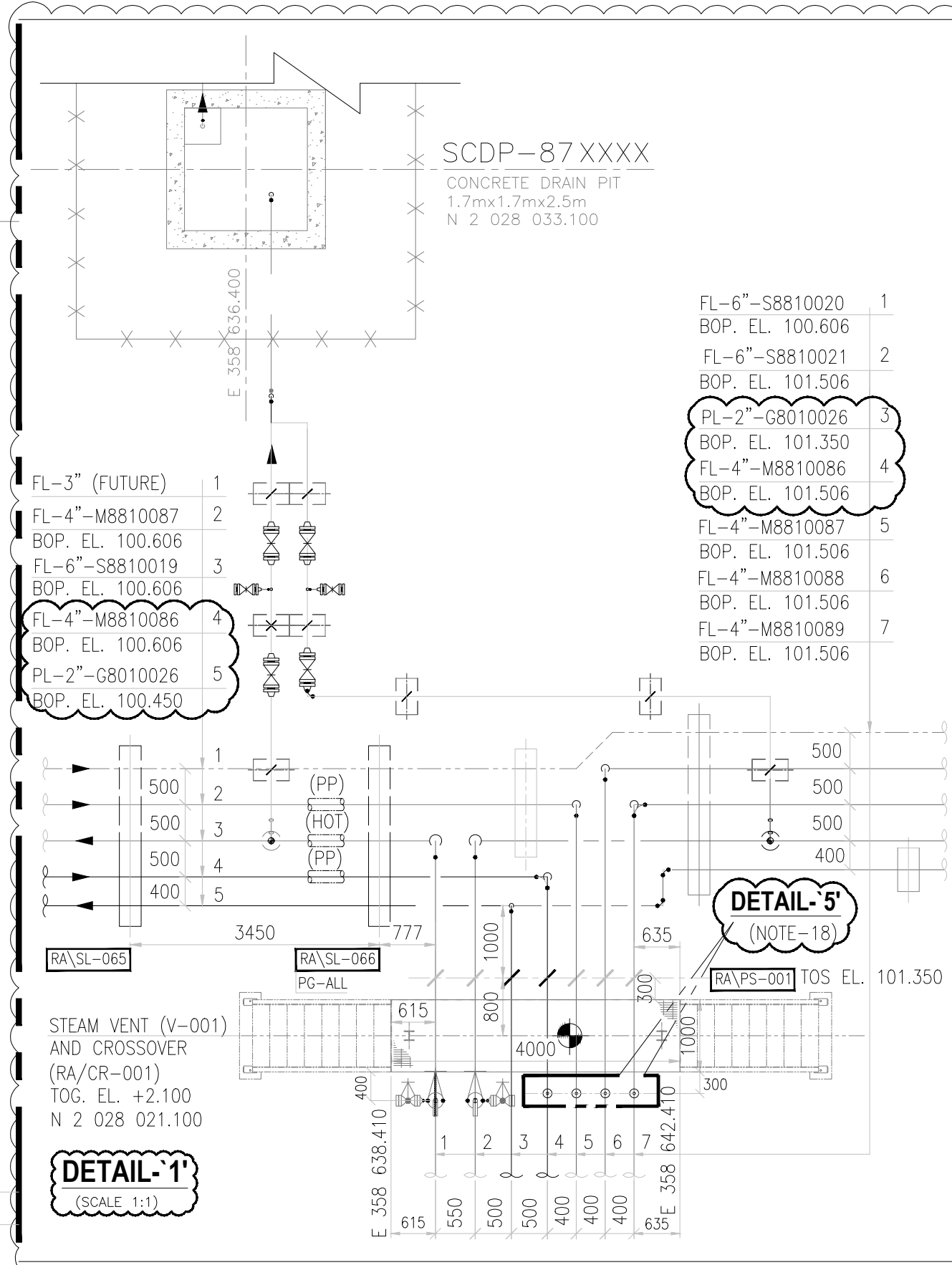
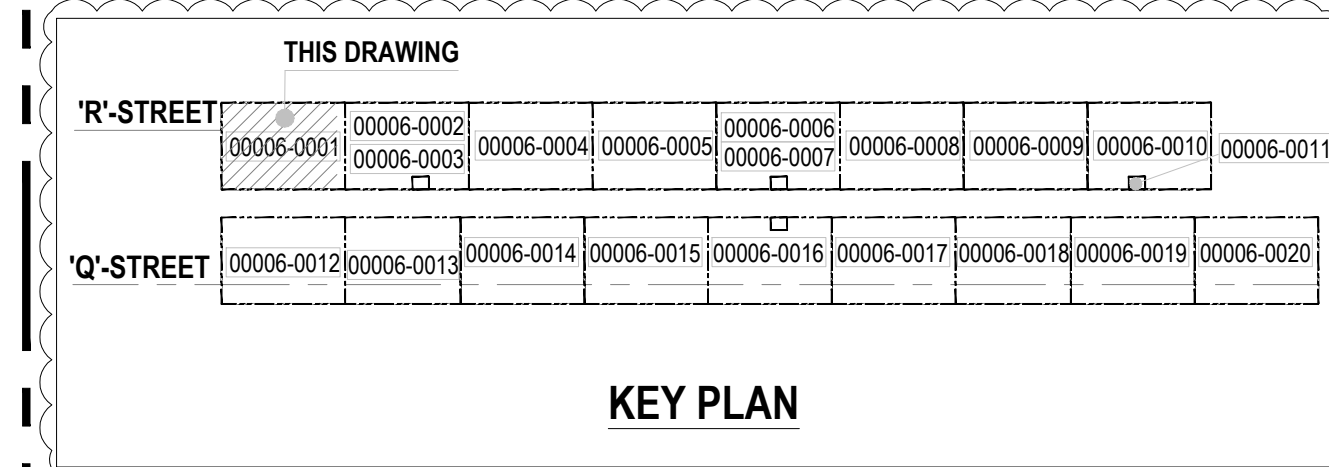
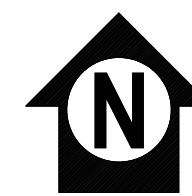
 TKAVIN ENGINEERING
CONSULTANCY LLC.
P.O. Box 2718, PC 130 MUSCAT, OMAN

التسليم للمشاريع المحدودة ش.م.م
AL TASNIM ENTERPRISES LLC

PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)

REV	DATE (DD.MM.YY)	APPROVED BY	DESCRIPTION	WBS NO 7 PROJECT ID	PROJ.ENG.	SPEC.ENG.
MASTER DRAWING REVISION IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO COASTAL DESIGN OFFICE						

LAST SAVED BY T-KAVIN ON 27/02/2025 4:13 PM PLOTTED BY T-KAVIN ON 2/27/2025 4:22 PM FILENAME: Z:\TKAVIN\PROJECTS\2024\TKP-24029 - DE FOR AMAL PHASE 2A HOOK UP-ATNM\8.0 DETAILED ENGINEERING FOR AMAL PH-2A BP WELL (300' FL) R-52-2-P (ID-28711)\7.0 PIPING\7.1 FLOWLINE PIPING PLAN\MASTER\REV.02\AMAL-88-106255-MP-4033-00006-0001_02.DWG

**LEGEND:-**

- 'RA' - R STREET\A MANIFOLD
 - 'SL' - SLEEPER
 - 'BP' - BEARING PLATE
 - 'PS' - PIPE SUPPORT
 - 'PG-X'
 - 'LS-X'
- LINE SIZE
PIPE GUIDE
LINE STOP

REVISION CLOUD

MATCH LINE N 2 028 004.100

C.O.D. AML-88-PDFEED-MP-4033-00008-0014

SPECIAL NOTE:

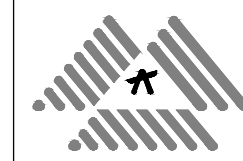
- R-STREET ENTIRE INSTRUMENT AIR MAIN HEADER SHALL BE CONSTRUCT UPTO INDIVIDUAL WELL HEAD ISOLATION VALVE AT SINGLE TIME.

SCOPE OF PROJECT

PROJECT ID: 'SST000AL3283'
ALW_PH2A_R_52_2_P
(WELL ID: 28711)

**THIS ENTIRE DRAWING IS NEW AND FORMS
PART OF PROJECT ID: 'SST000AL3283'**

Job No.: TKP-24029



**TKAVIN ENGINEERING
CONSULTANCY LLC.**
P.O. Box 2718, PC 130 MUSCAT, OMAN

التسليم للمشاريع المحدودة ش م م
AL TASNIM ENTERPRISES LLC



PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)

02	27-02-2025	THS	PMD/MP	APM	APPROVED FOR CONSTRUCTION		
01	22-08-2024	NVP	PMD/RR	APM	APPROVED FOR CONSTRUCTION		
REV	DATE	ORIGINATOR	CHECKER	APPROVER	DESCRIPTION	TA REF. NO.	SIGN
PROJECT DRAWING REVISION					PROJECT ID: SST000AL3283		
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO THE PROJECT ENGINEER							
AREA:	AMAL (PHASE-2A)				SCALE: 1:250	CAD DRAWING	
SUB AREA:	AMAL WEST				NO. 88	شركة تنمية نفط عمان Petroleum Development Oman	
DRWG. TYPE:	PIPING PLAN					AREA CODE	UNIT CODE
DESCRIPTION:	AMAL WEST OFF PLOT (R STREET)					ORIGINATOR CODE	DISC. CODE
						DOC. TYPE	SEQ. NO.
						SHT. NO.	REV. NO.
						AML 88	106255 MP 4033 00006 0002 02

UEFS	AML-80-PDFEED-PX-2364-00003-0001
PEFS	AML-88-PDFEED-PX-2365-00008-0001
GRADING LAYOUT AND DETAILS	AML-88-106255-CS-4017-00010-0001
KEY PLAN FOR PIPING PLAN	AML-88-106255-MP-4033-00005-0001
INDEX OF DOCUMENT	AML-88-106255-JA-6611-00006-0000

DESCRIPTION	DRAWING NUMBER
REFERENCE DRAWINGS	
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO COASTAL DESIGN OFFICE	

REV	DATE (DD.MM.YY)	APPROVED BY	DESCRIPTION	WBS NO / PROJECT ID	PROJ. ENG.	SPEC. ENG.
					PDO APPROVAL	

REFERENCE DRAWINGS

IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO COASTAL DESIGN OFFICE

MATCH LINE E 358 554.400
C.O.D. AML-88-106255-MP-4033-00006-0001MATCH LINE E 358 704.400
C.O.D. AML-88-106255-MP-4033-00006-0004

NODE: 4580

NODE: 4540

NODE: 4480

NODE: 4700

NODE: 4720

NODE: 4820

NODE: 4840

NODE: 4860

NODE: 4880

NODE: 4900

NODE: 4920

NODE: 4940

NODE: 4950

NODE: 5020

18. FOR SECTIONS & DETAILS REFER DRAWING NO. AML-87-106255-MP-4033-00006-0027.
17. 6mm THK. SS BEARING PLATE TO BE PROVIDED AT PIPE SUPPORT LOCATION FOR ALL STAINLESS STEEL PIPE.
16. HYDROSTATIC VENT & DRAIN LOCATION IS DECIDED AT SITE BY CONTRACTOR.
15. ALL PROCESS & STEAM, DRAIN, VENT & HEADER LINES ARE CATEGORIZED AS STRESS CRITICAL.
14. UNLESS OTHERWISE SPECIFIED, SHOE LENGTH FOR 4" TO 8" LINE SHALL BE 800mm & FOR LINE STOP 800mm. WEAR PAD LENGTH SHALL BE FURTHER INCREASED BASED ON SLEEPER / BEAM WIDTH.
13. FLOW LINE SUPPORTS SHALL BE PLACED AT EVERY 6 METERS FOR STRAIGHT SECTIONS AND 5 METERS IN LOOP AREAS UNLESS OTHERWISE SPECIFIED.
12. PRIMARY SUPPORT DETAILS FOR GUIDE / LINE STOP TO BE REFERRED AS PER FOLLOWING PDO STANDARD DRAWING FROM SP-2411:
- a) GUIDE SUPPORT TYPE SHALL BE SELECTED BASED ON LINE SIZE AS SHOWN IN DWG. STD-00-000000-MP-4337-0044-02
- b) LINE STOP DETAIL SHALL BE SHOWN IN DWG. STD-00-000000-MP-4337-00001-0067-02
- c) WEAR PAD DETAILS SHALL BE SHOWN IN DWG. STD-00-000000-MP-4337-00001-0025-02
- d) U-BOLT DETAILS SHALL BE SHOWN IN DWG. STD-00-000000-MP-4337-00001-0101-02.
11. CONSTRUCTABILITY OF PROPOSED ROUTE FROM THE NEW FLOWLINE HAS BEEN STUDIED FROM ENGINEERING SIDE. HOWEVER, DETAIL CONSTRUCTABILITY REVIEW SHALL BE PERFORMED BY CONSTRUCTION TEAM PRIOR TO START OF CONSTRUCTION. IN CASE OF ANY ISSUE OR OBSERVATIONS, CONSTRUCTION TEAM SHALL CONTACT ENGINEERING TEAM FOR FURTHER DISCUSSION AND SUITABLE RESOLUTIONS.
10. CONTRACTOR SHALL VERIFY SIZE, ELEVATION / DEPTH AND LOCATION OF EXISTING FACILITIES (PIPE, CABLE ETC)
9. LONGITUDINAL WELDING OF WEAR PAD TO FLOWLINE SHALL BE AVOIDED AND METAL PASTE TO BE APPLIED TO ARREST INGRESS OF WATER BETWEEN THE PIPE AND WEAR PAD.
8. WHEREVER WELDED SHOE IS USED, WEAR PAD (WP-A1) SHALL BE PROVIDED INLINE WITH SP-2411 REQUIREMENT.
7. INSULATION TYPE SHALL BE PERSONAL PROTECTION-PP (WIRE MESH TYPE). INSULATION TO BE PROVIDED ONLY AT ACCESSIBLE LOCATIONS.
6. THIS DRAWING IS PREPARED BASED ON PEFS NO. AML-88-PDFEED-PX-2365-00008-0001 AND URBAN PLAN DRAWING.
5. ALL LINES ARE CATEGORIZED AS STRESS CRITICAL. HENCE ANY MODIFICATION SHALL BE CARRIED OUT WITH PRIOR APPROVAL FROM ENGINEERING CONTRACTOR.
4. FINISHED GRADE LEVEL FOR FLOWLINE CORRIDOR SHALL MATCH WITH WELL PAD GRADE ELEVATION 254.300.
3. GRADE ELEVATION 0.000m IS EQUAL TO ELEVATION 254.300m BASED ON PSD93 DATUM.
2. TOP OF SLEEPER IS AT 450mm ABOVE LOCAL GRADE LEVEL (FOR NEW SLEEPER).
1. ALL CO-ORDINATES & ELEVATIONS ARE IN m. AND DIMENSIONS ARE IN mm.

NOTES

A1-PDO.DWG (SP-2065)

LAST SAVED BY T-KAVIN ON 27/02/2025 4:13 PM PLOTTED BY T-KAVIN ON 2/27/2025 4:21 PM FILENAME: Z:\TKAVIN\PROJECTS\2024\TKP-24029 - DE FOR AMAL PHASE 2A HOOK UP-ATNM8.0 DETAILED ENGINEERING FOR AMAL PH-2A BP WELL (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.1 FLOWLINE PIPING PLAN\MASTER\REV.02\AML-88-106255-MP-4033-00006-0002_02.DWG



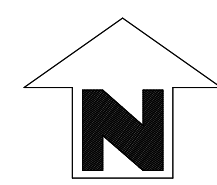
7. FOR INSTRUMENT AIR LINE PIPING CLASS 18011 IS CONSIDERED.
- 6mm THK. SS BEARING PLATE TO BE PROVIDED AT PIPE SUPPORT LOCATION FOR ALL STAINLESS STEEL PIPE.
5. ALL PROCESS & STEAM, DRAIN, VENT & HEADER LINES ARE CATEGORIZED AS STRESS CRITICAL.
4. INSULATION TYPE SHALL BE PERSONAL PROTECTION-PP (WIRE MESH TYPE). INSULATION TO BE PROVIDED ONLY AT ACCESSIBLE LOCATIONS.
3. FOR GRADE ELEVATION 100.000m IS EQUAL TO EL 254.600m BASED ON PSD/PHD 93 DATUM.
2. FOR CABLE TRAY DETAILS REFER ELECTRICAL & INSTRUMENT GA DRAWINGS.
1. ALL DIMENSIONS, ELEVATIONS AND COORDINATES ARE IN mm UNLESS OTHERWISE NOTED.

NOTES

LAST SAVED BY 348 ON 28/01/2025 4:28 PM PLOTTED BY T-KAVIN ON 2/27/2025 4:23 PM FILENAME: Z:\TKAVIN\PROJECTS\2024\TKP-24029 - DE FOR AMAL PHASE 2A HOOK UP-ATNM\8.0 DETAILED ENGINEERING FOR AMAL PH-2A BP WELL (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.1 FLOWLINE PIPING PLAN\MASTER\REV.02\AML-88-PDFEED-MP-4033-00008-0014-02.DWG

LIST OF SUPPORT					
SUPPORT TAG	PIPE SIZE	SUPPORT TYPE	LENGTH	TOS. EL.	QTY
PS1-01A	2"x 1"	Rest+Steel	0.5m	+100.977	1
BP-A1-2	2"	Rest	NOTE-6	-	1

PLANT NORTH



SN	TYPE	PS(REF)	GAP	BOP-EL	COORDINATES	SN	TYPE	PS(REF)	GAP	BOP-EL	COORDINATES
S1	REST	SL-5307		100550	E 155065 N 131478	S39	PG-50	PS-5308		100977	E 151815 N 129150
S2	PG-100	SL-5307		100550	E 154665 N 131478	S40	REST	PS-5308		101140	E 152364 N 129150
S3	SH-150	PS-5435		101100	E 153115 N 131395	S41	REST	SL-5132		100550	E 157115 N 128525
S4	SH-250	PS-5435		101140	E 152364 N 131395	S42	PG-100	SL-5212		100550	E 161915 N 128525
S5	REST	SL-5307		100550	E 155865 N 131478	S43	REST	SL-5212		100550	E 161915 N 128525
S6	W.PAD	PS-5435		101100	E 153115 N 131395	S44	REST	PS-5308		101100	E 153665 N 129150
S7	PG-100	SL-5307		100550	E 155865 N 131478	S45	REST	PS-5225		101140	E 150984 N 128475
S8	LS-100	SL-5307		100550	E 155065 N 131478	S46	REST	PS-5308		101696	E 152785 N 128875
S9	REST	SL-5307		100550	E 155465 N 131478	S47	REST	PS-5308		100550	E 155665 N 129000
S10	REST	PS-5435		100977	E 151815 N 131395	S48	SH-250	PS-5225		101140	E 150984 N 128475
S11	LS-150	PS-5435		101100	E 153115 N 131395	S49	SH-150	PS-5308		101100	E 153665 N 129150
S12	PG-150	PS-5435	25	101100	E 153665 N 131395	S50	REST	SL-5133		100550	E 154665 N 128075
S13	SH-150	PS-5435		101100	E 153665 N 131395	S51	REST	SL-5133		100550	E 155065 N 128075
S14	LS-100	SL-5307		100550	E 155865 N 131478	S52	SH-150	PS-5225		101100	E 150984 N 127925
S15	REST	PS-5435		101100	E 153665 N 131395	S53	LS-100	SL-5132		100550	E 157115 N 128125
S16	LS-250	PS-5435		101140	E 152364 N 131395	S54	REST	SL-5132		100550	E 157115 N 128125
S17	PG-50	PS-5435		100977	E 151815 N 131395	S55	LS-100	SL-5212		100550	E 161915 N 128525
S18	W.PAD	SL-5307		100550	E 154665 N 131478	S56	W.PAD	SL-5212		100550	E 161915 N 128525
S19	W.PAD	SL-5307		100550	E 155465 N 131478	S57	REST	PS-5225		101100	E 150984 N 127925
S20	PG-100	SL-5307		100550	E 155065 N 131478	S58	W.PAD	SL-5132		100550	E 157115 N 128125
S21	REST	SL-5307		100550	E 154665 N 131478	S59	LS-100	SL-5132		100550	E 157115 N 127725
S22	W.PAD	SL-5307		100550	E 155865 N 131478	S60	REST	PS-5226		100976	E 153065 N 127575
S23	REST	PS-5435		101140	E 152364 N 131395	S61	REST	SL-5208		100550	E 160665 N 128125
S24	W.PAD	PS-5435		101100	E 153665 N 131395	S62	PG-50	PS-5225		100976	E 150984 N 127575
S25	LS-100	SL-5307		100550	E 155465 N 131478	S63	REST	PS-5225		100976	E 150984 N 127575
S26	LS-100	SL-5307		100550	E 154665 N 131478	S64	REST	SL-5136		100550	E 159665 N 127277
S27	W.PAD	PS-5435		101140	E 152364 N 131395	S65	REST	SL-5207		100550	E 161665 N 127275
S28	W.PAD	SL-5307		100550	E 155065 N 131478	S66	REST	SL-5211		100550	E 163665 N 127275
S29	REST	PS-5435		101100	E 153115 N 131395	S67	W.PAD	SL-5132		100550	E 157115 N 127725
S30	LS-150	PS-5435		101100	E 153665 N 131395	S68	REST	SL-5132		100550	E 157115 N 127725
S31	PG-100	SL-5307		100550	E 155465 N 131478	S69	REST	PS-5350		101555	E 154459 N 127575
S32	PG-250	PS-5435	25	101140	E 152364 N 131395	S70	PG-50	PS-5350		101555	E 154459 N 127575
S33	PG-150	PS-5435	25	101100	E 153115 N 131395	S71	REST	PS-5226		101100	E 153065 N 127225
S34	REST	PS-5308		100977	E 151815 N 129150	S72	PG-100	PS-5396		100550	E 153415 N 126594
S35	SH-150	PS-5308		101100	E 153115 N 129150	S73	REST	PS-5429		101555	E 156409 N 127349
S36	REST	PS-5308		101100	E 153115 N 129150	S74	SH-150	PS-5226		101100	E 153065 N 127225
S37	SH-250	PS-5308		101140	E 152364 N 129150	S75	REST	PS-5396		100550	E 153415 N 126594
S38	REST	SL-5209		100550	E 155465 N 129000						

DRAWING LIMIT N 134500mm

C.O.D. AML-88-106255-MP-4033-00006-0003
UTM N 2 028 004.100

NODE: 5220

NODE: 5240

NODE: 5280

FGL.E.L. +100000
(NOTE-3)

PLAN AT EL. +100000 & ABOVE

LEGEND:

- RESTING SUPPORT
- GUIDE SUPPORT
- AXIAL STOP SUPPORT
- FIXED POINT (AXIAL STOP & GUIDE SUPPORT)
- ANCHOR (WELDED OR BOLTED)
- U BOLT

PIPE SUPPORT LEGEND		ELEVATION LEGEND	
PG	PIPE GUIDE	BOP	BOTTOM OF PIPE
LS	LIMIT STOP	CL	CENTER LINE
TR	TRUNNION	WP	WORK POINT
SH	SHOE		
UB	U BOLT		
WP	WEAR PAD		

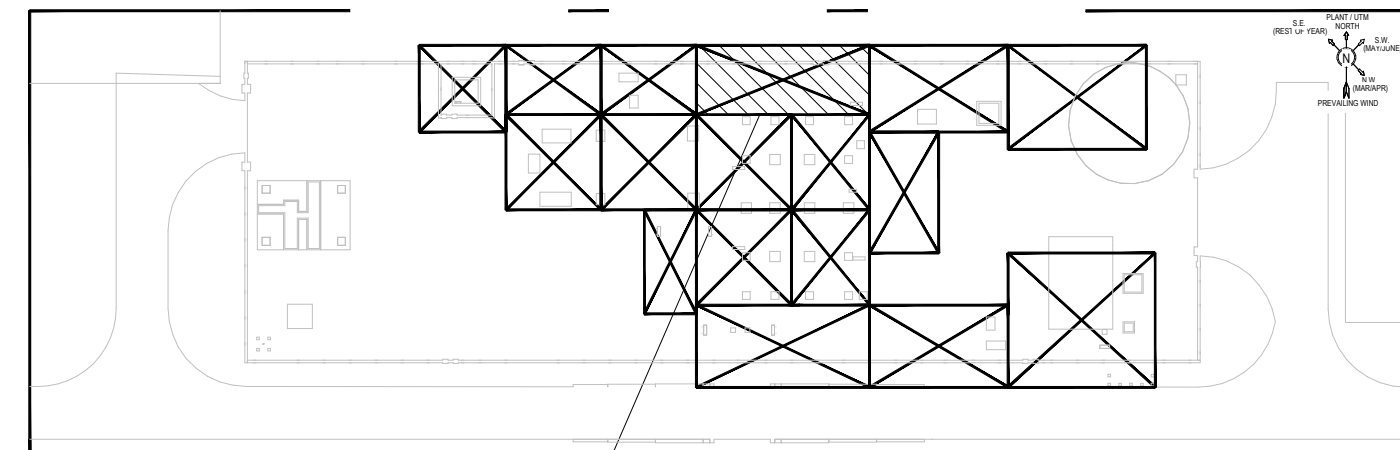
REVISION CLOUD

LEGEND:-

- 'RA' - R STREET/A MANIFOLD
 - 'SL' - SLEEPER
 - 'BP' - BEARING PLATE
 - 'PS' - PIPE SUPPORT
 - 'PG-X'
 - 'LS-X'
- LINE SIZE
PIPE GUIDE
LINE STOP

DEMOLITION SCOPE
PROJECT ID: SST000AL3283
ALW_PH2A_R_52_2_P
(WELL ID: (ID 28711))

SCOPE OF PROJECT
PROJECT ID: 'SST000AL3283'
ALW_PH2A_R_52_2_P
(WELL ID: 28711)

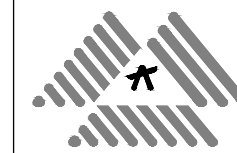


THIS DRAWING
KEY PLAN

THIS DRAWING IS BASED ON TKAVIN PROJECT
DRAWING NO. AML-88-PDFEED-MP-4033-00008-0014 REV.01
PROJECT ID: SST000AL3283 AND FORMS PART OF PROJECT ID: SST000AL3283

This document is issued for the party which commissioned it and for specific purposes connected with the captioned project only. It should not be relied upon by any other party or used for any other purpose. We accept no responsibility for the consequences of this document being relied upon by any other party, or being used for any other purpose, or containing any error or omission which is due to an error or omission in data supplied to us by other parties.

Job No.: TKP-24029



TKAVIN ENGINEERING
CONSULTANCY LLC.
P.O. Box 2718, PC 130 MUSCAT, OMAN

التسليم للمشروع المحدودة ش م م
AL TASNIM ENTERPRISES LLC

PROJECT:
OFF PLOT DELIVERY CONTRACT ODC SOUTH
(C-3100000177)



REV	DATE	ORIGINATOR	CHECKER	APPROVER	DESCRIPTION	TA	REF.IND.	SIGN
02	27-02-2025	THS	PMD/MP	APM	APPROVED FOR CONSTRUCTION			
01	22-08-2024	NVP	PMD/RR	APM	APPROVED FOR CONSTRUCTION			

PROJECT DRAWING REVISION PROJECT ID : SST000AL3283
IN CASE OF DEVIATIONS A MARKED UP PRINT SHALL BE SENT TO THE PROJECT ENGINEER

AREA: AMAL WEST OFFPLOT FACILITIES NO. 88

SUB AREA: AMAL PHASE 2A

DRWG. TYPE: PIPING PLAN

DESCRIPTION: Q-R STREET (88-MS-10005A)

SCALE : 1:50

CAD DRAWING

شركة تنمية نفط عمان
Petroleum Development Oman

AREA CODE UNIT CODE ORIGINATOR CODE DISC. CODE TYPE SEQ. NO. SHT. NO. REV. NO.

AML 88 PDFEED MP 4033 00008 0014 02

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 5

INPUT ECHO FROM CAESAR II

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

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CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS

- 1 (HYD) WW+HP
- 2 (OPE) W+D1+T1+P1
- 3 (Alt-SUS) W+P1
- 4 (OPE) W+T2+P2
- 5 (Alt-SUS) W+P2
- 6 (OPE) W+T3+P1
- 7 (Alt-SUS) W+P1
- 8 (SUS) W+P1
- 9 (SUS) W+P2
- 10 (EXP) L10=L2-L8
- 11 (EXP) L11=L4-L9
- 12 (EXP) L12=L6-L8
- 13 (EXP) L13=L10-L12
- 14 (EXP) L14=L11-L12

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

Job Description:

PROJECT:

CLIENT :

ANALYST:

NOTES :

CAESAR II Ver.13.00.00.0890, (Build 220811) MAR 26, 2025 8:45:56

PIPE DATA

From 1110 To 4020 DX= 88.900 mm.

PIPE

Dia= 114.300 mm. Wall= 6.020 mm. Cor= 3.0000 mm.

Element Name=FLANGE_FLG_300

GENERAL

T1= 21 C T2= 260 C T3= 5 C P1= 100.0000 KPa P2= 4,140.0000 KPa

PHyd= 7,665.0000 KPa Mat= (106)A106 Grade B E= 202,711,488 KPa

EH1= 202,711,488 KPa EH2= 188,226,864 KPa EH3= 203,523,824 KPa

EH4= 202,711,488 KPa EH5= 202,711,488 KPa EH6= 202,711,488 KPa

EH7= 202,711,488 KPa EH8= 202,711,488 KPa EH9= 202,711,488 KPa

v = .292 Pipe Den= .0078334 kg./cu.cm. Fluid Den= .0010000 kg./cu.cm.

Insul Thk= .000 mm. Insul Den= .0001400 kg./cu.cm.

RIGID Weight= 115.65 N. Type=Flange

ALLOWABLE STRESSES

B31.3 IX (2018) Sc= 160,648 KPa Sh1= 160,648 KPa Sh2= 131,000 KPa

Sh3= 160,648 KPa Sh4= 160,648 KPa Sh5= 160,648 KPa Sh6= 160,648 KPa

Sh7= 160,648 KPa Sh8= 160,648 KPa Sh9= 160,648 KPa Sy= 241,316 KPa

From 4020 To 4030 DX= 1,500.000 mm.

From 4030 To 4040 DX= 314.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4039

Angle/Node @2= .00 4038

From 4040 To 4060 DZ= -1,280.000 mm.

RESTRAINTS

Node 4060 +Y Mu = .30 Tag = RA\SL-001

From 4060 To 4080 DZ= -3,200.000 mm.

RESTRAINTS

Node 4080 +Y Mu = .30 Tag = RA\SL-002

From 4080 To 4100 DZ= -1,400.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4099

Angle/Node @2= .00 4098

From 4100 To 4120 DX= 1,389.000 mm.

RESTRAINTS

Node 4120 +Y Mu = .30 Tag = RA\SL-003

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INPUT LISTING

From 4120 To 4140 DX= 3,000.000 mm.

From 4140 To 4160 DX= 3,000.000 mm.

RESTRAINTS

Node 4160 +Y Mu = .30 Tag = RA\SL-005

Node 4160 Guide Mu = .30 Tag = RA\SL-005

From 4160 To 4180 DX= 5,500.000 mm.

RESTRAINTS

Node 4180 +Y Mu = .30 Tag = RA\SL-007

From 4180 To 4200 DX= 1,500.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4199

Angle/Node @2= .00 4198

From 4200 To 4220 DZ= -1,278.000 mm.

RESTRAINTS

Node 4220 +Y Mu = .30 Tag = RA\SL-009

From 4220 To 4240 DZ= -2,500.000 mm.

RESTRAINTS

Node 4240 +Y Mu = .30 Tag = RA\SL-010

From 4240 To 4260 DZ= -6,000.000 mm.

RESTRAINTS

Node 4260 +Y Mu = .30 Tag = RA\SL-012

Node 4260 Guide Mu = .30 Tag = RA\SL-012

From 4260 To 4280 DZ= -6,000.000 mm.

RESTRAINTS

Node 4280 +Y Mu = .30 Tag = RA\SL-014

From 4280 To 4300 DZ= -6,000.000 mm.

RESTRAINTS

Node 4300 +Y Mu = .30 Tag = RA\SL-016

From 4300 To 4320 DZ= -6,000.000 mm.

RESTRAINTS

Node 4320 +Y Mu = .30 Tag = RA\SL-018

Node 4320 Guide Tag = RA\SL-018

From 4320 To 4340 DZ= -6,000.000 mm.

RESTRAINTS

Node 4340 +Y Mu = .30 Tag = RA\SL-020

Node 4340 LIM Tag = RA\SL-020

From 4340 To 4360 DZ= -6,000.000 mm.

RESTRAINTS

Node 4360 +Y Mu = .30 Tag = RA\SL-022

From 4360 To 4380 DZ= -6,000.000 mm.

RESTRAINTS

Node 4380 +Y Mu = .30 Tag = RA\SL-024

Node 4380 Guide Tag = RA\SL-024

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INPUT LISTING

From 4380 To 4400 DZ= -6,000.000 mm.

RESTRAINTS

Node 4400 +Y Mu = .30 Tag = RA\SL-026

From 4400 To 4420 DZ= -6,000.000 mm.

RESTRAINTS

Node 4420 +Y Mu = .30 Tag = RA\SL-028

From 4420 To 4440 DZ= -6,000.000 mm.

RESTRAINTS

Node 4440 +Y Mu = .30 Tag = RA\SL-030

Node 4440 Guide Tag = RA\SL-030

From 4440 To 4460 DZ= -6,000.000 mm.

RESTRAINTS

Node 4460 +Y Mu = .30 Tag = RA\SL-032

From 4460 To 4480 DZ= -5,500.000 mm.

RESTRAINTS

Node 4480 +Y Mu = .30 Tag = RA\SL-034

From 4480 To 4500 DZ= -1,200.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4499

Angle/Node @2= .00 4498

From 4500 To 4520 DX= -1,000.000 mm.

From 4520 To 4540 DX= -1,576.000 mm.

RESTRAINTS

Node 4540 +Y Mu = .30 Tag = RA\SL-036

From 4540 To 4560 DX= -2,424.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4559

Angle/Node @2= .00 4558

From 4560 To 4580 DZ= -1,150.000 mm.

RESTRAINTS

Node 4580 +Y Mu = .30 Tag = RA\SL-037

From 4580 To 4600 DZ= -2,700.000 mm.

RESTRAINTS

Node 4600 +Y Mu = .30 Tag = RA\SL-038

From 4600 To 4620 DZ= -1,150.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4619

Angle/Node @2= .00 4618

From 4620 To 4640 DX= 2,424.000 mm.

RESTRAINTS

Node 4640 +Y Mu = .30 Tag = RA\SL-039

From 4640 To 4660 DX= 1,576.000 mm.

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INPUT LISTING

```

-----
From 4660 To 4680 DX= 1,000.000 mm.
BEND at "TO" end
  Radius= 152.400 mm. (LONG)   Bend Angle= 90.000   Angle/Node @1= 45.00 4679
  Angle/Node @2= .00 4678
-----
From 4680 To 4700 DZ= -1,200.000 mm.
RESTRAINTS
  Node 4700 +Y Mu = .30 Tag = RA\SL-040
-----
From 4700 To 4720 DZ= -5,500.000 mm.
RESTRAINTS
  Node 4720 +Y Mu = .30 Tag = RA\SL-042
  Node 4720 Guide Tag = RA\SL-042
-----
From 4720 To 4740 DZ= -6,000.000 mm.
RESTRAINTS
  Node 4740 +Y Mu = .30 Tag = RA\SL-044
-----
From 4740 To 4760 DZ= -6,000.000 mm.
RESTRAINTS
  Node 4760 +Y Mu = .30 Tag = RA\SL-046
-----
From 4760 To 4780 DZ= -6,000.000 mm.
RESTRAINTS
  Node 4780 +Y Mu = .30 Tag = RA\SL-048
  Node 4780 Guide Tag = RA\SL-048
-----
From 4780 To 4800 DZ= -5,500.000 mm.
RESTRAINTS
  Node 4800 +Y Mu = .30 Tag = RA\SL-050
-----
From 4800 To 4820 DZ= -3,500.000 mm.
RESTRAINTS
  Node 4820 +Y Mu = .30 Tag = RA\SL-051
-----
From 4820 To 4840 DZ= -6,000.000 mm.
RESTRAINTS
  Node 4840 +Y Mu = .30 Tag = RA\SL-053
  Node 4840 Guide Tag = RA\SL-053
-----
From 4840 To 4860 DZ= -4,500.000 mm.
RESTRAINTS
  Node 4860 +Y Mu = .30 Tag = RA\SL-055
  Node 4860 LIM Tag = RA\SL-055
-----
From 4860 To 4880 DZ= -4,500.000 mm.
RESTRAINTS
  Node 4880 +Y Mu = .30 Tag = R8\SL-057
-----
From 4880 To 4900 DZ= -6,000.000 mm.
RESTRAINTS
  Node 4900 +Y Mu = .30 Tag = RA\SL-059
  Node 4900 Guide Mu = .30 Tag = RA\SL-059
-----
From 4900 To 4920 DZ= -6,000.000 mm.

```

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INPUT LISTING

RESTRAINTS

Node 4920 +Y Mu = .30 Tag = RA/SL-061

From 4920 To 4940 DZ= -6,000.000 mm.

RESTRAINTS

Node 4940 +Y Mu = .30 Tag = RA/SL-063

From 4940 To 4950 DZ= -6,000.000 mm.

RESTRAINTS

Node 4950 +Y Mu = .30 Tag = RA/SL-065

Node 4950 Guide Tag = RA/SL-065

From 4950 To 4960 DZ= -3,450.000 mm.

RESTRAINTS

Node 4960 +Y Mu = .30 Tag = RA/SL-066

From 4960 To 4980 DZ= -2,327.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4979

Angle/Node @2= .00 4978

From 4980 To 5000 DY= 900.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 4999

Angle/Node @2= .00 4998

From 5000 To 5020 DX= 1,400.000 mm.

RESTRAINTS

Node 5020 +Y Mu = .30 Tag = RA/PS-001

From 5020 To 5040 DX= 3,600.000 mm.

RESTRAINTS

Node 5040 +Y Mu = .30 Tag = RA/PS-002

From 5040 To 5060 DX= 1,529.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 5059

Angle/Node @2= .00 5058

From 5060 To 5080 DY= -956.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 5079

Angle/Node @2= .00 5078

From 5080 To 5100 DX= 1,000.000 mm.

RESTRAINTS

Node 5100 +Y Mu = .30 Tag = RA/SL-68

From 5100 To 5120 DX= 5,000.000 mm.

RESTRAINTS

Node 5120 +Y Mu = .30 Tag = RA/SL-70

From 5120 To 5140 DX= 5,000.000 mm.

RESTRAINTS

Node 5140 +Y Mu = .30 Tag = RA/SL-72

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INPUT LISTING

From 5140 To 5150 DX= 2,800.000 mm.

From 5150 To 5160 DX= 216.000 mm.

From 5160 To 5180 DX= 88.900 mm.

Element Name=FLANGE_NOFLG_300

RIGID Weight= 115.65 N. Type=Flange

FLANGES

Location= To Method= NC-3658.3 G/C= 200.000 mm.

Bolt Area= 1,723.860 mm. SYC= 241,000 KPa SY1= 188,000 KPa

SY2= 188,000 KPa SY3= 188,000 KPa SY4= 188,000 KPa SY5= 188,000 KPa

SY6= 188,000 KPa SY7= 188,000 KPa SY8= 188,000 KPa SY9= 188,000 KPa

From 5180 To 5200 DX= 88.900 mm.

Element Name=FLANGE_NOFLG_300

RIGID Weight= 115.65 N. Type=Flange

From 5200 To 5220 DX= 1,530.000 mm.

RESTRAINTS

Node 5220 +Y Mu = .30 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)

Node 5220 Guide Mu = .30 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)

Node 5220 LIM Mu = .30 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)

From 5220 To 5240 DX= 3,405.000 mm.

RESTRAINTS

Node 5240 +Y Mu = .30 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)

From 5240 To 5260 DX= 1,130.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 5259

Angle/Node @2= .00 5258

From 5260 To 5270 DZ= 1,250.000 mm.

BEND at "TO" end

Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 5269

Angle/Node @2= .00 5268

From 5270 To 5280 DX= 353.000 mm.

RESTRAINTS

Node 5280 +Y Mu = .30 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)

From 5280 To 5300 DX= 825.000 mm.

From 5300 To 5320 DX= 450.000 mm.

From 5320 To 5340 DX= 88.900 mm.

Element Name=FLANGE_NOFLG_300

RIGID Weight= 115.65 N. Type=Flange

From 10 To 20 DX= 123.825 mm.

PIPE

Dia= 114.300 mm. Wall= 11.125 mm. Cor= 3.0000 mm.

Element Name=FLANGE_NOFLG_900

GENERAL

T1= 325 C T2= 260 C T3= 5 C P1=11,600.0000 KPa PHyd=22,980.0000 KPa

Mat= (106)A106 Grade B E= 203,401,072 KPa EH1= 182,683,472 KPa

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Job:: R-52-2-P (ID-28711)

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INPUT LISTING

EH2= 188,226,864 KPa EH3= 204,228,448 KPa EH4= 203,401,072 KPa

EH5= 203,401,072 KPa EH6= 203,401,072 KPa EH7= 203,401,072 KPa

EH8= 203,401,072 KPa EH9= 203,401,072 KPa v = .292

Pipe Den= .0078334 kg./cu.cm. Fluid Den= .0010000 kg./cu.cm.

Insul Thk= 80.000 mm. Insul Den= .0001800 kg./cu.cm.

RIGID Weight= 231.31 N. Type=Flange

ALLOWABLE STRESSES

B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa

Sh1= 122,010 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa

Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa

Sh9= 137,895 KPa Sy= 241,316 KPa

From 20 To 30 DX= 640.000 mm.

PIPE

Dia= 114.300 mm. Wall= 11.125 mm.

GENERAL

Insul Thk= 80.000 mm. Insul Den= .0001400 kg./cu.cm.

RESTRAINTS

Node 30 +Y Mu = .30 Tag = SL-001

Node 30 Guide Tag = SL-001

Node 30 LIM Tag = SL-001

ALLOWABLE STRESSES

B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa

Sh1= 122,010 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa

Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa

Sh9= 137,895 KPa Sy= 241,316 KPa

From 30 To 40 DX= 380.000 mm.

From 40 To 50 DX= 300.000 mm.

From 50 To 60 DX= 1,000.000 mm.

RESTRAINTS

Node 60 +Y Mu = .30 Tag = SL-004

From 60 To 70 DX= 1,372.000 mm.

From 70 To 80 DX= 123.825 mm.

Element Name=FLANGE_NOFLG_900

RIGID Weight= 231.31 N. Type=Flange

FLANGES

Location= To Method= NC-3658.3 G/C= 235.000 mm.

Bolt Area= 3,938.000 mm. SYC= 241,000 KPa SY1= 188,000 KPa

SY2= 188,000 KPa SY3= 188,000 KPa SY4= 188,000 KPa SY5= 188,000 KPa

SY6= 188,000 KPa SY7= 188,000 KPa SY8= 188,000 KPa SY9= 188,000 KPa

From 80 To 90 DX= 123.825 mm.

Element Name=FLANGE_NOFLG_900

RIGID Weight= 231.31 N. Type=Flange

From 90 To 100 DX= 2,381.000 mm.

RESTRAINTS

Node 100 +Y Mu = .30 Tag = SL-005 (4" PIPE)

From 100 To 110 DX= 4,000.000 mm.

RESTRAINTS

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Node 110 +Y Mu = .30 Tag = SL-006 (4" PIPE)
Node 110 Guide Tag = SL-006 (4" PIPE)
-----
From 110 To 120 DX= 3,377.000 mm.
-----
From 120 To 130 DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3
-----
From 130 To 140 DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 140 To 150 DX= 400.000 mm.
-----
From 150 To 160 DX= 276.000 mm.
RESTRAINTS
Node 160 +Y Mu = .30 Tag = SL-007
-----
From 160 To 170 DX= 1,791.000 mm.
SIF's & TEE's
Node 170 Welding Tee Srf.Node=1651
-----
From 170 To 180 DX= 105.000 mm.
-----
From 180 To 190 CHECK VALVE DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3
-----
From 190 CHECK VALVE To 200 DX= 457.200 mm.
Element Name=CHECK_NOFLG_900
RIGID Weight= 1,912.74 N. Type=Valve
-----
From 200 To 210 DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
Location= From Method= NC-3658.3
-----
From 210 To 220 DX= 739.000 mm.
RESTRAINTS
Node 220 +Y Mu = .30 Tag = PS-010 (4" PIPE)
Node 220 Guide Mu = .30 Tag = PS-010 (4" PIPE)
-----
From 220 To 230 DX= 650.000 mm.
SIF's & TEE's
Node 230 Welding Tee Srf.Node=2201
-----
From 230 To 240 DY= 120.000 mm.
PIPE
Dia= 88.900 mm. Wall= 11.125 mm.
Insul Thk= 80.000 mm.
```


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-----
From 240 To 250 DY= 25.000 mm.
-----
From 250 To 260 GATE VALVE DY= 111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
-----
From 260 GATE VALVE To 270 GATE VALVE DY= 381.000 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 1,156.54 N. Type=Unspecified
FLANGES
  Location= Both Method= NC-3658.3 G/C= 190.500 mm.
  Bolt Area= 2,384.000 mm.
-----
From 270 GATE VALVE To 280 DY= 111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
-----
From 280 To 290 DY= 289.000 mm.
BEND at "TO" end
  Radius= 114.300 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 289
  Angle/Node @2= .00 288
-----
From 290 To 300 DZ= -500.000 mm.
BEND at "TO" end
  Radius= 114.300 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 299
  Angle/Node @2= .00 298
-----
From 300 To 310 DX= -650.000 mm.
RESTRAINTS
  Node 310 +Y Mu = .30 Tag = PS-010 (3" BY PASS LINE)
  Node 310 Guide Tag = PS-010 (3" BY PASS LINE)
-----
From 310 To 320 DX= -1,548.850 mm.
BEND at "TO" end
  Radius= 114.300 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 319
  Angle/Node @2= .00 318
-----
From 320 To 330 DZ= 500.000 mm.
BEND at "TO" end
  Radius= 114.300 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 329
  Angle/Node @2= .00 328
-----
From 330 To 170 DY= -1,037.000 mm.
-----
From 230 To 340 DX= 879.000 mm.
PIPE
  Dia= 114.300 mm. Wall= 11.125 mm.
  Insul Thk= 80.000 mm.
SIF's & TEE's
  Node 340 Welding Tee Srf.Node=3101
-----
From 340 To 350 DX= 2,071.000 mm.
RESTRAINTS
  Node 350 +Y Mu = .30 Tag = SL-009
-----

```

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```

From 350 To 360 DX= 1,659.000 mm.
-----
From 360 To 370 UZV VALVE DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
  Location= To Method= NC-3658.3 G/C= 235.000 mm.
  Bolt Area= 3,938.000 mm.
-----
From 370 UZV VALVE To 380 DX= 457.200 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 2,375.00 N. Type=Unspecified
-----
From 380 To 390 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
  Location= From Method= NC-3658.3
-----
From 390 To 400 DX= 593.000 mm.
-----
From 400 To 410 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
  Location= To Method= NC-3658.3
-----
From 410 To 420 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 420 To 430 DX= 105.000 mm.
SIF's & TEE's
  Node 430 Welding Tee Srf.Node=4001
-----
From 430 To 440 DX= 105.000 mm.
-----
From 440 To 450 GATE VALVE DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
  Location= To Method= NC-3658.3
-----
From 450 GATE VALVE To 460 DX= 457.200 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N. Type=Unspecified
-----
From 460 To 470 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
  Location= From Method= NC-3658.3
-----
From 470 To 480 DX= 600.000 mm.
RESTRAINTS
  Node 480 +Y Mu = .30 Tag = SL-010

```

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```

Node 480 Guide Tag = SL-010
-----
From 480 To 490 DX= 940.000 mm.
-----
From 490 To 500 GATE VALVE DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 500 GATE VALVE To 510 DX= 457.200 mm.
Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N. Type=Unspecified
FLANGES
Location= Both Method= NC-3658.3
-----
From 510 To 520 DX= 123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 520 To 530 DX= 105.000 mm.
SIF's & TEE's
Node 530 Welding Tee Srf.Node=5001
-----
From 530 To 540 DZ= 800.000 mm.
SINGLE FLANGED BEND at "TO" end
Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 539
Angle/Node @2= .00 538
-----
From 540 To 550 DX= -172.000 mm.
-----
From 550 To 560 GATE VALVE DX= -123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 560 GATE VALVE To 570 DX= -457.200 mm.
Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N. Type=Unspecified
FLANGES
Location= Both Method= NC-3658.3
-----
From 570 To 580 DX= -123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 580 To 590 DX= -873.000 mm.
RESTRAINTS
Node 590 +Y Mu = .30 Tag = SL-010 (COMMON SLEEPER FOR FUTURE)
-----
From 590 To 600 DX= -553.000 mm.
-----
From 600 To 610 GATE VALVE DX= -123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3
-----
From 610 GATE VALVE To 620 DX= -457.200 mm.

```

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```

    Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N.    Type=Unspecified
-----
From 620 To 630 DX= -123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
FLANGES
    Location= From    Method= NC-3658.3
-----
From 630 To 640 DX= -152.000 mm.
SINGLE FLANGED BEND at "TO" end
    Radius= 152.400 mm. (LONG)    Bend Angle= 90.000    Angle/Node @1= 45.00 639
-----
From 640 To 430 DZ= -800.000 mm.
-----
From 530 To 650 DX= 105.000 mm.
-----
From 650 To 660 DX= 123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
FLANGES
    Location= To    Method= NC-3658.3
-----
From 660 To 670 DX= 123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 670 To 680 DX= 105.000 mm.
SIF's & TEE's
    Node 680
-----
From 680 To 690 DX= 500.000 mm.
-----
From 680 To 700 DZ= -805.000 mm.
-----
From 700 To 710 GATE VALVE DZ= -123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 710 GATE VALVE To 720 DZ= -457.200 mm.
    Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N.    Type=Unspecified
FLANGES
    Location= Both    Method= NC-3658.3
-----
From 720 To 730 DZ= -123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 730 To 740 DZ= -1,400.000 mm.
RESTRAINTS
    Node 740 +Y    Mu = .30    Tag = SL-016
-----
From 740 To 750 DZ= -600.000 mm.
-----
From 750 To 760 DZ= -123.825 mm.

```

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```

    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 760 To 770 DZ= -457.200 mm.
    Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N.    Type=Unspecified
FLANGES
    Location= Both    Method= NC-3658.3
-----
From 770 To 780 DZ= -123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 780 To 790 DZ= -600.000 mm.
RESTRAINTS
    Node 790 +Y    Mu = .30    Tag = SL-017
-----
From 790 To 800 DZ= -600.000 mm.
-----
From 800 To 810 DZ= -123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
FLANGES
    Location= To    Method= NC-3658.3
-----
From 810 To 820 DZ= -50.000 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 400.00 N.    Type=Flange
-----
From 690 To 830 DX= 205.000 mm.
RESTRAINTS
    Node 830 +Y    Mu = .30    Tag = SL-011
    Node 830 Guide    Tag = SL-011
-----
From 830 To 840 DX= 600.000 mm.
-----
From 840 To 850 GATE VALVE DX= 123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 850 GATE VALVE To 860 DX= 457.200 mm.
    Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N.    Type=Unspecified
FLANGES
    Location= Both    Method= NC-3658.3
-----
From 860 To 870 DX= 123.825 mm.
    Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.    Type=Flange
-----
From 870 To 880 DX= 293.500 mm.
-----
From 880 To 890 DX= 293.500 mm.
-----
From 890 To 900 SPECTACLE BLIND DX= 123.825 mm.
    Element Name=FLANGE_NOFLG_900

```


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```

RIGID Weight= 231.31 N.    Type=Flange
-----
From 900 SPECTACLE BLIND To 910 GATE VALVE DX= 43.000 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 70.00 N.    Type=Unspecified
-----
From 910 GATE VALVE To 920 DX= 457.200 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 1,912.74 N.  Type=Unspecified
FLANGES
  Location= Both Method= NC-3658.3
-----
From 920 To 930 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
GENERAL
  T1= 21 C P1=11,600.0000 KPa PHyd= 7,665.0000 KPa
RIGID Weight= 231.31 N.    Type=Flange
ALLOWABLE STRESSES
  B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa
  Sh1= 137,895 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
  Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
  Sh9= 137,895 KPa Sy= 241,316 KPa
-----
From 930 To 940 DX= 505.000 mm.
GENERAL
  P1= 100.0000 KPa P2= 4,140.0000 KPa
RESTRAINTS
  Node 940 +Y Mu = .30 Tag = SL-012
ALLOWABLE STRESSES
  B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa
  Sh1= 137,895 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
  Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
  Sh9= 137,895 KPa Sy= 241,316 KPa
-----
From 940 To 950 DX= 660.000 mm.
SIF's & TEE's
  Node 950 Welding Tee Srf.Node=9301
-----
From 950 To 960 DZ= -805.000 mm.
-----
From 960 To 970 GATE VALVE DZ= -88.900 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N.    Type=Unspecified
-----
From 970 GATE VALVE To 980 DZ= -304.800 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 733.96 N.    Type=Unspecified
FLANGES
  Location= Both Method= NC-3658.3
-----
From 980 To 990 DZ= -88.900 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N.    Type=Flange
-----
From 990 To 1000 DZ= -1,400.000 mm.
RESTRAINTS

```

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Node 1000 +Y Mu = .30 Tag = SL-014
-----
From 1000 To 1010 DZ= -600.000 mm.
-----
From 1010 To 1020 GATE VALVE DZ= -88.900 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
-----
From 1020 GATE VALVE To 1030 DZ= -304.800 mm.
Element Name=GATE_NOFLG_900
RIGID Weight= 733.96 N. Type=Valve
FLANGES
Location= Both Method= NC-3658.3
-----
From 1030 To 1040 DZ= -88.900 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
-----
From 1040 To 1050 DZ= -600.000 mm.
RESTRAINTS
Node 1050 +Y Mu = .30 Tag = SL-015
-----
From 1050 To 1060 DZ= -600.000 mm.
-----
From 1060 To 1070 DZ= -88.900 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3
-----
From 1070 To 1080 DZ= -50.000 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 400.00 N. Type=Flange
-----
From 950 To 1090 DX= 600.000 mm.
RESTRAINTS
Node 1090 +Y Mu = .30 Tag = SL-014
-----
From 1090 To 1100 DX= 450.000 mm.
-----
From 1100 To 1110 DX= 88.900 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3
-----
From 340 To 1120 DZ= -950.000 mm.
PIPE
Dia= 88.900 mm. Wall= 11.125 mm.
GENERAL
T1= 325 C P1=11,600.0000 KPa PHyd=22,980.0000 KPa
Insul Thk= 50.000 mm.
SINGLE FLANGED BEND at "TO" end
Radius= 114.300 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1119
Angle/Node @2= .00 1118
ALLOWABLE STRESSES

```

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    B31.3 (2020)          Cycle Max Switch = ---    Sc= 137,895 KPa
    Sh1= 122,010 KPa      Sh2= 131,000 KPa      Sh3= 137,895 KPa      Sh4= 137,895 KPa
    Sh5= 137,895 KPa      Sh6= 137,895 KPa      Sh7= 137,895 KPa      Sh8= 137,895 KPa
    Sh9= 137,895 KPa      Sy= 241,316 KPa
-----
From 1120 To 1130 DX= -115.000 mm.
-----
From 1130 To 1140 GATE VALVE DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N.    Type=Flange
-----
From 1140 GATE VALVE To 1150 DX= -398.000 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 1,156.54 N.  Type=Unspecified
FLANGES
  Location= Both    Method= NC-3658.3    G/C= 190.500 mm.
  Bolt Area= 2,384.000 mm.
-----
From 1150 To 1160 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N.    Type=Flange
-----
From 1160 To 1170 DX= -790.000 mm.
RESTRAINTS
  Node 1170 +Y    Mu = .30    Tag = PS-010 (3" PIPE)
-----
From 1170 To 1180 DX= -1,728.000 mm.
-----
From 1180 To 1190 DX= -1,309.000 mm.
-----
From 1190 To 1200 CHECK VALVE DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N.    Type=Flange
-----
From 1200 CHECK VALVE To 1210 DX= -381.000 mm.
  Element Name=CHECK_NOFLG_900
RIGID Weight= 800.68 N.    Type=Unspecified
FLANGES
  Location= Both    Method= NC-3658.3
-----
From 1210 To 1220 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N.    Type=Flange
-----
From 1220 To 1230 DX= -561.000 mm.
RESTRAINTS
  Node 1230 +Y    Mu = .30    Tag = SL-018 (3" PIPE)
  Node 1230 Guide    Tag = SL-018 (3" PIPE)
-----
From 1230 To 1240 DX= -541.000 mm.
-----
From 1240 To 1250 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N.    Type=Flange
FLANGES
  Location= Both    Method= NC-3658.3

```

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```

-----
From 1250 To 1260 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
-----
From 1260 To 1270 DX= -2,680.000 mm.
RESTRAINTS
  Node 1270 +Y Mu = .30 Tag = SL-006 (3" PIPE)
-----
From 1270 To 1280 DX= -4,000.000 mm.
RESTRAINTS
  Node 1280 +Y Mu = .30 Tag = SL-005 (3" PIPE)
-----
From 1280 To 1290 DX= -2,353.000 mm.
RESTRAINTS
  Node 1290 +Y Mu = .30 Tag = SL-019
  Node 1290 Guide Tag = SL-019
-----
From 1290 To 1300 DX= -646.000 mm.
-----
From 1300 To 1310 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
FLANGES
  Location= To Method= NC-3658.3
-----
From 1310 To 1320 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
-----
From 1320 To 1330 DX= -89.000 mm.
REDUCER
  Diam2= 48.260 mm. Wall2= 10.160 mm.
-----
From 1330 To 1340 DX= -367.000 mm.
PIPE
  Dia= 48.260 mm. Wall= 10.160 mm.
  Insul Thk= 50.000 mm.
-----
From 1340 To 1350 FLOW METER DX= -92.075 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 66.72 N. Type=Flange
-----
From 1350 FLOW METER To 1360 DX= -262.000 mm.
  Element Name=GATE_NOFLG_900
RIGID Weight= 400.00 N. Type=Unspecified
FLANGES
  Location= Both Method= NC-3658.3 G/C= 123.800 mm.
  Bolt Area= 1,563.600 mm.
-----
From 1360 To 1370 DX= -92.075 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 66.72 N. Type=Flange
-----
From 1370 To 1380 DX= -703.000 mm.
RESTRAINTS
  
```

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 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

```

Node 1380 +Y Mu = .30 Tag = SL-020
-----
From 1380 To 1390 DX= -400.000 mm.
-----
From 1390 To 1400 DX= -89.000 mm.
REDUCER
Diam2= 88.900 mm. Wall2= 11.125 mm.
-----
From 1400 To 1410 DX= -111.125 mm.
PIPE
Dia= 88.900 mm. Wall= 11.125 mm. Element Name=FLANGE_NOFLG_900
Insul Thk= 50.000 mm.
RIGID Weight= 146.79 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3 G/C= 190.500 mm.
Bolt Area= 2,384.000 mm.
-----
From 1410 To 1420 DX= -111.125 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 146.79 N. Type=Flange
-----
From 1420 To 1430 DX= -500.000 mm.
-----
From 1430 To 1460 DX= -2,421.000 mm.
RESTRAINTS
Node 1460 +Y Mu = .30 Tag = SL-003
-----
From 1460 To 1470 DX= -575.520 mm.
-----
From 1470 To 1480 DX= -114.000 mm. DY= -14.200 mm.
REDUCER
Diam2= 60.325 mm. Wall2= 11.074 mm.
-----
From 1480 To 1490 CHOKE VALVE DX= -110.273 mm.
PIPE
Dia= 60.325 mm. Wall= 11.074 mm. Element Name=FLANGE_NOFLG_900
Insul Thk= 50.000 mm.
RIGID Weight= 115.65 N. Type=Flange
FLANGES
Location= To Method= NC-3658.3 G/C= 165.100 mm.
-----
From 1490 CHOKE VALVE To 1500 CHOKE VALVE DX= -199.000 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 300.00 N. Type=Unspecified
-----
From 1500 CHOKE VALVE To 1510 DZ= 199.000 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 300.00 N. Type=Unspecified
-----
From 1510 To 1520 DZ= 111.125 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
FLANGES
Location= From Method= NC-3658.3
-----
From 1520 To 1530 DZ= 1,146.000 mm.
```

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INPUT LISTING

RESTRAINTS

Node 1530 Y Mu = .30 Tag = SL-002
 Node 1530 Guide Gap= 5.000 mm. Tag = SL-002
 Node 1530 LIM Gap= 5.000 mm. Tag = SL-002

 From 1540 To 1550 DX= 75.000 mm.

SIF's & TEE's

Node 1550 Welding Tee Srf.Node=2721

 From 1550 To 1560 DY= -407.000 mm.

BEND at "TO" end

Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1559
 Angle/Node @2= .00 1558

 From 1560 To 1570 DX= 250.000 mm.

RESTRAINTS

Node 1570 +Y Mu = .30 Tag = SL-001

 From 1570 To 1580 DX= 200.000 mm.

 From 1580 To 1590 DX= 100.000 mm.

REDUCER

Diam2= 88.900 mm. Wall2= 15.240 mm.

 From 1590 To 1595 DX= 241.000 mm.

PIPE

Dia= 88.900 mm. Wall= 11.125 mm.
 Insul Thk= 50.000 mm.

 From 1595 To 1600 DX= 100.000 mm.

REDUCER

Diam2= 60.325 mm. Wall2= 11.074 mm.

 From 1600 To 1605 DX= 76.200 mm.

PIPE

Dia= 60.325 mm. Wall= 11.125 mm.
 Insul Thk= 50.000 mm.

BEND at "TO" end

Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1604

 From 1605 To 1610 X-MAS CASING CONNECTION DZ= 274.000 mm.

PIPE

Dia= 60.325 mm. Wall= 11.074 mm.
 Insul Thk= 50.000 mm.

 From 1610 X-MAS CASING CONNECTION To 1620 DZ= 111.125 mm.

PIPE

Dia= 60.325 mm. Wall= 11.074 mm. Element Name=FLANGE_NOFLG_900

GENERAL

P2= 4,140.0000 KPa Insul Thk= 50.000 mm.

RIGID Weight= 115.65 N. Type=Flange

RESTRAINTS

Node 1620 ANC Cnode 1611

DISPLACEMENTS

Node 1611 DX1= .000 mm. DY1= 2,200.000 mm. DZ1= .000 mm. RX1= .000
 RY1= .000 RZ1= .000

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INPUT LISTING

```

-----
From 1550 To 1630 DY= 50.000 mm.
-----
From 1630 To 1640 DY= 76.000 mm.
REDUCER
  Diam2= 48.260 mm.  Wall2= 10.160 mm.
-----
From 1640 To 1650 DY= 92.075 mm.
PIPE
  Dia= 48.260 mm.  Wall= 10.160 mm.  Element Name=FLANGE_NOFLG_900
  Insul Thk= 50.000 mm.
RIGID Weight= 66.72 N.  Type=Flange
-----
From 1650 To 1660 DY= 305.000 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 400.00 N.  Type=Valve
-----
From 1660 To 1670 DY= 92.075 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 66.72 N.  Type=Flange
-----
From 1670 To 1680 DY= 595.000 mm.
BEND at "TO" end
  Radius= 57.150 mm. (LONG)  Bend Angle= 90.000  Angle/Node @1= 45.00 1679
  Angle/Node @2= .00 1678
-----
From 1680 To 1690 DX= 396.000 mm.
-----
From 1690 To 1700 DX= 92.075 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 66.72 N.  Type=Flange
-----
From 1700 To 1710 DX= 305.000 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 400.00 N.  Type=Valve
-----
From 1710 To 1720 DX= 92.075 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 66.72 N.  Type=Flange
-----
From 1720 To 1730 DX= 157.000 mm.
BEND at "TO" end
  Radius= 57.150 mm. (LONG)  Bend Angle= 90.000  Angle/Node @1= 45.00 1729
  Angle/Node @2= .00 1728
-----
From 1730 To 1740 DY= -130.552 mm.
-----
From 1740 To 1750 DY= -76.000 mm.
REDUCER
  Diam2= 60.325 mm.  Wall2= 11.074 mm.
-----
From 1750 To 1760 DX= .100 mm.  DY= -116.449 mm.  DZ= -.397 mm.
PIPE
  Dia= 60.325 mm.  Wall= 11.074 mm.
  Insul Thk= 50.000 mm.
SIF's & TEE's

```

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INPUT LISTING

```

Node 1760 Welding Tee Srf.Node=3141
-----
From 1760 To 1770 X-MAS TREE WING VALVE DZ= 685.000 mm.
PIPE
Dia= 114.300 mm. Wall= 11.125 mm.
Insul Thk= 80.000 mm.
-----
From 1770 X-MAS TREE WING VALVE To 1780 DZ= 123.825 mm.
PIPE
Dia= 114.300 mm. Wall= 11.125 mm. Element Name=FLANGE_NOFLG_900
GENERAL
P2= 4,140.0000 KPa Insul Thk= 80.000 mm.
RIGID Weight= 231.31 N. Type=Flange
RESTRAINTS
Node 1780 ANC Cnode 1781
DISPLACEMENTS
Node 1781 DX1= .000 mm. DY1=2,200.000 mm. DZ1= .000 mm. RX1= .000
RY1= .000 RZ1= .000
FLANGES
Location= To Method= NC-3658.3 G/C= 235.000 mm.
Bolt Area= 3,938.000 mm.
-----
From 1760 To 1790 DZ= -100.000 mm.
PIPE
Dia= 114.300 mm. Wall= 11.125 mm.
Insul Thk= 80.000 mm.
-----
From 1790 To 1800 DZ= -123.825 mm.
Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange
-----
From 1530 To 1810 DZ= 600.000 mm.
PIPE
Dia= 60.325 mm. Wall= 11.074 mm.
GENERAL
T1= 325 C P1=11,600.0000 KPa PHyd=22,980.0000 KPa
Insul Thk= 50.000 mm.
BEND at "TO" end
Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1809
Angle/Node @2= .00 1808
ALLOWABLE STRESSES
B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa Sy= 241,316 KPa
-----
From 1810 To 1820 DX= 76.000 mm.
-----
From 1820 To 1830 DX= 111.125 mm.
Element Name=FLANGE_FLG_300
RIGID Weight= 115.65 N. Type=Flange
RESTRAINTS
Node 1830 ANC Cnode 2220
-----
From 1840 To 1540 DX= 73.025 mm.
PIPE

```

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INPUT LISTING

```

Dia= 60.325 mm.   Wall= 11.074 mm.   Element Name=FLANGE_NOFLG_300
Insul Thk= 50.000 mm.
RIGID Weight= 44.48 N.   Type=Flange
FLANGES
  Location= From   Method= NC-3658.3   G/C= 165.100 mm.
  Bolt Area= 2,384.000 mm.
-----
From 1800 To 1850 DZ= -123.825 mm.
PIPE
  Dia= 114.300 mm.   Wall= 11.125 mm.   Cor= 3.0000 mm.
  Element Name=FLANGE_NOFLG_900
GENERAL
  T1= 325 C   T3= 5 C   P1=11,600.0000 KPa   PHyd=22,980.0000 KPa
  Mat= (106)A106 Grade B   E= 203,401,072 KPa   EH1= 182,683,472 KPa
  EH2= 188,226,864 KPa   EH3= 204,228,448 KPa   EH4= 203,401,072 KPa
  EH5= 203,401,072 KPa   EH6= 203,401,072 KPa   EH7= 203,401,072 KPa
  EH8= 203,401,072 KPa   EH9= 203,401,072 KPa   v = .292
  Pipe Den= .0078334 kg./cu.cm.   Fluid Den= .0009996 kg./cu.cm.
  Insul Thk= 80.000 mm.   Insul Den= .0001842 kg./cu.cm.
RIGID Weight= 231.31 N.   Type=Flange
ALLOWABLE STRESSES
  B31.3 (2020)   Cycle Max Switch = ---   Sc= 137,895 KPa
  Sh1= 122,010 KPa   Sh2= 131,000 KPa   Sh3= 137,895 KPa   Sh4= 137,895 KPa
  Sh5= 137,895 KPa   Sh6= 137,895 KPa   Sh7= 137,895 KPa   Sh8= 137,895 KPa
  Sh9= 137,895 KPa   Sy= 241,000 KPa
FLANGES
  Location= From   Method= NC-3658.3   G/C= 235.000 mm.
  Bolt Area= 3,938.000 mm.
-----
From 1850 To 1860 DZ= -152.000 mm.
SINGLE FLANGED BEND at "TO" end
  Radius= 152.400 mm. (LONG)   Bend Angle= 90.000   Angle/Node @1= 45.00 1859
-----
From 1860 To 1870 DX= 627.000 mm.
-----
From 1870 To 1880 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.   Type=Flange
FLANGES
  Location= To   Method= NC-3658.3
-----
From 1880 To 1890 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N.   Type=Flange
-----
From 1890 To 1900 DX= 121.000 mm.
RIGID Weight= 238.00 N.   Type=Valve
RESTRAINTS
  Node 1900 Z   Cnode 1910
  Node 1900 Y   Cnode 1910
  Node 1900 X   Cnode 1910
  Node 1910 RX2   Yield Kr= 1 N.m./deg   Yield Moment= 3,740 N.m.
-----
From 1910 To 1920 DX= 121.000 mm.
RIGID Weight= 238.00 N.   Type=Valve
RESTRAINTS

```

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INPUT LISTING

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Node 1900 RZ Cnode 1910
Node 1900 RY Cnode 1910
-----
From 1920 To 1930 DX= 153.000 mm.
BEND at "TO" end
  Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1929
-----
From 1930 To 1940 DY= 1,597.000 mm. DZ= -1,586.000 mm.
BEND at "TO" end
  Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1939
  Angle/Node @2= .00 1938
-----
From 1940 To 1950 DX= -153.000 mm.
-----
From 1950 To 1960 DX= -121.000 mm.
RIGID Weight= 238.00 N. Type=Valve
RESTRAINTS
  Node 1960 Z Cnode 1970
  Node 1960 Y Cnode 1970
  Node 1960 X Cnode 1970
  Node 1970 RX2 Yield Kr= 1 N.m./deg Yield Moment= 4,244 N.m.
-----
From 1970 To 1980 DX= -121.000 mm.
RIGID Weight= 238.00 N. Type=Valve
RESTRAINTS
  Node 1970 RZ2 Yield Kr= 1 N.m./deg Yield Moment= 4,244 N.m.
  Node 1970 RY2 Yield Kr= 1 N.m./deg Yield Moment= 4,244 N.m.
-----
From 1980 To 1990 DX= -153.000 mm.
BEND at "TO" end
  Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1989
-----
From 1990 To 2000 DY= -2,208.000 mm. DZ= -806.000 mm.
BEND at "TO" end
  Radius= 152.400 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 1999
  Angle/Node @2= .00 1998
-----
From 2000 To 2010 DX= 153.000 mm.
-----
From 2010 To 2020 DX= 121.000 mm.
RIGID Weight= 238.00 N. Type=Valve
RESTRAINTS
  Node 2020 Z Cnode 2030
  Node 2020 Y Cnode 2030
  Node 2020 X Cnode 2030
  Node 2030 RX2 Yield Kr= 1 N.m./deg Yield Moment= 3,745 N.m.
-----
From 2030 To 2040 DX= 121.000 mm.
RIGID Weight= 238.00 N. Type=Valve
RESTRAINTS
  Node 2020 RZ Cnode 2030
  Node 2020 RY Cnode 2030
-----
From 2040 To 10 DX= 123.825 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 231.31 N. Type=Flange

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INPUT LISTING

FLANGES

Location= To Method= NC-3658.3

From 2050 To 2060 DX= -111.125 mm.

PIPE

Dia= 60.325 mm. Wall= 11.074 mm. Cor= 3.0000 mm.

Element Name=FLANGE_NOFLG_900

GENERAL

T1= 325 C P1=11,600.0000 KPa PHyd=22,980.0000 KPa

Mat= (106)A106 Grade B E= 203,401,072 KPa EH1= 182,683,472 KPa

EH2= 188,226,864 KPa EH3= 204,228,448 KPa EH4= 203,401,072 KPa

EH5= 203,401,072 KPa EH6= 203,401,072 KPa EH7= 203,401,072 KPa

EH8= 203,401,072 KPa EH9= 203,401,072 KPa v = .292

Pipe Den= .0078334 kg./cu.cm. Fluid Den= .0001000 kg./cu.cm.

Insul Thk= 50.000 mm. Insul Den= .0001362 kg./cu.cm.

RIGID Weight= 115.65 N. Type=Flange

RESTRAINTS

Node 2050 ANC Cnode 1840

ALLOWABLE STRESSES

B31.3 (2020) Cycle Max Switch = --- Sc= 137,895 KPa

Sh1= 122,010 KPa Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa

Sh5= 137,895 KPa Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa

Sh9= 137,895 KPa Sy= 241,000 KPa

From 2060 To 2070 DX= -86.550 mm.

RIGID Weight= 65.00 N. Type=Valve

RESTRAINTS

Node 2070 Z Cnode 2080

Node 2070 Y Cnode 2080

Node 2070 X Cnode 2080

Node 2080 RX2 Yield Kr= 1 N.m./deg Yield Moment= 1,445 N.m.

From 2080 To 2090 DX= -86.550 mm.

RIGID Weight= 65.00 N. Type=Valve

RESTRAINTS

Node 2070 RZ Cnode 2080

Node 2070 RY Cnode 2080

From 2090 To 2100 DX= -76.200 mm.

BEND at "TO" end

Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 2099

From 2100 To 2110 DY= 2,208.322 mm. DZ= -435.000 mm.

BEND at "TO" end

Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 2109

From 2110 To 2120 DX= 76.200 mm.

From 2120 To 2130 DX= 86.550 mm.

RIGID Weight= 65.00 N. Type=Valve

RESTRAINTS

Node 2130 Z Cnode 2140

Node 2130 Y Cnode 2140

Node 2130 X Cnode 2140

Node 2140 RX2 Yield Kr= 1 N.m./deg Yield Moment= 1,445 N.m.

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INPUT LISTING

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From 2140 To 2150 DX= 86.550 mm.
RIGID Weight= 65.00 N. Type=Valve
RESTRAINTS
  Node 2140 RZ2 Yield Kr= 1 N.m./deg Yield Moment= 1,803 N.m.
  Node 2140 RY2 Yield Kr= 1 N.m./deg Yield Moment= 1,803 N.m.
-----
From 2150 To 2160 DX= 76.200 mm.
BEND at "TO" end
  Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 2159
-----
From 2160 To 2170 DY= -1,946.370 mm. DZ= -1,350.920 mm.
BEND at "TO" end
  Radius= 76.200 mm. (LONG) Bend Angle= 90.000 Angle/Node @1= 45.00 2169
-----
From 2170 To 2180 DX= -76.200 mm.
-----
From 2180 To 2190 DX= -86.550 mm.
RIGID Weight= 65.00 N. Type=Valve
RESTRAINTS
  Node 2190 Z Cnode 2200
  Node 2190 Y Cnode 2200
  Node 2190 X Cnode 2200
  Node 2200 RX2 Yield Kr= 1 N.m./deg Yield Moment= 1,445 N.m.
-----
From 2200 To 2210 DX= -86.550 mm.
RIGID Weight= 65.00 N. Type=Valve
RESTRAINTS
  Node 2190 RZ Cnode 2200
  Node 2190 RY Cnode 2200
-----
From 2210 To 2220 DX= -111.125 mm.
  Element Name=FLANGE_NOFLG_900
RIGID Weight= 115.65 N. Type=Flange
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NODENAMES

180	190	From=	To=	CHECK VALVE
190	200	From=	CHECK VALVE	To=
250	260	From=	To=	GATE VALVE
260	270	From=	GATE VALVE	To= GATE VALVE
270	280	From=	GATE VALVE	To=
360	370	From=	To=	UZV VALVE
370	380	From=	UZV VALVE	To=
440	450	From=	To=	GATE VALVE
450	460	From=	GATE VALVE	To=
490	500	From=	To=	GATE VALVE
500	510	From=	GATE VALVE	To=
550	560	From=	To=	GATE VALVE
560	570	From=	GATE VALVE	To=
600	610	From=	To=	GATE VALVE
610	620	From=	GATE VALVE	To=
700	710	From=	To=	GATE VALVE
710	720	From=	GATE VALVE	To=
840	850	From=	To=	GATE VALVE

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INPUT LISTING

850	860	From=	GATE VALVE	To=	
890	900	From=		To=	SPECTACLE BLIND
900	910	From=	SPECTACLE BLIND	To=	GATE VALVE
910	920	From=	GATE VALVE	To=	
960	970	From=		To=	GATE VALVE
970	980	From=	GATE VALVE	To=	
1010	1020	From=		To=	GATE VALVE
1020	1030	From=	GATE VALVE	To=	
1130	1140	From=		To=	GATE VALVE
1140	1150	From=	GATE VALVE	To=	
1190	1200	From=		To=	CHECK VALVE
1200	1210	From=	CHECK VALVE	To=	
1340	1350	From=		To=	FLOW METER
1350	1360	From=	FLOW METER	To=	
1480	1490	From=		To=	CHOKE VALVE
1490	1500	From=	CHOKE VALVE	To=	CHOKE VALVE
1500	1510	From=	CHOKE VALVE	To=	
1605	1610	From=		To=	X-MAS CASING CONNECTION
1610	1620	From=	X-MAS CASING CONNECTION	To=	
1760	1770	From=		To=	X-MAS TREE WING VALVE
1770	1780	From=	X-MAS TREE WING VALVE	To=	

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MATERIAL Changes:

1110	4020
Mat= (106)A106 Grade B E= 202,711,488 KPa v = .292	
Density= .0078 kg./cu.cm.	
10	20

1800	1850
Mat= (106)A106 Grade B E= 203,401,072 KPa v = .292	
Density= .0078 kg./cu.cm.	

2050	2060
Mat= (106)A106 Grade B E= 203,401,072 KPa v = .292	
Density= .0078 kg./cu.cm.	

Mat= (106)A106 Grade B E= 203,401,072 KPa v = .292	
Density= .0078 kg./cu.cm.	
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JOBNAME: X:\8.0 (300# FL) R-52-2-P (ID-28711)\7.0 PIPING\7.3 PIPE STRESS ANALY	

ALLOWABLE STRESS Changes

1110	4020		
B31.3 IX (2018)			
Sc= 160,648 KPa	Sh1= 160,648 KPa	Sh2= 131,000 KPa	
Sh3= 160,648 KPa	Sh4= 160,648 KPa	Sh5= 160,648 KPa	Sh6= 160,648 KPa
Sh7= 160,648 KPa	Sh8= 160,648 KPa	Sh9= 160,648 KPa	Sy= 241,316 KPa

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INPUT LISTING

10	20
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
20	30
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
920	930
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 137,895 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
930	940
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 137,895 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
340	1120
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
1530	1810
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,316 KPa
1800	1850
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,000 KPa
2050	2060
B31.3 (2020)	Cycle Max Switch = --- Sc= 137,895 KPa
Sh1= 122,010 KPa	Sh2= 131,000 KPa Sh3= 137,895 KPa Sh4= 137,895 KPa
Sh5= 137,895 KPa	Sh6= 137,895 KPa Sh7= 137,895 KPa Sh8= 137,895 KPa
Sh9= 137,895 KPa	Sy= 241,000 KPa

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BEND ELEMENTS

4030	4040	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4039
Angle/Node @2= .00 4038		
4080	4100	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4099
Angle/Node @2= .00 4098		
4180	4200	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4199
Angle/Node @2= .00 4198		
4480	4500	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4499
Angle/Node @2= .00 4498		
4540	4560	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4559
Angle/Node @2= .00 4558		
4600	4620	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4619
Angle/Node @2= .00 4618		
4660	4680	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4679
Angle/Node @2= .00 4678		
4960	4980	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4979
Angle/Node @2= .00 4978		
4980	5000	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 4999
Angle/Node @2= .00 4998		
5040	5060	
Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 5059
Angle/Node @2= .00 5058		
5060	5080	

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INPUT LISTING

Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 5079
Angle/Node @2= .00 5078		
5240	5260	

Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 5259
Angle/Node @2= .00 5258		
5260	5270	

Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 5269
Angle/Node @2= .00 5268		
280	290	

Radius= 114.300 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 289
Angle/Node @2= .00 288		
290	300	

Radius= 114.300 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 299
Angle/Node @2= .00 298		
310	320	

Radius= 114.300 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 319
Angle/Node @2= .00 318		
320	330	

Radius= 114.300 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 329
Angle/Node @2= .00 328		
530	540	

Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 539
Angle/Node @2= .00 538		
630	640	

Radius= 152.400 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 639
340	1120	

Radius= 114.300 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 1119
Angle/Node @2= .00 1118		
1550	1560	

Radius= 76.200 mm. (LONG)	Bend Angle= 90.000	Angle/Node @1= 45.00 1559
Angle/Node @2= .00 1558		
1600	1605	

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INPUT LISTING

Radius= 76.200 mm. (LONG) 1670	Bend Angle= 90.000 1680	Angle/Node @1= 45.00 1604
Radius= 57.150 mm. (LONG) Angle/Node @2= .00 1678 1720	Bend Angle= 90.000 1730	Angle/Node @1= 45.00 1679
Radius= 57.150 mm. (LONG) Angle/Node @2= .00 1728 1530	Bend Angle= 90.000 1810	Angle/Node @1= 45.00 1729
Radius= 76.200 mm. (LONG) Angle/Node @2= .00 1808 1850	Bend Angle= 90.000 1860	Angle/Node @1= 45.00 1809
Radius= 152.400 mm. (LONG) 1920	Bend Angle= 90.000 1930	Angle/Node @1= 45.00 1859
Radius= 152.400 mm. (LONG) 1930	Bend Angle= 90.000 1940	Angle/Node @1= 45.00 1929
Radius= 152.400 mm. (LONG) Angle/Node @2= .00 1938 1980	Bend Angle= 90.000 1990	Angle/Node @1= 45.00 1939
Radius= 152.400 mm. (LONG) 1990	Bend Angle= 90.000 2000	Angle/Node @1= 45.00 1989
Radius= 152.400 mm. (LONG) Angle/Node @2= .00 1998 2090	Bend Angle= 90.000 2100	Angle/Node @1= 45.00 1999
Radius= 76.200 mm. (LONG) 2100	Bend Angle= 90.000 2110	Angle/Node @1= 45.00 2099
Radius= 76.200 mm. (LONG) 2150	Bend Angle= 90.000 2160	Angle/Node @1= 45.00 2109
Radius= 76.200 mm. (LONG) 2160	Bend Angle= 90.000 2170	Angle/Node @1= 45.00 2159
Radius= 76.200 mm. (LONG) CAESAR II Ver.13.00.00.0890,	Bend Angle= 90.000 (Build 220811)	Angle/Node @1= 45.00 2169 MAR 26, 2025 8:45:56

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INPUT LISTING

RIGIDS		
1110		4020
Weight= 115.65 N.	Type=Flange	
5160		5180
Weight= 115.65 N.	Type=Flange	
5180		5200
Weight= 115.65 N.	Type=Flange	
5320		5340
Weight= 115.65 N.	Type=Flange	
10		20
Weight= 231.31 N.	Type=Flange	
70		80
Weight= 231.31 N.	Type=Flange	
80		90
Weight= 231.31 N.	Type=Flange	
120		130
Weight= 231.31 N.	Type=Flange	
130		140
Weight= 231.31 N.	Type=Flange	
180		190 CHECK VALVE
Weight= 231.31 N.	Type=Flange	
190 CHECK VALVE		200
Weight= 1,912.74 N.	Type=Valve	
200		210
Weight= 231.31 N.	Type=Flange	
250		260 GATE VALVE
Weight= 146.79 N.	Type=Flange	
260 GATE VALVE		270 GATE VALVE

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INPUT LISTING

Weight= 1,156.54 N. Type=Unspecified 270 GATE VALVE	280
Weight= 146.79 N. Type=Flange 360	370 UZV VALVE
Weight= 231.31 N. Type=Flange 370 UZV VALVE	380
Weight= 2,375.00 N. Type=Unspecified 380	390
Weight= 231.31 N. Type=Flange 400	410
Weight= 231.31 N. Type=Flange 410	420
Weight= 231.31 N. Type=Flange 440	450 GATE VALVE
Weight= 231.31 N. Type=Flange 450 GATE VALVE	460
Weight= 1,912.74 N. Type=Unspecified 460	470
Weight= 231.31 N. Type=Flange 490	500 GATE VALVE
Weight= 231.31 N. Type=Flange 500 GATE VALVE	510
Weight= 1,912.74 N. Type=Unspecified 510	520
Weight= 231.31 N. Type=Flange 550	560 GATE VALVE
Weight= 231.31 N. Type=Flange 560 GATE VALVE	570

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INPUT LISTING

Weight= 1,912.74 N.	Type=Unspecified	580
570		
Weight= 231.31 N.	Type=Flange	610 GATE VALVE
600		
Weight= 231.31 N.	Type=Flange	620
610 GATE VALVE		
Weight= 1,912.74 N.	Type=Unspecified	630
620		
Weight= 231.31 N.	Type=Flange	660
650		
Weight= 231.31 N.	Type=Flange	670
660		
Weight= 231.31 N.	Type=Flange	710 GATE VALVE
700		
Weight= 231.31 N.	Type=Flange	720
710 GATE VALVE		
Weight= 1,912.74 N.	Type=Unspecified	730
720		
Weight= 231.31 N.	Type=Flange	760
750		
Weight= 231.31 N.	Type=Flange	770
760		
Weight= 1,912.74 N.	Type=Unspecified	780
770		
Weight= 231.31 N.	Type=Flange	810
800		
Weight= 231.31 N.	Type=Flange	820
810		

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INPUT LISTING

Weight= 400.00 N. Type=Flange 840	850 GATE VALVE
Weight= 231.31 N. Type=Flange 850 GATE VALVE	860
Weight= 1,912.74 N. Type=Unspecified 860	870
Weight= 231.31 N. Type=Flange 890	900 SPECTACLE BLIND
Weight= 231.31 N. Type=Flange 900 SPECTACLE BLIND	910 GATE VALVE
Weight= 70.00 N. Type=Unspecified 910 GATE VALVE	920
Weight= 1,912.74 N. Type=Unspecified 920	930
Weight= 231.31 N. Type=Flange 960	970 GATE VALVE
Weight= 115.65 N. Type=Unspecified 970 GATE VALVE	980
Weight= 733.96 N. Type=Unspecified 980	990
Weight= 115.65 N. Type=Flange 1010	1020 GATE VALVE
Weight= 115.65 N. Type=Flange 1020 GATE VALVE	1030
Weight= 733.96 N. Type=Valve 1030	1040
Weight= 115.65 N. Type=Flange 1060	1070

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INPUT LISTING

Weight= 115.65 N.	Type=Flange	
1070		1080
Weight= 400.00 N.	Type=Flange	
1100		1110
Weight= 115.65 N.	Type=Flange	
1130		1140 GATE VALVE
Weight= 146.79 N.	Type=Flange	
1140 GATE VALVE		1150
Weight= 1,156.54 N.	Type=Unspecified	
1150		1160
Weight= 146.79 N.	Type=Flange	
1190		1200 CHECK VALVE
Weight= 146.79 N.	Type=Flange	
1200 CHECK VALVE		1210
Weight= 800.68 N.	Type=Unspecified	
1210		1220
Weight= 146.79 N.	Type=Flange	
1240		1250
Weight= 146.79 N.	Type=Flange	
1250		1260
Weight= 146.79 N.	Type=Flange	
1300		1310
Weight= 146.79 N.	Type=Flange	
1310		1320
Weight= 146.79 N.	Type=Flange	
1340		1350 FLOW METER
Weight= 66.72 N.	Type=Flange	
1350 FLOW METER		1360

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INPUT LISTING

Weight= 400.00 N.	Type=Unspecified	
1360		1370
Weight= 66.72 N.	Type=Flange	
1400		1410
Weight= 146.79 N.	Type=Flange	
1410		1420
Weight= 146.79 N.	Type=Flange	
1480		1490 CHOKE VALVE
Weight= 115.65 N.	Type=Flange	
1490 CHOKE VALVE		1500 CHOKE VALVE
Weight= 300.00 N.	Type=Unspecified	
1500 CHOKE VALVE		1510
Weight= 300.00 N.	Type=Unspecified	
1510		1520
Weight= 115.65 N.	Type=Flange	
1610 X-MAS CASING CONNECTION		1620
Weight= 115.65 N.	Type=Flange	
1640		1650
Weight= 66.72 N.	Type=Flange	
1650		1660
Weight= 400.00 N.	Type=Valve	
1660		1670
Weight= 66.72 N.	Type=Flange	
1690		1700
Weight= 66.72 N.	Type=Flange	
1700		1710
Weight= 400.00 N.	Type=Valve	
1710		1720

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INPUT LISTING

Weight= 66.72 N.	Type=Flange	
1770 X-MAS TREE WING VALVE		1780
Weight= 231.31 N.	Type=Flange	
1790		1800
Weight= 231.31 N.	Type=Flange	
1820		1830
Weight= 115.65 N.	Type=Flange	
1840		1540
Weight= 44.48 N.	Type=Flange	
1800		1850
Weight= 231.31 N.	Type=Flange	
1870		1880
Weight= 231.31 N.	Type=Flange	
1880		1890
Weight= 231.31 N.	Type=Flange	
1890		1900
Weight= 238.00 N.	Type=Valve	
1910		1920
Weight= 238.00 N.	Type=Valve	
1950		1960
Weight= 238.00 N.	Type=Valve	
1970		1980
Weight= 238.00 N.	Type=Valve	
2010		2020
Weight= 238.00 N.	Type=Valve	
2030		2040
Weight= 238.00 N.	Type=Valve	
2040		10

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INPUT LISTING

Weight= 231.31 N. Type=Flange
 2050 2060

Weight= 115.65 N. Type=Flange
 2060 2070

Weight= 65.00 N. Type=Valve
 2080 2090

Weight= 65.00 N. Type=Valve
 2120 2130

Weight= 65.00 N. Type=Valve
 2140 2150

Weight= 65.00 N. Type=Valve
 2180 2190

Weight= 65.00 N. Type=Valve
 2200 2210

Weight= 65.00 N. Type=Valve
 2210 2220

Weight= 115.65 N. Type=Flange
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SIF's & TEE's
 160 170

Node 170 Welding Tee Srf.Node=1651
 220 230

Node 230 Welding Tee Srf.Node=2201
 230 340

Node 340 Welding Tee Srf.Node=3101
 420 430

Node 430 Welding Tee Srf.Node=4001

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INPUT LISTING

520 530

Node 530 Welding Tee Srf.Node=5001
 670 680

Node 680
 940 950

Node 950 Welding Tee Srf.Node=9301
 1540 1550

Node 1550 Welding Tee Srf.Node=2721
 1750 1760

Node 1760 Welding Tee Srf.Node=3141
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REDUCERS

1320 1330

Diam2= 48.260 mm. Wall2= 10.160 mm.
 1390 1400

Diam2= 88.900 mm. Wall2= 11.125 mm.
 1470 1480

Diam2= 60.325 mm. Wall2= 11.074 mm.
 1580 1590

Diam2= 88.900 mm. Wall2= 15.240 mm.
 1595 1600

Diam2= 60.325 mm. Wall2= 11.074 mm.
 1630 1640

Diam2= 48.260 mm. Wall2= 10.160 mm.
 1740 1750

Diam2= 60.325 mm. Wall2= 11.074 mm.
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RESTRAINTS

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INPUT LISTING

4040

4060

RESTRAINTS

Node	4060	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-001	

4060

4080

RESTRAINTS

Node	4080	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-002	

4100

4120

RESTRAINTS

Node	4120	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-003	

4140

4160

RESTRAINTS

Node	4160	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-005	
Node	4160	Type	Guide	MU/YIELD/FORCE =	.30
Dir Vec=	.000	.000	.000	Tag = RA\SL-005	

4160

4180

RESTRAINTS

Node	4180	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-007	

4200

4220

RESTRAINTS

Node	4220	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-009	

4220

4240

RESTRAINTS

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INPUT LISTING

Node	4240	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-010	
4240		4260			

RESTRAINTS

Node	4260	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-012	
Node	4260	Type	Guide	MU/YIELD/FORCE =	.30
Dir Vec=	.000	.000	.000	Tag = RA\SL-012	
4260		4280			

RESTRAINTS

Node	4280	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-014	
4280		4300			

RESTRAINTS

Node	4300	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-016	
4300		4320			

RESTRAINTS

Node	4320	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-018	
Node	4320	Type	Guide	Dir Vec=	.000 .000 .000
Tag = RA\SL-018					
4320		4340			

RESTRAINTS

Node	4340	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-020	
Node	4340	Type	LIM	Dir Vec=	.000 .000 .000
Tag = RA\SL-020					
4340		4360			

RESTRAINTS

Node	4360	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-022	

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INPUT LISTING

4360

4380

RESTRAINTS

Node	4380	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-024	
Node	4380	Type	Guide	Dir Vec=	.000 .000 .000
Tag =	RA\SL-024				

4380

4400

RESTRAINTS

Node	4400	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-026	

4400

4420

RESTRAINTS

Node	4420	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-028	

4420

4440

RESTRAINTS

Node	4440	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-030	
Node	4440	Type	Guide	Dir Vec=	.000 .000 .000
Tag =	RA\SL-030				

4440

4460

RESTRAINTS

Node	4460	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-032	

4460

4480

RESTRAINTS

Node	4480	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-034	

4520

4540

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INPUT LISTING

RESTRAINTS

Node	4540	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-036	
4560				4580	

RESTRAINTS

Node	4580	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-037	
4580				4600	

RESTRAINTS

Node	4600	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-038	
4620				4640	

RESTRAINTS

Node	4640	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-039	
4680				4700	

RESTRAINTS

Node	4700	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-040	
4700				4720	

RESTRAINTS

Node	4720	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-042	
Node	4720	Type	Guide	Dir Vec=	.000 .000 .000
Tag = RA\SL-042					
4720				4740	

RESTRAINTS

Node	4740	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-044	
4740				4760	

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INPUT LISTING

RESTRAINTS

Node	4760	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-046	
4760				4780	

RESTRAINTS

Node	4780	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-048	
Node	4780	Type	Guide	Dir Vec=	.000 .000 .000
Tag = RA\SL-048					
4780				4800	

RESTRAINTS

Node	4800	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-050	
4800				4820	

RESTRAINTS

Node	4820	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-051	
4820				4840	

RESTRAINTS

Node	4840	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-053	
Node	4840	Type	Guide	Dir Vec=	.000 .000 .000
Tag = RA\SL-053					
4840				4860	

RESTRAINTS

Node	4860	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-055	
Node	4860	Type	LIM	Dir Vec=	.000 .000 .000
Tag = RA\SL-055					
4860				4880	

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INPUT LISTING

RESTRAINTS

Node	4880	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = R8\SL-057	
4880		4900			

RESTRAINTS

Node	4900	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA\SL-059	
Node	4900	Type	Guide	MU/YIELD/FORCE =	.30
Dir Vec=	.000	.000	.000	Tag = RA\SL-059	
4900		4920			

RESTRAINTS

Node	4920	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA/SL-061	
4920		4940			

RESTRAINTS

Node	4940	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA/SL-063	
4940		4950			

RESTRAINTS

Node	4950	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA/SL-065	
Node	4950	Type	Guide	Dir Vec=	.000 .000 .000
Tag = RA/SL-065					
4950		4960			

RESTRAINTS

Node	4960	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA/SL-066	
5000		5020			

RESTRAINTS

Node	5020	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = RA/PS-001	

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INPUT LISTING

5020

5040

RESTRAINTS

Node	5040	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = RA/PS-002	

5080

5100

RESTRAINTS

Node	5100	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = RA/SL-68	

5100

5120

RESTRAINTS

Node	5120	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = RA/SL-70	

5120

5140

RESTRAINTS

Node	5140	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = RA/SL-72	

5200

5220

RESTRAINTS

Node	5220	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)	
Node	5220	Type	Guide	MU/YIELD/FORCE =	.30
Dir Vec=	.000	.000		.000 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)	
Node	5220	Type	LIM	MU/YIELD/FORCE =	.30
Dir Vec=	.000	.000		.000 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)	

5220

5240

RESTRAINTS

Node	5240	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000		.000 Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)	

5270

5280

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INPUT LISTING

RESTRAINTS

Node	5280	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = MANIFOLD SUPPORT (NOT IN ODC SCOPE)	
20				30	

RESTRAINTS

Node	30	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-001	
Node	30	Type	Guide	Dir Vec=	.000 .000 .000
Tag = SL-001					
Node	30	Type	LIM	Dir Vec=	.000 .000 .000
Tag = SL-001					
50				60	

RESTRAINTS

Node	60	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-004	
90				100	

RESTRAINTS

Node	100	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-005 (4" PIPE)	
100				110	

RESTRAINTS

Node	110	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-006 (4" PIPE)	
Node	110	Type	Guide	Dir Vec=	.000 .000 .000
Tag = SL-006 (4" PIPE)					
150				160	

RESTRAINTS

Node	160	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-007	
210				220	

RESTRAINTS

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INPUT LISTING

```

Node      220  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = PS-010 (4" PIPE)
Node      220  Type  Guide    MU/YIELD/FORCE =      .30
Dir Vec=   .000   .000   .000   Tag = PS-010 (4" PIPE)

      300                                310

```

RESTRAINTS

```

Node      310  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = PS-010 (3" BY PASS LINE)
Node      310  Type  Guide    Dir Vec=   .000   .000   .000
Tag = PS-010 (3" BY PASS LINE)

      340                                350

```

RESTRAINTS

```

Node      350  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = SL-009

      470                                480

```

RESTRAINTS

```

Node      480  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = SL-010
Node      480  Type  Guide    Dir Vec=   .000   .000   .000
Tag = SL-010

      580                                590

```

RESTRAINTS

```

Node      590  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = SL-010 (COMMON SLEEPER FOR FUTURE)

      730                                740

```

RESTRAINTS

```

Node      740  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = SL-016

      780                                790

```

RESTRAINTS

```

Node      790  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=   .000   1.000   .000   Tag = SL-017

```


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INPUT LISTING

690

830

RESTRAINTS

Node	830	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-011	
Node	830	Type	Guide	Dir Vec=	.000 .000 .000
Tag =	SL-011				

930

940

RESTRAINTS

Node	940	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-012	

990

1000

RESTRAINTS

Node	1000	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-014	

1040

1050

RESTRAINTS

Node	1050	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-015	

950

1090

RESTRAINTS

Node	1090	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-014	

1160

1170

RESTRAINTS

Node	1170	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = PS-010 (3" PIPE)	

1220

1230

RESTRAINTS

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INPUT LISTING

```

Node      1230  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-018 (3" PIPE)
Node      1230  Type  Guide   Dir Vec=    .000   .000   .000
Tag = SL-018 (3" PIPE)

```

1260

1270

RESTRAINTS

```

Node      1270  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-006 (3" PIPE)

```

1270

1280

RESTRAINTS

```

Node      1280  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-005 (3" PIPE)

```

1280

1290

RESTRAINTS

```

Node      1290  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-019
Node      1290  Type  Guide   Dir Vec=    .000   .000   .000
Tag = SL-019

```

1370

1380

RESTRAINTS

```

Node      1380  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-020

```

1430

1460

RESTRAINTS

```

Node      1460  Type  +Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-003

```

1520

1530

RESTRAINTS

```

Node      1530  Type  Y      MU/YIELD/FORCE =      .30
Dir Vec=    .000   1.000   .000   Tag = SL-002
Node      1530  Type  Guide   Len/GAP/STIF2 =      5.00 mm.
Dir Vec=    .000   .000   .000   Tag = SL-002

```

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INPUT LISTING

Node	1530	Type	LIM	Len/GAP/STIF2 =	5.00 mm.
Dir Vec=	.000	.000	.000	Tag = SL-002	
1560				1570	

RESTRAINTS

Node	1570	Type	+Y	MU/YIELD/FORCE =	.30
Dir Vec=	.000	1.000	.000	Tag = SL-001	
1610	X-MAS CASING CONNECTION			1620	

RESTRAINTS

Node	1620	Type	ANC	Cnode	1611	Dir Vec=	.000	.000
.000								
1770	X-MAS TREE WING VALVE			1780				

RESTRAINTS

Node	1780	Type	ANC	Cnode	1781	Dir Vec=	.000	.000
.000								
1820				1830				

RESTRAINTS

Node	1830	Type	ANC	Cnode	2220	Dir Vec=	.000	.000
.000								
1890				1900				

RESTRAINTS

Node	1900	Type	Z	Cnode	1910	Dir Vec=	.000	.000
1.000								
Node	1900	Type	Y	Cnode	1910	Dir Vec=	.000	1.000
.000								
Node	1900	Type	X	Cnode	1910	Dir Vec=	1.000	.000
.000								
Node	1910	Type	RX2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	3,740.00			Dir Vec=	1.000	.000	.000	
1910				1920				

RESTRAINTS

Node	1900	Type	RZ	Cnode	1910	Dir Vec=	.000	.000
1.000								

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INPUT LISTING

Node	1900	Type	RY	Cnode	1910	Dir Vec=	.000	1.000
.000								
1950				1960				

RESTRAINTS

Node	1960	Type	Z	Cnode	1970	Dir Vec=	.000	.000
1.000								
Node	1960	Type	Y	Cnode	1970	Dir Vec=	.000	1.000
.000								
Node	1960	Type	X	Cnode	1970	Dir Vec=	1.000	.000
.000								
Node	1970	Type	RX2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	4,243.50			Dir Vec=	1.000	.000	.000	
1970				1980				

RESTRAINTS

Node	1970	Type	RZ2	Len/GAP/STIF2 =	1.00 mm.
MU/YIELD/FORCE =	4,243.50			Dir Vec=	.000 .000 1.000
Node	1970	Type	RY2	Len/GAP/STIF2 =	1.00 mm.
MU/YIELD/FORCE =	4,243.50			Dir Vec=	.000 1.000 .000
2010				2020	

RESTRAINTS

Node	2020	Type	Z	Cnode	2030	Dir Vec=	.000	.000
1.000								
Node	2020	Type	Y	Cnode	2030	Dir Vec=	.000	1.000
.000								
Node	2020	Type	X	Cnode	2030	Dir Vec=	1.000	.000
.000								
Node	2030	Type	RX2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	3,745.00			Dir Vec=	1.000 .000 .000			
2030				2040				

RESTRAINTS

Node	2020	Type	RZ	Cnode	2030	Dir Vec=	.000	.000
1.000								
Node	2020	Type	RY	Cnode	2030	Dir Vec=	.000	1.000
.000								
2050				2060				

RESTRAINTS

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INPUT LISTING

Node	2050	Type	ANC	Cnode	1840	Dir Vec=	.000	.000
.000								

2060

2070

RESTRAINTS

Node	2070	Type	Z	Cnode	2080	Dir Vec=	.000	.000
1.000								
Node	2070	Type	Y	Cnode	2080	Dir Vec=	.000	1.000
.000								
Node	2070	Type	X	Cnode	2080	Dir Vec=	1.000	.000
.000								
Node	2080	Type	RX2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	1,445.00			Dir Vec=	1.000	.000	.000	

2080

2090

RESTRAINTS

Node	2070	Type	RZ	Cnode	2080	Dir Vec=	.000	.000
1.000								
Node	2070	Type	RY	Cnode	2080	Dir Vec=	.000	1.000
.000								

2120

2130

RESTRAINTS

Node	2130	Type	Z	Cnode	2140	Dir Vec=	.000	.000
1.000								
Node	2130	Type	Y	Cnode	2140	Dir Vec=	.000	1.000
.000								
Node	2130	Type	X	Cnode	2140	Dir Vec=	1.000	.000
.000								
Node	2140	Type	RX2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	1,445.00			Dir Vec=	1.000	.000	.000	

2140

2150

RESTRAINTS

Node	2140	Type	RZ2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	1,803.00			Dir Vec=	.000	.000	1.000	
Node	2140	Type	RY2	Len/GAP/STIF2 =	1.00 mm.			
MU/YIELD/FORCE =	1,803.00			Dir Vec=	.000	1.000	.000	

2180

2190

RESTRAINTS

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INPUT LISTING

Node	2190	Type	Z	Cnode	2200	Dir Vec=	.000	.000
1.000								
Node	2190	Type	Y	Cnode	2200	Dir Vec=	.000	1.000
.000								
Node	2190	Type	X	Cnode	2200	Dir Vec=	1.000	.000
.000								
Node	2200	Type	RX2	Len/GAP/STIF2 =		1.00 mm.		
MU/YIELD/FORCE =	1,445.00			Dir Vec=	1.000	.000	.000	
	2200				2210			

RESTRAINTS

Node	2190	Type	RZ	Cnode	2200	Dir Vec=	.000	.000
1.000								
Node	2190	Type	RY	Cnode	2200	Dir Vec=	.000	1.000
.000								

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DISPLACEMENTS

1610 X-MAS CASING CONNECTION 1620

Node	1611	DX1=	.000 mm.	DY1=	2,200.000 mm.	DZ1=	.000 mm.	RX1=	.000
		RY1=	.000	RZ1=	.000				
	1770	X-MAS TREE WING VALVE			1780				

Node	1781	DX1=	.000 mm.	DY1=	2,200.000 mm.	DZ1=	.000 mm.	RX1=	.000
		RY1=	.000	RZ1=	.000				

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FLANGES

5160 5180

Location= To	Method= NC-3658.3	G/C= 200.000 mm.
Bolt Area= 1,723.860 mm.	SYC= 241,000 KPa	SY1= 188,000 KPa
SY2= 188,000 KPa	SY3= 188,000 KPa	SY4= 188,000 KPa
SY5= 188,000 KPa	SY6= 188,000 KPa	SY7= 188,000 KPa
SY8= 188,000 KPa	SY9= 188,000 KPa	
70	80	

Location= To	Method= NC-3658.3	G/C= 235.000 mm.
Bolt Area= 3,938.000 mm.	SYC= 241,000 KPa	SY1= 188,000 KPa
SY2= 188,000 KPa	SY3= 188,000 KPa	SY4= 188,000 KPa
SY5= 188,000 KPa	SY6= 188,000 KPa	SY7= 188,000 KPa
SY8= 188,000 KPa	SY9= 188,000 KPa	
120	130	

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INPUT LISTING

Location= To	Method= NC-3658.3	
180		190 CHECK VALVE
Location= To	Method= NC-3658.3	
200		210
Location= From	Method= NC-3658.3	
260 GATE VALVE		270 GATE VALVE
Location= Both	Method= NC-3658.3	G/C= 190.500 mm.
Bolt Area= 2,384.000 mm.		
360		370 UZV VALVE
Location= To	Method= NC-3658.3	G/C= 235.000 mm.
Bolt Area= 3,938.000 mm.		
380		390
Location= From	Method= NC-3658.3	
400		410
Location= To	Method= NC-3658.3	
440		450 GATE VALVE
Location= To	Method= NC-3658.3	
460		470
Location= From	Method= NC-3658.3	
500 GATE VALVE		510
Location= Both	Method= NC-3658.3	
560 GATE VALVE		570
Location= Both	Method= NC-3658.3	
600		610 GATE VALVE
Location= To	Method= NC-3658.3	
620		630
Location= From	Method= NC-3658.3	
650		660
Location= To	Method= NC-3658.3	
710 GATE VALVE		720

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INPUT LISTING

Location= Both	Method= NC-3658.3	
760		770
Location= Both	Method= NC-3658.3	
800		810
Location= To	Method= NC-3658.3	
850 GATE VALVE		860
Location= Both	Method= NC-3658.3	
910 GATE VALVE		920
Location= Both	Method= NC-3658.3	
970 GATE VALVE		980
Location= Both	Method= NC-3658.3	
1020 GATE VALVE		1030
Location= Both	Method= NC-3658.3	
1060		1070
Location= To	Method= NC-3658.3	
1100		1110
Location= To	Method= NC-3658.3	
1140 GATE VALVE		1150
Location= Both	Method= NC-3658.3	G/C= 190.500 mm.
Bolt Area= 2,384.000 mm.		
1200 CHECK VALVE		1210
Location= Both	Method= NC-3658.3	
1240		1250
Location= Both	Method= NC-3658.3	
1300		1310
Location= To	Method= NC-3658.3	
1350 FLOW METER		1360
Location= Both	Method= NC-3658.3	G/C= 123.800 mm.

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INPUT LISTING

Bolt Area= 1,563.600 mm.
 1400

1410

Location= To Method= NC-3658.3 G/C= 190.500 mm.
 Bolt Area= 2,384.000 mm.
 1480

1490 CHOKE VALVE

Location= To Method= NC-3658.3 G/C= 165.100 mm.
 1510

1520

Location= From Method= NC-3658.3
 1770 X-MAS TREE WING VALVE

1780

Location= To Method= NC-3658.3 G/C= 235.000 mm.
 Bolt Area= 3,938.000 mm.
 1840

1540

Location= From Method= NC-3658.3 G/C= 165.100 mm.
 Bolt Area= 2,384.000 mm.
 1800

1850

Location= From Method= NC-3658.3 G/C= 235.000 mm.
 Bolt Area= 3,938.000 mm.
 1870

1880

Location= To Method= NC-3658.3
 2040

10

Location= To Method= NC-3658.3
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INPUT UNITS USED...

UNITS=	SI (m NOM/SCH	INPUT= ON				
LENGTH	inches	x	25.400	=	mm.	
FORCE	pounds	x	4.448	=	N.	
MASS (dynamics)	pounds	x	0.454	=	Kg.	
MOMENTS (INPUT)	inch-pounds	x	0.113	=	N.m.	
MOMENTS (OUTPUT)	inch-pounds	x	0.113	=	N.m.	
STRESS	lbs./sq.in.	x	6.895	=	KPa	
TEMP. SCALE	degrees F.	x	0.556	=	C	
PRESSURE	psig	x	6.895	=	KPa	
ELASTIC MODULUS	lbs./sq.in.	x	6.895	=	KPa	
PIPE DENSITY	lbs./cu.in.	x	0.028	=	kg./cu.cm.	
INSULATION DENS.	lbs./cu.in.	x	0.028	=	kg./cu.cm.	
FLUID DENSITY	lbs./cu.in.	x	0.028	=	kg./cu.cm.	
TRANSL. STIF	lbs./in.	x	1.751	=	N./cm.	
ROTATIONAL STIF	in.lb./deg.	x	0.113	=	N.m./deg	

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

UNIFORM LOAD	lb./in.	x	1.751	=	N./cm.
G LOAD	g's	x	1.000	=	g's
WIND LOAD	lbs./sq.in.	x	6.895	=	KPa
ELEVATION	inches	x	0.025	=	m.
COMPOUND LENGTH	inches	x	25.400	=	mm.
DIAMETER	inches	x	25.400	=	mm.
WALL THICKNESS	inches	x	25.400	=	mm.

CAESAR II Ver.13.00.00.0890, (Build 220811) MAR 26, 2025 8:45:56

SETUP FILE PARAMETERS

```

CONNECT GEOMETRY THRU CNODES =      YES
MIN ALLOWED BEND ANGLE =             5.00000
MAX ALLOWED BEND ANGLE =             95.0000
BEND LENGTH ATTACHMENT PERCENT =     1.00000
MIN ANGLE TO ADJACENT BEND PT =      5.00000
LOOP CLOSURE TOLERANCE =             25.4000      mm.
THERMAL BOWING HORZ TOLERANCE =      0.100000E-03
AUTO NODE NUMBER INCREMENT=         20.0000
Z AXIS UP=                           NO
USE PRESSURE STIFFENING =             DEFAULT
ALPHA TOLERANCE =                     0.500000E-01
RESLD-FORCE =                         NO
HGR DEF RESWGT STIF =                 0.175127E+13 N./cm.
DECOMP SNG TOL =                     0.100000E+11
BEND AXIAL SHAPE =                    YES
FRICT STIF =                          0.175127E+07 N./cm.
FRICT NORM FORCE VAR =                 0.150000
FRICT ANGLE VAR =                     15.0000
FRICT SLIDE MULT =                    1.00000
ROD TOLERANCE =                       1.00000
ROD INC =                             2.00000
INCORE NUMERICAL CHECK =              NO
OUTCORE NUMERICAL CHECK =             NO
DEFAULT TRANS RESTRAINT STIFF=        0.175127E+13      N./cm.
DEFAULT ROT RESTRAINT STIFF=          0.112985E+12      N.m./deg
IGNORE SPRING HANGER STIFFNESS =      NO
MISSING MASS ZPA =                    EXTRACTED
MIN WALL MILL TOLERANCE =             12.5000
WRC-107 VERSION =                     MAR 79 1B1/2B1
WRC-107 INTERPOLATION =               LAST VALUE
DEFAULT AMBIENT TEMPERATURE=          21.1111      C
BOURDON PRESSURE=                     NONE
COEFFICIENT OF FRICTION (MU) =        0.300000
INCLUDE SPRG STIF IN HGR OPE =        NO
INCLUDE INSULATION IN HYDROTEST =     NO
REDUCED INTERSECTION =                B31.1 (POST1980)
USE WRC329                            NO
NO REDUCED SIF FOR RFT AND WLT        NO
B31.1 REDUCED Z FIX =                 YES
CLASS 1 BRANCH FLEX                   NO
ALL STRESS CASES CORRODED =           NO
ADD TORSION IN SL STRESS =             DEFAULT
ADD F/A IN STRESS =                   DEFAULT
OCCASIONAL LOAD FACTOR =               0.000000
DEFAULT CODE =                        B31.3 2020

```

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

```

B31.3 SUS CASE SIF FACTOR =                0.000000
ALLOW USERS BEND SIF =                    NO
USE SCHNEIDER                             NO
YIELD CRITERION STRESS =                  MAX 3D SHEAR
USE PD/4T                                  NO
BASE HOOP STRESS ON ? =                   ID
EN13480 USE IN OUTPLANE SIFS=              NO
LIBERAL EXPANSION ALLOWABLE=              NO
B31.3 SEC 319.2.3C SAXIAL=                 Default
B31.3 WELDING/CONTOUR TEE ISB16.9 FALSE
PRESSURE VARIATION IN EXP CASE=           DEFAULT
IMPLEMENT B313 APP-P                      NO
IMPLEMENT B313 CODE CASE 178              YES
IGNORE B31.1/B31.3 Wc FACTOR=             YES
APPLY B31J SIFS & FLEX=                   DEFAULT
ENFORCE B31J SIFS ONLY=                   FALSE
USE FRP SIF =                             YES
USE FRP FLEX =                             YES
BS 7159 Pressure Stiffening=              Design Strain
FRP Property Data File=                   CAESAR.FRP
FRP Emod (axial) =                        0.220632E+08      KPa
FRP Ratio Gmod/Emod (axial) =             0.250000
FRP Ea/Eh*Vh/a =                         0.152730
FRP Laminate Type =                       THREE
FRP Alpha =                              21.6000              C
FRP Density =                             0.166079E-02 kg./cu.cm.
EXCLUDE f2 FROM UKOOA BENDING =          NO
CAESAR II Ver.13.00.00.0890, (Build 220811)      MAR 26,2025    8:45:56

```

EXECUTION CONTROL PARAMETERS

```

Rigid/ExpJt Print Flag ..... 1.000
Bourdon Option ..... 1.000
Loop Closure Flag ..... 2.000
Thermal Bowing Delta Temp .. .000 C
Liberal Allowable Flag ..... .000
Uniform Load Option ..... .000

Ambient Temperature ..... 21.000 C
Plastic (FRP) Alpha ..... 21.600
Plastic (FRP) GMOD/EMODa ... .250
Plastic (FRP) Laminate Type. 3.000
Eqn Optimizer ..... .000
Node Selection ..... .000
Eqn Ordering ..... .000
Collins ..... .000
Degree Determination ..... .000
User Eqn Control ..... .000
North Direction ..... -X
CAESAR II Ver.13.00.00.0890, (Build 220811)      MAR 26,2025    8:45:56

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COORDINATE REPORT

```

/----- (mm.) -----/
NODE          X          Y          Z

```

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

1110	34260.406	.000	.000
4020	34349.305	.000	.000
4030	35849.305	.000	.000
4040	36163.305	.000	.000
4060	36163.305	.000	-1280.000
4080	36163.305	.000	-4480.000
4100	36163.305	.000	-5880.000
4120	37552.305	.000	-5880.000
4140	40552.305	.000	-5880.000
4160	43552.305	.000	-5880.000
4180	49052.305	.000	-5880.000
4200	50552.305	.000	-5880.000
4220	50552.305	.000	-7158.000
4240	50552.305	.000	-9658.000
4260	50552.305	.000	-15658.000
4280	50552.305	.000	-21658.000
4300	50552.305	.000	-27658.000
4320	50552.305	.000	-33658.000
4340	50552.305	.000	-39658.000
4360	50552.305	.000	-45658.000
4380	50552.305	.000	-51658.000
4400	50552.305	.000	-57658.000
4420	50552.305	.000	-63658.000
4440	50552.305	.000	-69658.000
4460	50552.305	.000	-75658.000
4480	50552.305	.000	-81158.000
4500	50552.305	.000	-82358.000
4520	49552.305	.000	-82358.000
4540	47976.305	.000	-82358.000
4560	45552.305	.000	-82358.000
4580	45552.305	.000	-83508.000
4600	45552.305	.000	-86208.000
4620	45552.305	.000	-87358.000
4640	47976.305	.000	-87358.000
4660	49552.305	.000	-87358.000
4680	50552.305	.000	-87358.000
4700	50552.305	.000	-88558.000
4720	50552.305	.000	-94058.000
4740	50552.305	.000	-100058.000
4760	50552.305	.000	-106058.000
4780	50552.305	.000	-112058.000
4800	50552.305	.000	-117558.000
4820	50552.305	.000	-121058.000
4840	50552.305	.000	-127058.000
4860	50552.305	.000	-131558.000
4880	50552.305	.000	-136058.000
4900	50552.305	.000	-142058.000
4920	50552.305	.000	-148058.000
4940	50552.305	.000	-154058.000
4950	50552.305	.000	-160058.000
4960	50552.305	.000	-163508.000
4980	50552.305	.000	-165835.000
5000	50552.305	900.000	-165835.000
5020	51952.305	900.000	-165835.000
5040	55552.305	900.000	-165835.000
5060	57081.305	900.000	-165835.000

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

5080	57081.305	-56.000	-165835.000
5100	58081.305	-56.000	-165835.000
5120	63081.305	-56.000	-165835.000
5140	68081.305	-56.000	-165835.000
5150	70881.305	-56.000	-165835.000
5160	71097.305	-56.000	-165835.000
5180	71186.203	-56.000	-165835.000
5200	71275.102	-56.000	-165835.000
5220	72805.102	-56.000	-165835.000
5240	76210.102	-56.000	-165835.000
5260	77340.102	-56.000	-165835.000
5270	77340.102	-56.000	-164585.000
5280	77693.102	-56.000	-164585.000
5300	78518.102	-56.000	-164585.000
5320	78968.102	-56.000	-164585.000
5340	79057.000	-56.000	-164585.000
10	-.000	.000	.000
20	123.825	.000	.000
30	763.825	.000	.000
40	1143.825	.000	.000
50	1443.825	.000	.000
60	2443.825	.000	.000
70	3815.825	.000	.000
80	3939.650	.000	.000
90	4063.475	.000	.000
100	6444.475	.000	.000
110	10444.475	.000	.000
120	13821.475	.000	.000
130	13945.300	.000	.000
140	14069.125	.000	.000
150	14469.125	.000	.000
160	14745.125	.000	.000
170	16536.125	.000	.000
180	16641.125	.000	.000
190	16764.949	.000	.000
200	17222.148	.000	.000
210	17345.973	.000	.000
220	18084.973	.000	.000
230	18734.973	.000	.000
240	18734.973	120.000	.000
250	18734.973	145.000	.000
260	18734.973	256.125	.000
270	18734.973	637.125	.000
280	18734.973	748.250	.000
290	18734.973	1037.250	.000
300	18734.973	1037.250	-500.000
310	18084.973	1037.250	-500.000
320	16536.123	1037.250	-500.000
330	16536.123	1037.250	.000
330	16536.125	1037.000	.000
170	16536.125	.000	.000
230	18734.973	.000	.000
340	19613.973	.000	.000
350	21684.973	.000	.000
360	23343.973	.000	.000
370	23467.797	.000	.000

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

380	23924.996	.000	.000
390	24048.820	.000	.000
400	24641.820	.000	.000
410	24765.645	.000	.000
420	24889.469	.000	.000
430	24994.469	.000	.000
440	25099.469	.000	.000
450	25223.293	.000	.000
460	25680.492	.000	.000
470	25804.316	.000	.000
480	26404.316	.000	.000
490	27344.316	.000	.000
500	27468.141	.000	.000
510	27925.340	.000	.000
520	28049.164	.000	.000
530	28154.164	.000	.000
540	28154.164	.000	800.000
550	27982.164	.000	800.000
560	27858.340	.000	800.000
570	27401.141	.000	800.000
580	27277.316	.000	800.000
590	26404.316	.000	800.000
600	25851.316	.000	800.000
610	25727.492	.000	800.000
620	25270.293	.000	800.000
630	25146.469	.000	800.000
640	24994.469	.000	800.000
430	24994.469	.000	.000
530	28154.164	.000	.000
650	28259.164	.000	.000
660	28382.988	.000	.000
670	28506.812	.000	.000
680	28611.812	.000	.000
690	29111.812	.000	.000
680	28611.812	.000	.000
700	28611.812	.000	-805.000
710	28611.812	.000	-928.825
720	28611.812	.000	-1386.025
730	28611.812	.000	-1509.850
740	28611.812	.000	-2909.850
750	28611.812	.000	-3509.850
760	28611.812	.000	-3633.675
770	28611.812	.000	-4090.875
780	28611.812	.000	-4214.700
790	28611.812	.000	-4814.700
800	28611.812	.000	-5414.700
810	28611.812	.000	-5538.525
820	28611.812	.000	-5588.525
690	29111.812	.000	.000
830	29316.812	.000	.000
840	29916.812	.000	.000
850	30040.637	.000	.000
860	30497.836	.000	.000
870	30621.660	.000	.000
880	30915.160	.000	.000
890	31208.660	.000	.000

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

900	31332.484	.000	.000
910	31375.484	.000	.000
920	31832.684	.000	.000
930	31956.508	.000	.000
940	32461.508	.000	.000
950	33121.508	.000	.000
960	33121.508	.000	-805.000
970	33121.508	.000	-893.900
980	33121.508	.000	-1198.700
990	33121.508	.000	-1287.600
1000	33121.508	.000	-2687.600
1010	33121.508	.000	-3287.600
1020	33121.508	.000	-3376.500
1030	33121.508	.000	-3681.300
1040	33121.508	.000	-3770.200
1050	33121.508	.000	-4370.200
1060	33121.508	.000	-4970.200
1070	33121.508	.000	-5059.100
1080	33121.508	.000	-5109.100
950	33121.508	.000	.000
1090	33721.508	.000	.000
1100	34171.508	.000	.000
1110	34260.406	.000	.000
340	19613.973	.000	.000
1120	19613.973	.000	-950.000
1130	19498.973	.000	-950.000
1140	19387.848	.000	-950.000
1150	18989.848	.000	-950.000
1160	18878.723	.000	-950.000
1170	18088.723	.000	-950.000
1180	16360.723	.000	-950.000
1190	15051.723	.000	-950.000
1200	14940.598	.000	-950.000
1210	14559.598	.000	-950.000
1220	14448.473	.000	-950.000
1230	13887.473	.000	-950.000
1240	13346.473	.000	-950.000
1250	13235.348	.000	-950.000
1260	13124.223	.000	-950.000
1270	10444.223	.000	-950.000
1280	6444.223	.000	-950.000
1290	4091.223	.000	-950.000
1300	3445.223	.000	-950.000
1310	3334.098	.000	-950.000
1320	3222.973	.000	-950.000
1330	3133.973	.000	-950.000
1340	2766.973	.000	-950.000
1350	2674.898	.000	-950.000
1360	2412.898	.000	-950.000
1370	2320.823	.000	-950.000
1380	1617.823	.000	-950.000
1390	1217.823	.000	-950.000
1400	1128.823	.000	-950.000
1410	1017.698	.000	-950.000
1420	906.573	.000	-950.000
1430	406.573	.000	-950.000

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

1460	-2014.427	.000	-950.000
1470	-2589.947	.000	-950.000
1480	-2703.947	-14.200	-950.000
1490	-2814.220	-14.200	-950.000
1500	-3013.220	-14.200	-950.000
1510	-3013.220	-14.200	-751.000
1520	-3013.220	-14.200	-639.875
1530	-3013.220	-14.200	506.125
1540	-2357.725	-276.149	2892.047
1550	-2282.725	-276.149	2892.047
1560	-2282.725	-683.149	2892.047
1570	-2032.725	-683.149	2892.047
1580	-1832.725	-683.149	2892.047
1590	-1732.725	-683.149	2892.047
1595	-1491.725	-683.149	2892.047
1600	-1391.725	-683.149	2892.047
1605	-1315.525	-683.149	2892.047
1610	-1315.525	-683.149	3166.047
1620	-1315.525	-683.149	3277.172
1550	-2282.725	-276.149	2892.047
1630	-2282.725	-226.149	2892.047
1640	-2282.725	-150.149	2892.047
1650	-2282.725	-58.074	2892.047
1660	-2282.725	246.926	2892.047
1670	-2282.725	339.001	2892.047
1680	-2282.725	934.001	2892.047
1690	-1886.725	934.001	2892.047
1700	-1794.650	934.001	2892.047
1710	-1489.650	934.001	2892.047
1720	-1397.575	934.001	2892.047
1730	-1240.575	934.001	2892.047
1740	-1240.575	803.449	2892.047
1750	-1240.575	727.449	2892.047
1760	-1240.475	611.000	2891.650
1770	-1240.475	611.000	3576.650
1780	-1240.475	611.000	3700.475
1760	-1240.475	611.000	2891.650
1790	-1240.475	611.000	2791.650
1800	-1240.475	611.000	2667.825
1530	-3013.220	-14.200	506.125
1810	-3013.220	-14.200	1106.125
1820	-2937.220	-14.200	1106.125
1830	-2826.095	-14.200	1106.125
2050	-2430.750	-276.149	2892.047
1540	-2357.725	-276.149	2892.047
1800	-1240.475	611.000	2667.825
1850	-1240.475	611.000	2544.000
1860	-1240.475	611.000	2392.000
1870	-613.475	611.000	2392.000
1880	-489.650	611.000	2392.000
1890	-365.825	611.000	2392.000
1900	-244.825	611.000	2392.000
1920	-123.825	611.000	2392.000
1930	29.175	611.000	2392.000
1940	29.175	2208.000	806.000
1950	-123.825	2208.000	806.000

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

INPUT LISTING

1960	-244.825	2208.000	806.000
1980	-365.825	2208.000	806.000
1990	-518.825	2208.000	806.000
2000	-518.825	.000	.000
2010	-365.825	.000	.000
2020	-244.825	.000	.000
2040	-123.825	.000	.000
10	-.000	.000	.000
2050	-2430.750	-276.149	2892.047
2060	-2541.875	-276.149	2892.047
2070	-2628.425	-276.149	2892.047
2090	-2714.975	-276.149	2892.047
2100	-2791.175	-276.149	2892.047
2110	-2791.175	1932.173	2457.047
2120	-2714.975	1932.173	2457.047
2130	-2628.425	1932.173	2457.047
2150	-2541.875	1932.173	2457.047
2160	-2465.675	1932.173	2457.047
2170	-2465.675	-14.197	1106.127
2180	-2541.875	-14.197	1106.127
2190	-2628.425	-14.197	1106.127
2210	-2714.975	-14.197	1106.127
2210	-2714.970	-14.200	1106.125
1830	-2826.095	-14.200	1106.125

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 6

LOAD CASE REPORT

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

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LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS2
CAESAR II LOAD CASE REPORT (DATE:MAR 26,2025)3

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS

1 (HYD) WW+HP
2 (OPE) W+D1+T1+P1
3 (Alt-SUS) W+P1
4 (OPE) W+T2+P2
5 (Alt-SUS) W+P2
6 (OPE) W+T3+P1
7 (Alt-SUS) W+P1
8 (SUS) W+P1
9 (SUS) W+P2
10 (EXP) L10=L2-L8
11 (EXP) L11=L4-L9
12 (EXP) L12=L6-L8
13 (EXP) L13=L10-L12
14 (EXP) L14=L11-L12
26,2025 @ 8:45 Pg: 18

CAESAR II Ver.13.00.00.0890, (Build 220811)

Date:MAR

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

CAESAR II LOAD CASE REPORT

DATE:MAR 26,2025

CASE 1 (HYD) WW+HP

HYDRO TEST CASE

Keep/Discard: Keep
 Display: Disp/Force/Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 Flg Analysis Temp: T Amb

CASE 2 (OPE) W+D1+T1+P1

OPERATING CASE CONTIDION 1

Keep/Discard: Keep
 Display: Disp/Force
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 Flg Analysis Temp: T1

CASE 3 (Alt-SUS) W+P1

ALTERNATE STRESS FOR OPERATING CASE CONTIDION 1

Keep/Discard: Keep
 Display: Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 SUS/OCC case SH: SH_MIN
 Flg Analysis Temp: None
 Alternate SUS/OCC: Enabled

CASE 4 (OPE) W+T2+P2

OPERATING CASE CONTIDION 2

Keep/Discard: Keep
 Display: Disp/Force
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 Flg Analysis Temp: T2

CASE 5 (Alt-SUS) W+P2

ALTERNATE STRESS FOR OPERATING CASE CONTIDION 2

Keep/Discard: Keep
 Display: Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 SUS/OCC case SH: SH_MIN
 Flg Analysis Temp: None
 Alternate SUS/OCC: Enabled

CASE 6 (OPE) W+T3+P1

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CAESAR II LOAD CASE REPORT

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OPERATING CASE CONTIDION 3

Keep/Discard: Keep
 Display: Disp/Force
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 Flg Analysis Temp: T3

CASE 7 (Alt-SUS) W+P1

ALTERNATE STRESS FOR OPERATING CASE CONTIDION 3

Keep/Discard: Keep
 Display: Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 SUS/OCC case SH: SH_MIN
 Flg Analysis Temp: None
 Alternate SUS/OCC: Enabled

CASE 8 (SUS) W+P1

SUSTAIN CASE

Keep/Discard: Keep
 Display: Disp/Force/Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 SUS/OCC case SH: SH_MIN
 Flg Analysis Temp: None

CASE 9 (SUS) W+P2

SUSTAIN CASE

Keep/Discard: Keep
 Display: Disp/Force/Stress
 Elastic Modulus: EC
 Friction Mult.: 1.0000
 Fluid Den Mult.: 1.0000
 SUS/OCC case SH: SH_MIN
 Flg Analysis Temp: None

CASE 10 (EXP) L10=L2-L8

EXPANSION CASE DUE TO T1

Keep/Discard: Keep
 Display: Stress
 Combination Method: ALG

CASE 11 (EXP) L11=L4-L9

EXPANSION CASE DUE TO T2

Keep/Discard: Keep
 Display: Stress
 Combination Method: ALG

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CAESAR II LOAD CASE REPORT

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CASE 12 (EXP) L12=L6-L8
EXPANSION CASE DUE TO T3

Keep/Discard: Keep
Display: Stress
Combination Method: ALG

CASE 13 (EXP) L13=L10-L12
FULL EXPANSION STRESS RANGE T1 TO T3

Keep/Discard: Keep
Display: Stress
Combination Method: ALG

CASE 14 (EXP) L14=L11-L12
FULL EXPANSION STRESS RANGE T2 TO T3

Keep/Discard: Keep
Display: Stress
Combination Method: ALG

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 7
STRESS SUMMARY REPORT

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
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LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS2
Stress Summary (Legacy) : Multiple.....3

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LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS

- 1 (HYD) WW+HP
- 2 (OPE) W+D1+T1+P1
- 3 (Alt-SUS) W+P1
- 4 (OPE) W+T2+P2
- 5 (Alt-SUS) W+P2
- 6 (OPE) W+T3+P1
- 7 (Alt-SUS) W+P1
- 8 (SUS) W+P1
- 9 (SUS) W+P2
- 10 (EXP) L10=L2-L8
- 11 (EXP) L11=L4-L9
- 12 (EXP) L12=L6-L8
- 13 (EXP) L13=L10-L12
- 14 (EXP) L14=L11-L12

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STRESS SUMMARY (LEGACY) REPORT: Highest Stresses Mini Statement Various Load Cases

LOAD CASE DEFINITION KEY

CASE 1 (HYD) WW+HP
 CASE 2 (OPE) W+D1+T1+P1
 CASE 3 (Alt-SUS) W+P1
 CASE 4 (OPE) W+T2+P2
 CASE 5 (Alt-SUS) W+P2
 CASE 6 (OPE) W+T3+P1
 CASE 7 (Alt-SUS) W+P1
 CASE 8 (SUS) W+P1
 CASE 9 (SUS) W+P2
 CASE 10 (EXP) L10=L2-L8
 CASE 11 (EXP) L11=L4-L9
 CASE 12 (EXP) L12=L6-L8
 CASE 13 (EXP) L13=L10-L12
 CASE 14 (EXP) L14=L11-L12

Note: This report indicates generic stresses summary. Currently, CAESAR II does not have summary for code-defined stresses.

Piping Code: Multiple Codes

B31.3-IX-18 = B31.3 Chapter IX -2018, Aug 30, 2019

B31.3-2020 = B31.3 -2020, June 18, 2021

CODE STRESS CHECK PASSED : LOADCASE 1 (HYD) WW+HP

Highest Stresses: (	KPa	)	LOADCASE 1 (HYD) WW+HP	
Ratio (%):	32.8		@Node 1770	
Code Stress:	79232.8		Allowable Stress:	241316.5
Axial Stress:	42707.9		@Node 540	
Bending Stress:	39663.5		@Node 1640	
Torsion Stress:	6414.7		@Node 1790	
Hoop Stress:	95068.0		@Node 30	
Max Stress Intensity:	131310.2		@Node 1770	

CODE STRESS CHECK PASSED : LOADCASE 3 (Alt-SUS) W+P1

Highest Stresses: (	KPa	)	LOADCASE 3 (Alt-SUS) W+P1	
Ratio (%):	91.1		@Node 60	
Code Stress:	111110.2		Allowable Stress:	122009.6
Axial Stress:	32609.7		@Node 1980	
Bending Stress:	85733.8		@Node 1740	
Torsion Stress:	24183.4		@Node 1610	
Hoop Stress:	69990.6		@Node 30	
Max Stress Intensity:	111653.2		@Node 60	

CODE STRESS CHECK PASSED : LOADCASE 5 (Alt-SUS) W+P2

Highest Stresses: (	KPa	)	LOADCASE 5 (Alt-SUS) W+P2	
Ratio (%):	54.0		@Node 1770	
Code Stress:	65849.1		Allowable Stress:	122009.6
Axial Stress:	36446.5		@Node 4998	

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STRESS SUMMARY (LEGACY) REPORT: Highest Stresses Mini Statement
Various Load Cases

Bending Stress:	52352.1	@Node	1770
Torsion Stress:	9598.8	@Node	1790
Hoop Stress:	74209.9	@Node	4030
Max Stress Intensity:	81059.4	@Node	5260

CODE STRESS CHECK PASSED : LOADCASE 7 (Alt-SUS) W+P1

Highest Stresses: (KPa)	LOADCASE 7 (Alt-SUS) W+P1	
Ratio (%):	70.5	@Node 1770
Code Stress:	86014.5	Allowable Stress: 122009.6
Axial Stress:	32487.5	@Node 1940
Bending Stress:	52254.1	@Node 1770
Torsion Stress:	9553.5	@Node 1790
Hoop Stress:	69990.6	@Node 30
Max Stress Intensity:	95291.6	@Node 1770

CODE STRESS CHECK PASSED : LOADCASE 8 (SUS) W+P1

Highest Stresses: (KPa)	LOADCASE 8 (SUS) W+P1	
Ratio (%):	70.5	@Node 1770
Code Stress:	85980.9	Allowable Stress: 122009.6
Axial Stress:	32685.9	@Node 540
Bending Stress:	52216.6	@Node 1770
Torsion Stress:	9549.4	@Node 1790
Hoop Stress:	69990.6	@Node 30
Max Stress Intensity:	95282.6	@Node 1770

CODE STRESS CHECK PASSED : LOADCASE 9 (SUS) W+P2

Highest Stresses: (KPa)	LOADCASE 9 (SUS) W+P2	
Ratio (%):	53.9	@Node 1770
Code Stress:	65779.5	Allowable Stress: 122009.6
Axial Stress:	36323.7	@Node 4998
Bending Stress:	52274.9	@Node 1770
Torsion Stress:	9600.5	@Node 1790
Hoop Stress:	74209.9	@Node 4030
Max Stress Intensity:	81084.9	@Node 5260

CODE STRESS CHECK PASSED : LOADCASE 10 (EXP) L10=L2-L8

Highest Stresses: (KPa)	LOADCASE 10 (EXP) L10=L2-L8	
Ratio (%):	81.5	@Node 1640
Code Stress:	165304.4	Allowable Stress: 202871.3
Axial Stress:	1288.9	@Node 530
Bending Stress:	164568.3	@Node 1640
Torsion Stress:	47160.4	@Node 1610
Hoop Stress:	0.0	@Node 4020
Max Stress Intensity:	249067.1	@Node 4099

CODE STRESS CHECK PASSED : LOADCASE 11 (EXP) L11=L4-L9

Highest Stresses: (KPa)	LOADCASE 11 (EXP) L11=L4-L9	
Ratio (%):	65.0	@Node 4618

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STRESS SUMMARY (LEGACY) REPORT: Highest Stresses Mini Statement
Various Load Cases

Code Stress:	151831.3	Allowable Stress:	233559.9
Axial Stress:	1770.9	@Node	40
Bending Stress:	151093.9	@Node	4618
Torsion Stress:	21734.5	@Node	4980
Hoop Stress:	0.0	@Node	4020
Max Stress Intensity:	279571.2	@Node	4618

CODE STRESS CHECK PASSED : LOADCASE 12 (EXP) L12=L6-L8

Highest Stresses: (KPa)	LOADCASE 12 (EXP) L12=L6-L8	
Ratio (%):	18.2	@Node 530
Code Stress:	37559.8	Allowable Stress: 206842.7
Axial Stress:	7966.9	@Node 40
Bending Stress:	35032.9	@Node 530
Torsion Stress:	2920.2	@Node 4980
Hoop Stress:	0.0	@Node 4020
Max Stress Intensity:	47630.2	@Node 530

CODE STRESS CHECK PASSED : LOADCASE 13 (EXP) L13=L10-L12

Highest Stresses: (KPa)	LOADCASE 13 (EXP) L13=L10-L12	
Ratio (%):	83.6	@Node 1480
Code Stress:	169638.5	Allowable Stress: 202871.3
Axial Stress:	9178.7	@Node 40
Bending Stress:	167777.5	@Node 1480
Torsion Stress:	47128.8	@Node 1610
Hoop Stress:	0.0	@Node 4020
Max Stress Intensity:	264014.3	@Node 4099

CODE STRESS CHECK PASSED : LOADCASE 14 (EXP) L14=L11-L12

Highest Stresses: (KPa)	LOADCASE 14 (EXP) L14=L11-L12	
Ratio (%):	68.9	@Node 4618
Code Stress:	160879.3	Allowable Stress: 233559.9
Axial Stress:	9737.8	@Node 40
Bending Stress:	160132.7	@Node 4618
Torsion Stress:	24654.7	@Node 4980
Hoop Stress:	0.0	@Node 4020
Max Stress Intensity:	296228.0	@Node 4618

		
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ANNEXURE – 8

DISPLACEMENT FOR SUSTAINED & DESIGN TEMPERATURE

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LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS	2
Displacements : 2 (OPE) W+D1+T1+P1.....	3
Displacements : 4 (OPE) W+T2+P2	11
Displacements : 6 (OPE) W+T3+P1	19
Displacements : 8 (SUS) W+P1.....	27
Displacements : 9 (SUS) W+P2.....	35

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LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS

- 1 (HYD) WW+HP
- 2 (OPE) W+D1+T1+P1
- 3 (Alt-SUS) W+P1
- 4 (OPE) W+T2+P2
- 5 (Alt-SUS) W+P2
- 6 (OPE) W+T3+P1
- 7 (Alt-SUS) W+P1
- 8 (SUS) W+P1
- 9 (SUS) W+P2
- 10 (EXP) L10=L2-L8
- 11 (EXP) L11=L4-L9
- 12 (EXP) L12=L6-L8
- 13 (EXP) L13=L10-L12
- 14 (EXP) L14=L11-L12

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
10	-3.134	5.933	-1.150	0.0000	-0.0882	-0.1913
20	-2.630	5.520	-0.959	0.0000	-0.0882	-0.1912
30	-0.000	3.538	-0.000	0.0039	-0.0778	-0.1623
40	1.553	2.525	0.468	0.0062	-0.0638	-0.1429
50	2.779	1.818	0.775	0.0080	-0.0537	-0.1275
60	6.865	-0.000	1.458	0.0140	-0.0263	-0.0841
70	12.473	-1.247	1.753	0.0223	0.0000	-0.0120
80	12.977	-1.273	1.753	0.0223	0.0000	-0.0119
90	13.480	-1.299	1.753	0.0223	0.0001	-0.0118
100	23.212	-0.000	1.229	0.0367	0.0203	0.0269
110	39.564	-0.000	0.000	0.0609	0.0070	-0.0090
120	53.371	-0.171	0.160	0.0813	-0.0066	0.0174
130	53.875	-0.133	0.174	0.0813	-0.0066	0.0174
140	54.379	-0.095	0.188	0.0813	-0.0066	0.0174
150	56.014	-0.002	0.231	0.0837	-0.0056	0.0065
160	57.143	-0.000	0.256	0.0854	-0.0046	-0.0075
170	64.468	-0.552	0.235	0.0962	0.0077	0.0129
180	64.897	-0.521	0.219	0.0955	0.0091	0.0206
190	65.401	-0.477	0.199	0.0955	0.0092	0.0207
200	67.261	-0.310	0.126	0.0955	0.0092	0.0210
210	67.765	-0.264	0.106	0.0955	0.0092	0.0211
220	70.787	-0.000	0.000	0.0907	0.0056	0.0048
230	73.448	-0.110	-0.065	0.0864	0.0107	-0.0110
240	73.470	0.380	0.120	0.0924	0.0128	-0.0098
250	73.474	0.482	0.161	0.0946	0.0136	-0.0094
260	73.492	0.934	0.344	0.0947	0.0136	-0.0093
270	73.554	2.484	0.974	0.0948	0.0137	-0.0093
280	73.572	2.936	1.158	0.0948	0.0137	-0.0093
288	73.595	3.652	1.449	0.0944	0.0194	-0.0056
289	73.589	4.038	1.444	0.0918	0.0226	-0.0026
290	73.555	4.302	1.166	0.0887	0.0261	0.0002
298	73.415	4.711	0.054	0.0853	0.0331	0.0077
299	73.228	4.825	-0.256	0.0848	0.0365	0.0102
300	72.875	4.858	-0.339	0.0847	0.0389	0.0126
310	70.680	4.718	-0.000	0.0855	0.0264	0.0154
318	64.801	4.488	-0.005	0.0875	-0.0111	0.0042
319	64.464	4.431	0.117	0.0877	-0.0100	0.0047
320	64.313	4.304	0.442	0.0883	-0.0086	0.0054
328	64.279	3.881	1.554	0.0913	-0.0058	0.0074
329	64.277	3.614	1.831	0.0940	-0.0044	0.0082
330	64.286	3.227	1.833	0.0968	-0.0032	0.0090
340	77.045	-0.124	-0.338	0.0285	0.0250	0.0113
350	85.522	-0.000	-0.919	0.0179	0.0069	-0.0599
360	92.316	-2.441	-0.891	0.0094	-0.0093	-0.0333
370	92.820	-2.513	-0.871	0.0094	-0.0093	-0.0331
380	94.681	-2.774	-0.796	0.0093	-0.0094	-0.0323

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Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
390	95.184	-2.844	-0.776	0.0093	-0.0094	-0.0321
400	97.613	-2.640	-0.643	0.0063	-0.0164	0.0685
410	98.117	-2.492	-0.607	0.0063	-0.0164	0.0687
420	98.621	-2.343	-0.572	0.0063	-0.0164	0.0689
430	99.051	-2.203	-0.540	0.0057	-0.0177	0.0834
440	99.481	-2.037	-0.506	0.0052	-0.0197	0.0954
450	99.984	-1.831	-0.463	0.0052	-0.0198	0.0956
460	101.845	-1.067	-0.305	0.0052	-0.0199	0.0959
470	102.349	-0.859	-0.262	0.0052	-0.0199	0.0959
480	104.806	-0.000	0.000	0.0020	-0.0301	0.0505
490	108.658	0.076	0.429	-0.0029	-0.0088	-0.0144
500	109.162	0.045	0.447	-0.0029	-0.0087	-0.0144
510	111.022	-0.070	0.515	-0.0029	-0.0083	-0.0142
520	111.526	-0.100	0.533	-0.0029	-0.0081	-0.0141
530	111.956	-0.120	0.537	-0.0034	0.0050	-0.0072
538	111.825	-0.138	3.197	0.0050	-0.0229	-0.0135
539	111.596	-0.137	3.621	0.0052	-0.0242	-0.0149
540	111.133	-0.111	3.760	0.0050	-0.0234	-0.0171
550	111.053	-0.106	3.752	0.0050	-0.0232	-0.0174
560	110.549	-0.068	3.702	0.0050	-0.0232	-0.0174
570	108.688	0.071	3.516	0.0050	-0.0232	-0.0174
580	108.184	0.109	3.466	0.0050	-0.0232	-0.0174
590	104.592	-0.000	3.149	0.0046	-0.0185	0.0503
600	102.318	-0.778	2.984	0.0043	-0.0159	0.1004
610	101.814	-0.995	2.949	0.0043	-0.0159	0.1004
620	99.954	-1.797	2.822	0.0043	-0.0159	0.1005
630	99.450	-2.015	2.788	0.0043	-0.0159	0.1005
639	99.019	-2.197	2.575	0.0045	-0.0157	0.0975
640	98.865	-2.264	2.120	0.0050	-0.0157	0.0950
650	112.387	-0.126	0.508	-0.0078	0.0283	0.0003
660	112.891	-0.125	0.447	-0.0079	0.0285	0.0004
670	113.395	-0.124	0.385	-0.0079	0.0289	0.0005
680	113.825	-0.117	0.309	-0.0123	0.0552	0.0060
690	115.877	-0.012	0.024	-0.0130	0.0137	0.0078
700	111.176	-0.321	-3.001	-0.0049	0.3102	0.0060
710	110.505	-0.332	-3.504	-0.0048	0.3106	0.0060
720	108.022	-0.369	-5.365	-0.0045	0.3117	0.0060
730	107.348	-0.378	-5.869	-0.0044	0.3120	0.0060
740	96.859	-0.000	-11.625	0.0102	0.5141	0.0060
750	91.254	0.016	-14.091	-0.0008	0.5533	0.0060
760	90.058	0.014	-14.595	-0.0007	0.5534	0.0060
770	85.641	0.009	-16.456	-0.0006	0.5536	0.0060
780	84.444	0.007	-16.960	-0.0006	0.5536	0.0060
790	78.593	-0.000	-19.427	-0.0062	0.5610	0.0060
800	72.718	-0.137	-21.894	-0.0166	0.5610	0.0060
810	71.506	-0.173	-22.398	-0.0166	0.5610	0.0060
820	71.016	-0.187	-22.601	-0.0166	0.5610	0.0060
830	116.718	-0.000	0.000	-0.0133	0.0014	-0.0052

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DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
840	119.181	-0.220	0.112	-0.0142	-0.0177	-0.0246
850	119.685	-0.273	0.150	-0.0142	-0.0177	-0.0245
860	121.546	-0.468	0.292	-0.0142	-0.0176	-0.0242
870	122.050	-0.520	0.330	-0.0142	-0.0176	-0.0241
880	123.255	-0.585	0.388	-0.0146	-0.0032	-0.0008
890	124.460	-0.528	0.357	-0.0150	0.0173	0.0232
900	124.964	-0.478	0.319	-0.0150	0.0174	0.0233
910	125.139	-0.460	0.306	-0.0150	0.0174	0.0233
920	126.999	-0.273	0.166	-0.0150	0.0179	0.0236
930	126.999	-0.221	0.127	-0.0150	0.0181	0.0237
940	126.997	-0.000	-0.350	-0.0157	0.0939	0.0178
950	126.994	0.060	-2.123	-0.0166	0.2200	-0.0022
960	122.947	-0.136	-2.123	-0.0059	0.3599	-0.0022
970	122.388	-0.146	-2.123	-0.0059	0.3601	-0.0022
980	120.471	-0.177	-2.123	-0.0058	0.3606	-0.0022
990	119.911	-0.185	-2.123	-0.0057	0.3607	-0.0022
1000	109.266	-0.000	-2.124	0.0071	0.4899	-0.0022
1010	103.996	0.024	-2.124	0.0002	0.5145	-0.0022
1020	103.197	0.025	-2.124	0.0002	0.5145	-0.0022
1030	100.460	0.025	-2.124	0.0002	0.5146	-0.0022
1040	99.662	0.026	-2.124	0.0002	0.5146	-0.0022
1050	94.233	-0.000	-2.124	-0.0080	0.5199	-0.0022
1060	88.789	-0.142	-2.124	-0.0163	0.5199	-0.0022
1070	87.982	-0.167	-2.124	-0.0163	0.5199	-0.0022
1080	87.528	-0.182	-2.124	-0.0163	0.5199	-0.0022
1090	126.993	-0.000	-4.468	-0.0119	0.2330	-0.0098
1100	126.992	-0.098	-6.387	-0.0084	0.2588	-0.0137
1110	126.992	-0.120	-6.788	-0.0084	0.2589	-0.0137
1118	76.468	-0.087	-3.762	-0.0148	0.0496	-0.0023
1119	76.260	-0.107	-4.063	-0.0159	0.0503	-0.0040
1120	75.899	-0.110	-4.129	-0.0162	0.0505	-0.0066
1130	75.894	-0.109	-4.128	-0.0162	0.0505	-0.0066
1140	75.442	-0.097	-4.031	-0.0162	0.0505	-0.0066
1150	73.823	-0.050	-3.680	-0.0162	0.0505	-0.0067
1160	73.370	-0.037	-3.582	-0.0162	0.0505	-0.0068
1170	70.132	-0.000	-2.899	-0.0185	0.0482	0.0153
1180	63.051	-1.206	-1.555	-0.0234	0.0408	0.0285
1190	57.687	-1.002	-0.692	-0.0272	0.0347	-0.0511
1200	57.235	-0.903	-0.625	-0.0272	0.0347	-0.0511
1210	55.684	-0.563	-0.394	-0.0272	0.0347	-0.0512
1220	55.232	-0.463	-0.326	-0.0272	0.0347	-0.0513
1230	52.933	-0.000	-0.000	-0.0288	0.0319	-0.0330
1240	50.717	0.144	0.289	-0.0303	0.0292	-0.0026
1250	50.265	0.149	0.345	-0.0303	0.0292	-0.0026
1260	49.812	0.154	0.402	-0.0303	0.0292	-0.0026
1270	38.833	-0.000	1.449	-0.0380	0.0155	0.0199
1280	22.448	-0.000	1.495	-0.0495	-0.0187	-0.0267
1290	12.811	-0.000	-0.000	-0.0562	-0.0577	0.0355

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1300	10.165	-0.529	-0.722	-0.0580	-0.0696	0.0516
1310	9.713	-0.629	-0.857	-0.0580	-0.0696	0.0516
1320	9.261	-0.729	-0.992	-0.0580	-0.0696	0.0516
1330	8.897	-0.807	-1.102	-0.0589	-0.0736	0.0467
1340	7.403	-0.895	-1.695	-0.0682	-0.1099	-0.0256
1350	7.028	-0.853	-1.872	-0.0682	-0.1099	-0.0257
1360	5.962	-0.736	-2.374	-0.0683	-0.1099	-0.0258
1370	5.588	-0.694	-2.551	-0.0683	-0.1099	-0.0258
1380	2.724	-0.000	-4.035	-0.0861	-0.1255	-0.0464
1390	1.096	0.193	-4.842	-0.0963	-0.0959	-0.0204
1400	0.732	0.225	-4.986	-0.0971	-0.0907	-0.0212
1410	0.280	0.266	-5.162	-0.0971	-0.0907	-0.0212
1420	-0.172	0.307	-5.338	-0.0971	-0.0907	-0.0212
1430	-2.220	0.510	-6.056	-0.0986	-0.0718	-0.0249
1460	-12.133	-0.000	-5.078	-0.1055	0.1561	0.1129
1470	-14.491	-1.509	-3.069	-0.1072	0.2482	0.1796
1480	-14.910	-1.939	-2.516	-0.1065	0.2918	0.1964
1490	-15.359	-2.317	-1.955	-0.1065	0.2920	0.1964
1500	-16.168	-2.999	-0.940	-0.1065	0.2924	0.1965
1510	-15.152	-2.630	-0.130	-0.1065	0.2928	0.1966
1520	-14.584	-2.423	0.322	-0.1065	0.2929	0.1966
1530	-5.000	0.000	5.000	-0.1219	0.5136	0.4155
1540	-5.453	2195.070	-3.524	-0.0661	-0.1117	0.1267
1550	-5.147	2195.222	-3.379	-0.0661	-0.1097	0.1096
1558	-4.149	2193.874	-2.986	-0.0682	-0.0988	0.2524
1559	-3.806	2193.761	-2.884	-0.0677	-0.0970	0.2864
1560	-3.470	2193.958	-2.768	-0.0668	-0.0952	0.3187
1570	-2.761	2195.041	-2.488	-0.0635	-0.0883	0.3836
1580	-1.945	2196.438	-2.198	-0.0598	-0.0772	0.4018
1590	-1.537	2197.137	-2.067	-0.0588	-0.0738	0.3960
1595	-0.550	2198.765	-1.766	-0.0576	-0.0686	0.3709
1600	-0.141	2199.392	-1.650	-0.0566	-0.0633	0.3341
1604	0.055	2199.710	-1.501	-0.0591	-0.0567	0.2845
1605	0.096	2199.872	-1.260	-0.0582	-0.0481	0.2217
1610	0.000	2200.000	-0.452	-0.0001	-0.0001	0.0003
1611	0.000	2200.000	-0.000	-0.0000	-0.0000	0.0000
1620	0.000	2200.000	-0.000	-0.0000	-0.0000	0.0000
1630	-5.225	2195.426	-3.436	-0.0652	-0.1079	0.0727
1640	-5.274	2195.736	-3.522	-0.0634	-0.1036	-0.0080
1650	-5.260	2196.111	-3.623	-0.0634	-0.1036	-0.0082
1651	64.455	-0.319	0.331	0.0962	0.0077	0.0129
1660	-5.215	2197.351	-3.961	-0.0634	-0.1035	-0.0088
1670	-5.201	2197.726	-4.063	-0.0634	-0.1035	-0.0089
1678	-4.203	2199.921	-4.810	-0.1027	-0.0585	-0.1159
1679	-4.057	2200.054	-4.868	-0.1077	-0.0553	-0.1022
1680	-3.863	2200.056	-4.862	-0.1139	-0.0525	-0.0867
1690	-2.479	2199.842	-4.639	-0.1653	-0.0201	0.0045
1700	-2.104	2199.849	-4.606	-0.1653	-0.0200	0.0045

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1710	-0.863	2199.873	-4.500	-0.1654	-0.0200	0.0045
1720	-0.489	2199.880	-4.468	-0.1655	-0.0199	0.0045
1728	-0.081	2199.868	-4.449	-0.1806	-0.0008	-0.0223
1729	0.075	2199.780	-4.398	-0.1879	0.0093	-0.0385
1730	0.111	2199.603	-4.266	-0.1941	0.0212	-0.0587
1740	0.012	2199.303	-4.013	-0.2018	0.0408	-0.0996
1750	-0.144	2198.994	-3.741	-0.2067	0.0544	-0.1342
1760	-0.446	2198.517	-3.319	-0.2089	0.0613	-0.1549
1770	-0.000	2199.999	-0.504	-0.0006	0.0002	-0.0003
1780	-0.000	2200.000	0.000	-0.0000	0.0000	-0.0000
1781	-0.000	2200.000	0.000	-0.0000	0.0000	-0.0000
1790	-0.558	2198.135	-3.730	-0.2237	0.0661	-0.1835
1800	-0.701	2197.651	-4.234	-0.2238	0.0661	-0.1839
1808	-0.778	0.857	7.139	-0.0430	0.4104	0.5156
1809	-0.306	1.094	7.203	-0.0241	0.3988	0.5279
1810	0.067	1.603	6.925	-0.0006	0.3875	0.5375
1820	0.070	1.610	6.919	-0.0003	0.3874	0.5376
1830	0.523	2.653	6.168	-0.0002	0.3874	0.5376
1840	-5.750	2194.908	-3.666	-0.0661	-0.1117	0.1267
1850	-0.844	2197.167	-4.738	-0.2239	0.0662	-0.1842
1859	-0.791	2196.570	-5.236	-0.2350	0.0724	-0.2209
1860	-0.406	2195.925	-5.561	-0.2401	0.0790	-0.2612
1870	1.545	2193.499	-6.297	-0.2401	0.1022	-0.3088
1880	2.048	2192.832	-6.518	-0.2401	0.1023	-0.3088
1890	2.552	2192.164	-6.739	-0.2401	0.1024	-0.3087
1900	3.045	2191.512	-6.955	-0.2401	0.1025	-0.3087
1910	3.045	2191.512	-6.955	-57.2489	0.1025	-0.3087
1920	3.537	2190.860	-7.172	-57.2493	0.1026	-0.3086
1929	4.076	2158.988	-39.176	-57.2861	0.1317	-0.2847
1930	4.425	2083.130	-116.082	-57.3326	0.1621	-0.2578
1938	2.400	713.575	-1506.478	-57.5306	0.2800	-0.0655
1939	1.920	637.684	-1583.346	-57.5329	0.2861	-0.0544
1940	1.347	606.321	-1614.736	-57.5335	0.2894	-0.0468
1950	1.341	606.322	-1614.728	-57.5335	0.2894	-0.0468
1960	0.848	606.421	-1614.117	-57.5335	0.2894	-0.0468
1970	0.848	606.421	-1614.117	-41.6740	0.0000	-0.0000
1980	0.356	606.421	-1614.117	-41.6744	0.0000	-0.0001
1989	-0.100	595.129	-1583.657	-41.7153	0.0073	-0.0249
1990	-0.359	567.817	-1510.050	-41.7674	0.0151	-0.0505
1998	-4.841	46.584	-106.696	-42.0127	-0.0693	-0.1929
1999	-4.953	18.920	-32.559	-42.0160	-0.0780	-0.1931
2000	-4.630	7.160	-1.715	-42.0174	-0.0880	-0.1914
2010	-4.623	7.155	-1.713	-42.0174	-0.0881	-0.1913
2020	-4.131	6.751	-1.527	-42.0174	-0.0881	-0.1913
2030	-4.131	6.751	-1.527	0.0000	-0.0881	-0.1913
2040	-3.638	6.347	-1.340	0.0000	-0.0882	-0.1913
2050	-5.750	2194.908	-3.666	-0.0661	-0.1117	0.1267
2060	-6.202	2194.662	-3.883	-0.0661	-0.1117	0.1268

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Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
2070	-6.554	2194.471	-4.052	-0.0661	-0.1117	0.1268
2080	-6.554	2194.471	-4.052	-62.7865	-0.1117	0.1268
2090	-6.906	2194.279	-4.221	-62.7867	-0.1117	0.1268
2099	-7.165	2189.521	-28.353	-62.8444	-0.1066	0.1189
2100	-7.342	2178.264	-86.446	-62.8926	-0.1006	0.1081
2109	-4.067	1726.302	-2425.793	-63.6951	-0.0203	-0.2514
2110	-3.748	1721.354	-2450.136	-63.6950	-0.0192	-0.2580
2120	-3.744	1721.351	-2450.136	-63.6950	-0.0192	-0.2580
2130	-3.392	1720.961	-2450.107	-63.6950	-0.0192	-0.2580
2140	-3.392	1720.961	-2450.107	-71.7543	0.0000	0.0000
2150	-3.040	1720.961	-2450.107	-71.7546	0.0000	0.0000
2159	-2.820	1704.943	-2427.188	-71.8134	0.0107	0.0120
2160	-2.723	1666.247	-2371.818	-71.8627	0.0228	0.0273
2169	1.816	22.047	-19.325	-72.6795	0.3794	0.5283
2170	1.682	5.327	4.242	-72.6786	0.3872	0.5375
2180	1.679	5.320	4.247	-72.6786	0.3873	0.5376
2190	1.327	4.508	4.832	-72.6786	0.3873	0.5376
2200	1.327	4.508	4.832	-0.0000	0.3873	0.5376
2201	73.459	0.122	0.021	0.0864	0.0108	-0.0110
2210	0.975	3.695	5.417	-0.0001	0.3873	0.5376
2220	0.523	2.653	6.168	-0.0002	0.3874	0.5376
2721	-5.269	2195.164	-3.437	-0.0661	-0.1097	0.1096
3101	77.020	-0.095	-0.571	0.0285	0.0251	0.0113
3141	-0.292	2198.751	-3.527	-0.2089	0.0613	-0.1549
4001	99.033	-2.208	-0.308	0.0057	-0.0177	0.0834
4020	126.992	-0.141	-7.190	-0.0084	0.2592	-0.0137
4030	126.988	-0.358	-17.734	0.0106	0.5893	-0.0023
4038	126.987	-0.364	-19.464	0.0127	0.6396	-0.0018
4039	126.387	-0.354	-20.842	0.0150	0.8386	-0.0000
4040	124.611	-0.322	-21.548	0.0189	1.0358	0.0010
4060	101.321	-0.000	-21.546	0.0055	1.2978	-0.0010
4080	26.233	-0.000	-21.540	0.0149	1.1952	-0.0065
4098	3.691	0.397	-21.538	0.0213	0.8465	-0.0087
4099	2.288	0.431	-22.088	0.0216	0.6321	-0.0090
4100	1.852	0.429	-23.062	0.0210	0.4173	-0.0107
4120	1.850	-0.000	-27.960	0.0197	0.0624	-0.0382
4140	1.844	-2.008	-17.959	0.0165	-0.3550	0.0061
4160	1.839	-0.000	-0.000	0.0133	-0.2416	0.0137
4180	1.840	-0.000	1.833	0.0075	0.0781	0.0122
4198	1.841	-0.124	0.161	0.0060	0.0638	-0.0131
4199	1.794	-0.145	0.047	0.0058	0.0570	-0.0137
4200	1.694	-0.145	0.005	0.0060	0.0499	-0.0136
4220	0.857	-0.000	0.004	0.0081	0.0346	-0.0132
4240	-0.000	-0.000	0.002	-0.0286	0.0090	-0.0122
4260	-0.000	-0.000	0.000	0.0076	-0.0024	-0.0100
4280	0.000	-0.000	0.000	-0.0020	0.0006	-0.0077
4300	-0.000	-0.000	0.000	0.0005	-0.0002	-0.0054
4320	0.000	-0.000	0.000	-0.0001	0.0000	-0.0031

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements

CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
4340	-0.000	-0.000	0.000	0.0000	-0.0000	-0.0008
4360	0.000	-0.000	-0.000	-0.0000	0.0000	0.0015
4380	-0.000	-0.000	-0.000	0.0000	-0.0000	0.0038
4400	0.000	-0.000	-0.000	-0.0000	-0.0000	0.0060
4420	-0.000	-0.000	-0.000	0.0000	0.0000	0.0083
4440	0.000	-0.000	-0.000	-0.0001	-0.0000	0.0106
4460	-0.000	-0.000	-0.000	0.0002	0.0000	0.0129
4480	0.000	-0.000	-0.001	0.0287	-0.0001	0.0150
4498	0.002	0.315	-0.002	0.0124	-0.0001	0.0154
4499	0.002	0.326	-0.002	0.0132	-0.0000	0.0151
4500	0.002	0.309	-0.003	0.0140	0.0000	0.0138
4520	0.001	0.142	-0.002	0.0177	0.0001	0.0078
4540	0.000	-0.000	-0.000	0.0244	0.0001	0.0109
4558	-0.002	-0.626	0.002	0.0340	-0.0000	0.0112
4559	-0.002	-0.619	0.002	0.0349	-0.0001	0.0099
4560	-0.002	-0.559	0.002	0.0358	-0.0001	0.0095
4580	-0.000	-0.000	0.001	0.0228	-0.0001	0.0096
4600	-0.000	-0.000	-0.001	-0.0228	0.0001	0.0096
4618	-0.002	-0.560	-0.002	-0.0359	0.0001	0.0096
4619	-0.002	-0.620	-0.002	-0.0351	0.0001	0.0099
4620	-0.002	-0.627	-0.002	-0.0342	0.0000	0.0113
4640	0.000	-0.000	0.000	-0.0247	-0.0001	0.0109
4660	0.001	0.143	0.002	-0.0182	-0.0001	0.0081
4678	0.002	0.317	0.003	-0.0147	-0.0000	0.0145
4679	0.002	0.335	0.002	-0.0137	0.0000	0.0161
4680	0.002	0.323	0.002	-0.0129	0.0001	0.0165
4700	0.000	-0.000	0.001	-0.0291	0.0001	0.0170
4720	-0.000	-0.000	0.000	-0.0001	-0.0000	0.0195
4740	0.000	-0.000	0.000	0.0000	0.0000	0.0222
4760	-0.000	-0.000	0.000	0.0000	-0.0000	0.0250
4780	0.000	-0.000	0.000	-0.0001	0.0000	0.0277
4800	-0.000	-0.000	0.000	0.0298	-0.0000	0.0302
4820	0.000	-0.000	0.000	-0.0390	-0.0000	0.0318
4840	-0.000	-0.000	0.000	0.0279	0.0000	0.0346
4860	0.000	-0.000	-0.000	-0.0021	-0.0000	0.0366
4880	-0.000	-0.000	-0.000	-0.0193	0.0000	0.0387
4900	0.000	-0.000	-0.000	0.0046	-0.0001	0.0414
4920	-0.000	-0.000	-0.000	0.0010	0.0003	0.0441
4940	0.000	-0.000	-0.000	-0.0085	-0.0011	0.0469
4950	-0.000	-0.000	-0.001	0.0332	0.0042	0.0496
4960	0.001	-0.000	-0.002	-0.0221	-0.0126	0.0512
4978	0.792	-1.202	-0.003	-0.0288	-0.0240	0.0522
4979	0.796	-1.253	-0.024	-0.0263	-0.0229	0.0526
4980	0.714	-1.273	-0.072	-0.0244	-0.0222	0.0535
4998	0.151	-1.272	-0.319	-0.0235	-0.0202	0.0554
4999	0.044	-1.227	-0.348	-0.0237	-0.0196	0.0585
5000	-0.002	-1.115	-0.331	-0.0241	-0.0180	0.0603
5001	111.961	-0.117	0.770	-0.0034	0.0049	-0.0072

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 2 (OPE) W+D1+T1+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
5020	-0.001	-0.000	-0.000	-0.0241	-0.0114	0.0292
5040	0.000	-0.000	-0.000	-0.0241	0.0091	0.0013
5058	0.001	0.011	-0.258	-0.0242	0.0114	0.0011
5059	0.002	0.013	-0.260	-0.0243	0.0110	0.0003
5060	0.001	0.012	-0.223	-0.0246	0.0107	-0.0007
5078	-0.006	0.012	0.061	-0.0256	0.0099	0.0000
5079	-0.004	0.014	0.103	-0.0271	0.0093	0.0023
5080	-0.002	0.020	0.108	-0.0280	0.0082	0.0033
5100	-0.001	-0.000	0.000	-0.0313	0.0059	-0.0122
5120	-0.000	-0.000	-0.000	-0.0505	-0.0016	0.0055
5140	-0.000	-0.000	0.000	-0.0698	0.0004	-0.0099
5150	-0.000	-1.010	-0.005	-0.0805	-0.0001	0.0114
5160	-0.000	-0.952	-0.004	-0.0814	-0.0001	0.0196
5180	-0.000	-0.921	-0.004	-0.0814	-0.0001	0.0197
5200	-0.000	-0.890	-0.004	-0.0814	-0.0001	0.0198
5220	0.000	-0.000	-0.000	-0.0873	-0.0001	0.0291
5240	0.001	-0.000	-0.000	-0.1004	0.0001	-0.0409
5258	0.002	-1.056	-0.002	-0.1042	0.0001	-0.0787
5259	0.002	-1.139	-0.002	-0.1009	-0.0000	-0.0930
5260	0.002	-1.026	-0.001	-0.0991	-0.0001	-0.0988
5268	0.000	0.651	-0.001	-0.1039	-0.0001	-0.1315
5269	-0.000	0.740	-0.000	-0.1008	-0.0001	-0.1377
5270	-0.000	0.545	-0.000	-0.0967	-0.0001	-0.1527
5280	0.000	-0.000	0.000	-0.0967	-0.0001	-0.1590
5300	0.001	-2.433	0.001	-0.0967	-0.0001	-0.1750
5320	0.001	-3.817	0.002	-0.0967	-0.0001	-0.1767
5340	0.001	-4.091	0.002	-0.0967	-0.0001	-0.1767
9301	126.775	0.043	-2.123	-0.0166	0.2201	-0.0022

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
10	-2.408	-1.206	-1.659	0.0000	-0.1272	0.0983
20	-2.019	-0.993	-1.384	0.0000	-0.1272	0.0984
30	-0.000	-0.000	-0.000	0.0030	-0.1132	0.0639
40	1.190	0.288	0.688	0.0047	-0.0945	0.0258
50	2.130	0.358	1.146	0.0061	-0.0808	0.0030
60	5.262	-0.000	2.195	0.0107	-0.0411	-0.0346
70	9.559	-0.704	2.663	0.0170	-0.0005	-0.0115
80	9.948	-0.729	2.664	0.0170	-0.0005	-0.0115
90	10.337	-0.754	2.665	0.0170	-0.0005	-0.0114
100	17.794	-0.000	1.885	0.0280	0.0306	0.0138
110	30.326	-0.000	0.000	0.0464	0.0123	-0.0053
120	40.908	-0.149	0.115	0.0619	-0.0085	0.0160
130	41.297	-0.115	0.133	0.0620	-0.0085	0.0160
140	41.686	-0.080	0.152	0.0620	-0.0085	0.0160
150	42.939	0.003	0.209	0.0638	-0.0076	0.0053
160	43.804	-0.000	0.243	0.0651	-0.0066	-0.0087
170	49.418	-0.566	0.260	0.0733	0.0079	0.0133
180	49.747	-0.535	0.244	0.0727	0.0094	0.0210
190	50.136	-0.490	0.224	0.0727	0.0094	0.0211
200	51.572	-0.320	0.148	0.0726	0.0095	0.0214
210	51.961	-0.273	0.128	0.0726	0.0095	0.0215
220	54.277	-0.000	0.000	0.0681	0.0089	0.0058
230	56.317	-0.097	-0.118	0.0640	0.0164	-0.0100
240	56.337	0.280	0.019	0.0693	0.0180	-0.0088
250	56.341	0.359	0.050	0.0712	0.0187	-0.0084
260	56.357	0.708	0.188	0.0712	0.0187	-0.0084
270	56.413	1.904	0.663	0.0713	0.0188	-0.0083
280	56.429	2.253	0.801	0.0713	0.0188	-0.0083
288	56.449	2.804	1.020	0.0705	0.0234	-0.0047
289	56.439	3.098	1.012	0.0678	0.0260	-0.0017
290	56.400	3.297	0.797	0.0646	0.0289	0.0011
298	56.250	3.592	-0.058	0.0612	0.0347	0.0087
299	56.093	3.672	-0.291	0.0607	0.0375	0.0112
300	55.816	3.689	-0.342	0.0606	0.0394	0.0136
310	54.128	3.539	-0.000	0.0613	0.0271	0.0165
318	49.608	3.283	0.056	0.0632	-0.0081	0.0053
319	49.349	3.238	0.151	0.0634	-0.0071	0.0057
320	49.234	3.145	0.401	0.0640	-0.0059	0.0064
328	49.212	2.837	1.256	0.0670	-0.0034	0.0084
329	49.213	2.636	1.471	0.0697	-0.0023	0.0092
330	49.226	2.340	1.476	0.0724	-0.0012	0.0100
340	59.075	-0.103	-0.507	0.0128	0.0346	0.0109
350	65.575	-0.000	-1.201	0.0085	0.0054	-0.0605
360	70.785	-2.450	-1.080	0.0050	-0.0134	-0.0334
370	71.174	-2.522	-1.051	0.0050	-0.0134	-0.0332
380	72.610	-2.784	-0.944	0.0050	-0.0135	-0.0324

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
390	72.998	-2.854	-0.915	0.0050	-0.0135	-0.0322
400	74.861	-2.650	-0.744	0.0037	-0.0195	0.0687
410	75.250	-2.501	-0.702	0.0037	-0.0195	0.0689
420	75.638	-2.352	-0.660	0.0037	-0.0195	0.0691
430	75.968	-2.211	-0.623	0.0035	-0.0205	0.0836
440	76.298	-2.045	-0.584	0.0031	-0.0228	0.0958
450	76.687	-1.838	-0.534	0.0031	-0.0228	0.0959
460	78.123	-1.071	-0.352	0.0031	-0.0229	0.0962
470	78.511	-0.863	-0.303	0.0031	-0.0229	0.0962
480	80.395	-0.000	0.000	0.0005	-0.0347	0.0507
490	83.350	0.077	0.506	-0.0034	-0.0121	-0.0145
500	83.739	0.045	0.532	-0.0034	-0.0120	-0.0145
510	85.175	-0.070	0.627	-0.0034	-0.0116	-0.0143
520	85.564	-0.100	0.652	-0.0034	-0.0114	-0.0142
530	85.894	-0.120	0.661	-0.0039	0.0030	-0.0072
538	85.738	-0.134	2.703	0.0045	-0.0255	-0.0132
539	85.546	-0.132	3.022	0.0047	-0.0273	-0.0146
540	85.184	-0.106	3.111	0.0044	-0.0271	-0.0168
550	85.122	-0.101	3.102	0.0044	-0.0270	-0.0171
560	84.733	-0.064	3.044	0.0044	-0.0270	-0.0171
570	83.297	0.073	2.828	0.0044	-0.0270	-0.0171
580	82.908	0.110	2.770	0.0044	-0.0270	-0.0171
590	80.150	-0.000	2.390	0.0035	-0.0225	0.0499
600	78.405	-0.773	2.188	0.0029	-0.0194	0.0996
610	78.016	-0.988	2.147	0.0029	-0.0194	0.0997
620	76.580	-1.784	1.992	0.0029	-0.0194	0.0997
630	76.191	-1.999	1.950	0.0029	-0.0194	0.0997
639	75.865	-2.182	1.774	0.0031	-0.0184	0.0969
640	75.758	-2.251	1.420	0.0034	-0.0178	0.0945
650	86.224	-0.126	0.634	-0.0082	0.0277	0.0003
660	86.613	-0.125	0.574	-0.0082	0.0280	0.0004
670	87.002	-0.124	0.513	-0.0083	0.0283	0.0005
680	87.332	-0.117	0.436	-0.0126	0.0565	0.0060
690	88.906	-0.012	0.086	-0.0133	0.0280	0.0078
700	84.699	-0.323	-2.104	-0.0048	0.3066	0.0060
710	84.036	-0.333	-2.493	-0.0048	0.3070	0.0060
720	81.581	-0.370	-3.929	-0.0045	0.3081	0.0060
730	80.915	-0.380	-4.318	-0.0044	0.3084	0.0060
740	70.629	-0.000	-8.735	0.0103	0.5010	0.0060
750	65.182	0.016	-10.629	-0.0008	0.5366	0.0060
760	64.022	0.015	-11.018	-0.0008	0.5367	0.0060
770	59.739	0.009	-12.454	-0.0007	0.5368	0.0060
780	58.578	0.008	-12.843	-0.0006	0.5369	0.0060
790	52.907	-0.000	-14.736	-0.0062	0.5435	0.0060
800	47.215	-0.137	-16.630	-0.0166	0.5435	0.0060
810	46.040	-0.173	-17.019	-0.0166	0.5435	0.0060
820	45.566	-0.188	-17.176	-0.0166	0.5435	0.0060
830	89.551	-0.000	-0.000	-0.0136	0.0218	-0.0052

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
840	91.442	-0.220	-0.195	-0.0143	0.0203	-0.0246
850	91.831	-0.273	-0.239	-0.0143	0.0204	-0.0245
860	93.267	-0.468	-0.402	-0.0144	0.0205	-0.0242
870	93.656	-0.520	-0.447	-0.0144	0.0206	-0.0241
880	94.580	-0.585	-0.603	-0.0147	0.0421	-0.0008
890	95.505	-0.528	-0.883	-0.0151	0.0694	0.0232
900	95.894	-0.478	-1.033	-0.0151	0.0695	0.0233
910	96.029	-0.460	-1.086	-0.0151	0.0695	0.0233
920	97.465	-0.273	-1.643	-0.0151	0.0701	0.0236
930	97.854	-0.221	-1.794	-0.0151	0.0703	0.0237
940	99.445	-0.000	-2.772	-0.0158	0.1553	0.0178
950	101.525	0.060	-5.312	-0.0167	0.2912	-0.0022
960	96.519	-0.137	-7.852	-0.0059	0.4250	-0.0022
970	95.860	-0.146	-8.131	-0.0059	0.4251	-0.0022
980	93.597	-0.177	-9.088	-0.0058	0.4256	-0.0022
990	92.937	-0.186	-9.368	-0.0057	0.4257	-0.0022
1000	80.786	-0.000	-13.785	0.0071	0.5491	-0.0022
1010	74.903	0.024	-15.679	0.0002	0.5724	-0.0022
1020	74.015	0.025	-15.958	0.0002	0.5725	-0.0022
1030	70.969	0.025	-16.915	0.0002	0.5725	-0.0022
1040	70.081	0.026	-17.195	0.0002	0.5726	-0.0022
1050	64.047	-0.000	-19.088	-0.0080	0.5776	-0.0022
1060	57.999	-0.142	-20.982	-0.0163	0.5776	-0.0022
1070	57.102	-0.168	-21.261	-0.0163	0.5776	-0.0022
1080	56.598	-0.182	-21.418	-0.0163	0.5776	-0.0022
1090	103.418	-0.000	-8.464	-0.0120	0.3143	-0.0098
1100	104.837	-0.098	-11.038	-0.0084	0.3434	-0.0137
1110	105.116	-0.120	-11.570	-0.0084	0.3435	-0.0137
1118	58.281	-0.262	-3.140	-0.0265	0.0651	-0.0086
1119	58.083	-0.295	-3.357	-0.0274	0.0640	-0.0110
1120	57.792	-0.294	-3.373	-0.0274	0.0621	-0.0143
1130	57.788	-0.293	-3.372	-0.0274	0.0620	-0.0144
1140	57.439	-0.265	-3.252	-0.0274	0.0620	-0.0144
1150	56.189	-0.165	-2.821	-0.0274	0.0620	-0.0145
1160	55.840	-0.137	-2.701	-0.0274	0.0620	-0.0145
1170	53.349	-0.000	-1.932	-0.0282	0.0495	0.0093
1180	47.903	-1.134	-0.800	-0.0299	0.0271	0.0289
1190	43.778	-0.963	-0.317	-0.0312	0.0161	-0.0490
1200	43.429	-0.868	-0.286	-0.0312	0.0161	-0.0490
1210	42.232	-0.541	-0.178	-0.0312	0.0161	-0.0492
1220	41.883	-0.445	-0.147	-0.0312	0.0161	-0.0492
1230	40.115	-0.000	0.000	-0.0317	0.0141	-0.0315
1240	38.411	0.133	0.126	-0.0322	0.0126	-0.0019
1250	38.062	0.137	0.151	-0.0322	0.0126	-0.0019
1260	37.713	0.141	0.175	-0.0322	0.0126	-0.0019
1270	29.270	-0.000	0.612	-0.0349	0.0063	0.0193
1280	16.670	-0.000	0.628	-0.0388	-0.0077	-0.0257
1290	9.260	-0.000	-0.000	-0.0411	-0.0248	0.0326

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1300	7.226	-0.485	-0.313	-0.0417	-0.0306	0.0470
1310	6.877	-0.576	-0.372	-0.0417	-0.0306	0.0470
1320	6.528	-0.668	-0.431	-0.0417	-0.0306	0.0470
1330	6.248	-0.739	-0.480	-0.0420	-0.0330	0.0421
1340	5.096	-0.799	-0.770	-0.0452	-0.0572	-0.0285
1350	4.807	-0.753	-0.862	-0.0452	-0.0572	-0.0285
1360	3.984	-0.622	-1.124	-0.0452	-0.0572	-0.0286
1370	3.695	-0.576	-1.216	-0.0452	-0.0572	-0.0287
1380	1.488	-0.000	-2.140	-0.0512	-0.0920	-0.0213
1390	0.232	-0.024	-2.794	-0.0547	-0.0874	0.0118
1400	-0.047	-0.041	-2.927	-0.0550	-0.0849	0.0102
1410	-0.396	-0.061	-3.092	-0.0550	-0.0849	0.0102
1420	-0.745	-0.081	-3.257	-0.0550	-0.0848	0.0102
1430	-2.320	-0.128	-3.949	-0.0555	-0.0724	0.0002
1460	-9.942	-0.000	-4.057	-0.0579	0.0979	0.0386
1470	-11.755	-0.585	-2.734	-0.0584	0.1690	0.0709
1480	-12.095	-0.775	-2.358	-0.0576	0.2029	0.0751
1490	-12.441	-0.920	-1.967	-0.0576	0.2031	0.0751
1500	-13.066	-1.181	-1.262	-0.0576	0.2034	0.0751
1510	-12.359	-0.981	-0.637	-0.0576	0.2036	0.0751
1520	-11.964	-0.869	-0.288	-0.0576	0.2038	0.0750
1530	-5.000	-0.000	3.316	-0.0012	0.3731	0.0246
1540	-3.716	0.932	-0.463	-0.0369	0.0638	0.0340
1550	-3.480	0.980	-0.547	-0.0369	0.0642	0.0366
1558	-3.181	-0.058	-0.381	-0.0219	0.0634	0.0665
1559	-3.046	-0.200	-0.386	-0.0198	0.0630	0.0711
1560	-2.848	-0.200	-0.438	-0.0175	0.0625	0.0724
1570	-2.301	-0.000	-0.625	-0.0108	0.0604	0.0482
1580	-1.673	0.085	-0.826	-0.0031	0.0524	0.0047
1590	-1.358	0.083	-0.914	-0.0011	0.0488	-0.0033
1595	-0.599	0.054	-1.107	0.0013	0.0419	-0.0086
1600	-0.285	0.037	-1.175	0.0033	0.0338	-0.0094
1604	-0.105	0.026	-1.132	0.0054	0.0241	-0.0087
1605	-0.017	0.017	-0.971	0.0064	0.0153	-0.0068
1610	0.000	0.000	-0.349	0.0000	-0.0000	-0.0000
1611	0.000	0.000	0.000	0.0000	-0.0000	-0.0000
1620	0.000	0.000	0.000	0.0000	-0.0000	-0.0000
1630	-3.511	1.137	-0.580	-0.0396	0.0646	0.0336
1640	-3.551	1.376	-0.636	-0.0457	0.0654	0.0255
1650	-3.592	1.665	-0.710	-0.0457	0.0654	0.0255
1651	49.405	-0.387	0.333	0.0733	0.0079	0.0133
1660	-3.728	2.622	-0.953	-0.0458	0.0654	0.0254
1670	-3.768	2.912	-1.027	-0.0458	0.0654	0.0253
1678	-3.177	4.600	-1.575	-0.0659	0.0740	-0.1635
1679	-3.003	4.676	-1.644	-0.0662	0.0746	-0.1821
1680	-2.821	4.593	-1.716	-0.0664	0.0750	-0.1994
1690	-1.756	3.193	-2.155	-0.0680	0.0713	-0.2475
1700	-1.466	2.795	-2.270	-0.0680	0.0713	-0.2475

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1710	-0.509	1.478	-2.649	-0.0680	0.0712	-0.2473
1720	-0.219	1.080	-2.764	-0.0680	0.0712	-0.2473
1728	0.094	0.695	-2.881	-0.0685	0.0628	-0.1931
1729	0.170	0.514	-2.904	-0.0683	0.0585	-0.1669
1730	0.115	0.341	-2.872	-0.0683	0.0532	-0.1396
1740	-0.035	0.110	-2.784	-0.0692	0.0441	-0.0937
1750	-0.135	-0.129	-2.692	-0.0703	0.0378	-0.0609
1760	-0.233	-0.496	-2.549	-0.0710	0.0347	-0.0439
1770	-0.000	-0.000	-0.389	-0.0002	0.0001	-0.0001
1780	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1781	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1790	-0.297	-0.627	-2.865	-0.0766	0.0374	-0.0460
1800	-0.377	-0.792	-3.253	-0.0766	0.0374	-0.0460
1808	-1.935	-0.141	4.964	0.0130	0.3020	0.0015
1809	-1.584	-0.152	5.017	0.0076	0.2954	-0.0013
1810	-1.300	-0.156	4.813	0.0002	0.2889	-0.0029
1820	-1.298	-0.156	4.809	0.0001	0.2888	-0.0029
1830	-0.949	-0.162	4.249	0.0001	0.2887	-0.0029
1840	-3.945	0.889	-0.381	-0.0369	0.0638	0.0340
1850	-0.458	-0.958	-3.642	-0.0766	0.0374	-0.0461
1859	-0.390	-1.144	-4.012	-0.0792	0.0393	-0.0487
1860	-0.081	-1.300	-4.228	-0.0796	0.0417	-0.0499
1870	1.416	-1.687	-4.626	-0.0796	0.0591	-0.0383
1880	1.805	-1.770	-4.754	-0.0796	0.0591	-0.0383
1890	2.194	-1.853	-4.882	-0.0796	0.0592	-0.0382
1900	2.574	-1.933	-5.007	-0.0796	0.0593	-0.0381
1910	2.574	-1.933	-5.007	0.0000	0.0593	-0.0381
1920	2.954	-2.013	-5.132	0.0002	0.0595	-0.0380
1929	3.267	-1.958	-5.355	0.0233	0.0822	-0.0158
1930	3.287	-1.671	-5.612	0.0531	0.1051	0.0070
1938	-2.563	6.251	-6.346	0.2051	0.2519	0.0716
1939	-3.138	6.711	-6.110	0.2075	0.2575	0.0716
1940	-3.660	6.791	-5.604	0.2081	0.2614	0.0706
1950	-3.665	6.789	-5.597	0.2081	0.2614	0.0706
1960	-4.045	6.640	-5.045	0.2081	0.2614	0.0706
1970	-4.045	6.640	-5.045	-0.0000	0.0000	-0.0000
1980	-4.425	6.640	-5.045	-0.0002	0.0000	-0.0001
1989	-4.770	6.512	-5.082	-0.0212	0.0010	-0.0069
1990	-4.929	6.177	-5.134	-0.0478	0.0013	-0.0110
1998	-4.320	-1.451	-3.050	-0.1696	-0.1005	0.0546
1999	-3.996	-1.826	-2.780	-0.1713	-0.1138	0.0772
2000	-3.562	-1.836	-2.474	-0.1721	-0.1270	0.0980
2010	-3.557	-1.834	-2.471	-0.1721	-0.1270	0.0981
2020	-3.177	-1.626	-2.202	-0.1721	-0.1271	0.0982
2030	-3.177	-1.626	-2.202	0.0000	-0.1271	0.0982
2040	-2.797	-1.419	-1.934	0.0000	-0.1271	0.0983
2050	-3.945	0.889	-0.381	-0.0369	0.0638	0.0340
2060	-4.294	0.823	-0.258	-0.0369	0.0638	0.0340

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
2070	-4.566	0.772	-0.161	-0.0369	0.0638	0.0339
2080	-4.566	0.772	-0.161	0.0000	0.0638	0.0339
2090	-4.837	0.720	-0.065	0.0000	0.0638	0.0339
2099	-5.022	0.763	-0.017	0.0114	0.0597	0.0215
2100	-5.116	0.926	-0.012	0.0210	0.0553	0.0087
2109	0.569	8.452	3.670	0.1961	-0.0349	-0.2431
2110	0.835	8.306	3.765	0.1961	-0.0357	-0.2438
2120	0.837	8.303	3.765	0.1961	-0.0357	-0.2438
2130	1.109	7.935	3.819	0.1961	-0.0357	-0.2438
2140	1.109	7.935	3.819	-0.0000	0.0000	0.0000
2150	1.381	7.935	3.819	-0.0000	0.0000	0.0000
2159	1.551	7.878	3.780	-0.0099	0.0027	0.0051
2160	1.625	7.733	3.694	-0.0181	0.0064	0.0105
2169	0.178	-0.095	2.551	-0.1093	0.2789	0.0029
2170	-0.054	-0.176	2.813	-0.1088	0.2885	-0.0028
2180	-0.056	-0.176	2.817	-0.1088	0.2886	-0.0028
2190	-0.328	-0.172	3.253	-0.1088	0.2887	-0.0028
2200	-0.328	-0.172	3.253	0.0000	0.2887	-0.0028
2201	56.327	0.082	-0.054	0.0640	0.0164	-0.0100
2210	-0.600	-0.168	3.689	0.0000	0.2887	-0.0029
2220	-0.949	-0.162	4.249	0.0001	0.2887	-0.0029
2721	-3.575	0.961	-0.513	-0.0369	0.0642	0.0366
3101	59.040	-0.090	-0.687	0.0128	0.0347	0.0109
3141	-0.189	-0.316	-2.619	-0.0710	0.0347	-0.0439
4001	75.948	-2.215	-0.444	0.0035	-0.0205	0.0836
4020	105.396	-0.141	-12.104	-0.0084	0.3438	-0.0137
4030	110.152	-0.358	-24.553	0.0106	0.6383	-0.0023
4038	110.665	-0.364	-26.411	0.0127	0.6806	-0.0018
4039	110.391	-0.354	-27.980	0.0150	0.8470	-0.0000
4040	108.774	-0.322	-29.026	0.0189	1.0110	0.0010
4060	86.492	-0.000	-32.603	0.0054	1.2243	-0.0010
4080	16.864	-0.000	-42.755	0.0149	1.1021	-0.0065
4098	-3.998	0.397	-46.714	0.0213	0.7885	-0.0087
4099	-5.170	0.431	-47.571	0.0216	0.5962	-0.0090
4100	-5.241	0.429	-48.641	0.0210	0.4021	-0.0107
4120	-1.319	-0.000	-53.427	0.0197	0.0569	-0.0382
4140	8.197	-2.008	-39.413	0.0165	-0.5509	0.0061
4160	17.713	-0.000	0.000	0.0133	-0.9132	0.0137
4180	35.164	-0.000	90.242	0.0075	-0.7734	0.0122
4198	39.441	-0.124	105.902	0.0060	-0.5373	-0.0131
4199	40.131	-0.145	106.651	0.0058	-0.4008	-0.0137
4200	40.890	-0.145	106.585	0.0060	-0.2630	-0.0136
4220	43.754	-0.000	103.014	0.0081	-0.0380	-0.0132
4240	37.542	-0.000	95.082	-0.0286	0.2817	-0.0122
4260	0.000	-0.000	76.052	0.0076	0.2738	-0.0100
4280	-9.622	-0.000	57.030	-0.0020	-0.0260	-0.0077
4300	-3.861	-0.000	38.014	0.0005	-0.0590	-0.0054
4320	-0.000	-0.000	19.004	-0.0001	-0.0136	-0.0031

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
4340	0.000	-0.000	0.000	0.0000	0.0028	-0.0008
4360	0.242	-0.000	-18.982	-0.0000	-0.0045	0.0015
4380	-0.000	-0.000	-37.971	0.0000	0.0151	0.0038
4400	-3.224	-0.000	-56.966	-0.0000	0.0434	0.0060
4420	-7.024	-0.000	-75.967	0.0000	0.0124	0.0083
4440	0.000	-0.000	-94.974	-0.0001	-0.1856	0.0106
4460	20.324	-0.000	-113.987	0.0002	-0.0529	0.0129
4480	-15.647	-0.000	-131.422	0.0287	0.9646	0.0150
4498	-35.992	0.315	-134.744	0.0125	1.2666	0.0154
4499	-38.682	0.327	-134.003	0.0132	1.4522	0.0151
4500	-40.204	0.310	-131.231	0.0141	1.6321	0.0138
4520	-42.894	0.142	-105.534	0.0177	1.8228	0.0078
4540	-47.895	-0.000	-53.441	0.0244	1.9034	0.0109
4558	-55.105	-0.627	14.911	0.0341	1.4326	0.0113
4559	-56.451	-0.620	17.270	0.0350	1.2121	0.0100
4560	-58.649	-0.561	17.813	0.0359	0.9859	0.0096
4580	-72.644	-0.000	14.650	0.0229	0.6243	0.0097
4600	-79.185	-0.000	6.088	-0.0230	-0.3529	0.0098
4618	-69.767	-0.565	2.925	-0.0363	-0.7336	0.0099
4619	-68.030	-0.625	3.279	-0.0356	-0.9750	0.0102
4620	-66.865	-0.633	5.208	-0.0348	-1.2111	0.0115
4640	-59.656	-0.000	65.722	-0.0263	-1.7247	0.0108
4660	-54.654	0.149	113.119	-0.0204	-1.6580	0.0090
4678	-51.965	0.348	136.404	-0.0172	-1.4700	0.0170
4679	-50.569	0.370	138.870	-0.0160	-1.2906	0.0199
4680	-48.182	0.358	139.486	-0.0149	-1.1059	0.0209
4700	-30.722	-0.000	136.164	-0.0307	-0.8116	0.0248
4720	0.000	-0.000	118.730	0.0003	-0.0547	0.0453
4740	2.219	-0.000	99.717	-0.0001	0.0021	0.0677
4760	1.133	-0.000	80.710	0.0000	0.0138	0.0900
4780	0.000	-0.000	61.709	-0.0001	0.0064	0.1124
4800	-0.227	-0.000	44.298	0.0298	-0.0003	0.1329
4820	-0.170	-0.000	33.220	-0.0390	-0.0013	0.1459
4840	0.000	-0.000	14.235	0.0279	-0.0022	0.1683
4860	-0.000	-0.000	-0.000	-0.0022	0.0051	0.1851
4880	-0.782	-0.000	-14.253	-0.0192	0.0117	0.2019
4900	0.000	-0.000	-33.263	0.0041	-0.0440	0.2242
4920	9.353	-0.000	-52.279	0.0026	-0.1261	0.2466
4940	20.468	-0.000	-71.300	-0.0147	-0.0376	0.2689
4950	-0.000	-0.000	-90.328	0.0564	0.5453	0.2913
4960	-41.924	-0.000	-101.272	-0.0920	0.7289	0.3042
4978	-65.363	-5.485	-108.171	-0.1535	0.4465	0.3123
4979	-66.326	-5.620	-108.626	-0.1396	0.3329	0.2811
4980	-67.056	-5.383	-109.017	-0.1271	0.2799	0.2465
4998	-69.551	-3.494	-110.313	-0.1266	0.1091	0.2283
4999	-69.809	-2.991	-110.601	-0.1042	0.0594	0.1934
5000	-69.608	-2.523	-110.682	-0.0779	-0.0520	0.1553
5001	85.897	-0.116	0.840	-0.0039	0.0029	-0.0073

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 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 4 (OPE) W+T2+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
5020	-65.650	-0.000	-106.954	-0.1169	-0.2750	0.0742
5040	-54.231	2.103	-77.627	-0.2292	-0.5904	0.0403
5058	-49.864	2.977	-63.159	-0.2722	-0.6036	0.0213
5059	-49.519	2.853	-61.814	-0.2757	-0.5999	-0.0044
5060	-49.411	2.501	-60.820	-0.2845	-0.5992	-0.0283
5078	-49.797	0.435	-57.537	-0.2905	-0.5852	-0.0319
5079	-49.702	0.076	-56.538	-0.2866	-0.5806	-0.0150
5080	-49.367	-0.075	-55.234	-0.2825	-0.5694	0.0028
5100	-46.678	-0.000	-46.998	-0.2762	-0.5399	-0.0039
5120	-30.821	-0.000	-11.441	-0.2388	-0.2625	0.0035
5140	-14.968	-0.000	0.312	-0.2014	-0.0367	-0.0102
5150	-6.093	-1.047	0.638	-0.1805	0.0149	0.0111
5160	-5.408	-0.989	0.578	-0.1788	0.0168	0.0197
5180	-5.129	-0.958	0.552	-0.1788	0.0168	0.0198
5200	-4.850	-0.927	0.526	-0.1788	0.0168	0.0198
5220	0.000	-0.000	0.000	-0.1674	0.0200	0.0321
5240	10.805	-0.000	-0.555	-0.1419	-0.0073	-0.0513
5258	13.908	-1.302	-0.309	-0.1346	-0.0224	-0.0959
5259	14.227	-1.397	-0.116	-0.1288	-0.0331	-0.1131
5260	14.296	-1.253	0.255	-0.1224	-0.0434	-0.1208
5268	13.494	0.765	3.255	-0.1230	-0.0512	-0.1565
5269	13.539	0.869	3.637	-0.1192	-0.0513	-0.1634
5270	13.841	0.638	3.875	-0.1149	-0.0507	-0.1793
5280	14.478	-0.000	4.052	-0.1149	-0.0506	-0.1857
5300	17.097	-2.818	4.781	-0.1149	-0.0506	-0.2017
5320	18.525	-4.412	5.179	-0.1149	-0.0506	-0.2035
5340	18.804	-4.728	5.257	-0.1149	-0.0506	-0.2035
9301	101.235	0.043	-5.492	-0.0167	0.2914	-0.0022

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
10	0.112	-1.424	-0.226	-0.0000	-0.0170	0.1206
20	0.089	-1.163	-0.190	-0.0000	-0.0170	0.1205
30	0.000	-0.000	-0.000	-0.0004	-0.0152	0.0730
40	-0.045	0.328	0.089	-0.0007	-0.0118	0.0291
50	-0.081	0.406	0.145	-0.0009	-0.0094	0.0030
60	-0.200	-0.000	0.247	-0.0016	-0.0027	-0.0379
70	-0.363	-0.741	0.236	-0.0025	0.0030	-0.0117
80	-0.385	-0.767	0.230	-0.0025	0.0030	-0.0116
90	-0.407	-0.792	0.223	-0.0025	0.0030	-0.0116
100	-0.691	-0.000	0.055	-0.0041	0.0032	0.0150
110	-1.169	-0.000	-0.000	-0.0069	0.0004	-0.0069
120	-1.576	-0.199	-0.090	-0.0092	0.0022	0.0182
130	-1.598	-0.160	-0.095	-0.0092	0.0022	0.0182
140	-1.621	-0.121	-0.099	-0.0092	0.0022	0.0182
150	-1.669	-0.017	-0.115	-0.0094	0.0023	0.0089
160	-1.702	-0.000	-0.126	-0.0096	0.0023	-0.0039
170	-1.920	-0.483	-0.132	-0.0108	-0.0042	0.0076
180	-1.933	-0.463	-0.124	-0.0111	-0.0044	0.0154
190	-1.955	-0.429	-0.114	-0.0111	-0.0045	0.0155
200	-2.037	-0.305	-0.079	-0.0111	-0.0045	0.0158
210	-2.060	-0.270	-0.069	-0.0111	-0.0045	0.0158
220	-2.150	-0.000	0.000	-0.0128	-0.0063	0.0126
230	-2.231	0.013	0.078	-0.0142	-0.0072	-0.0031
240	-2.224	-0.007	0.050	-0.0108	-0.0069	-0.0040
250	-2.222	-0.011	0.046	-0.0094	-0.0068	-0.0044
260	-2.213	-0.031	0.028	-0.0094	-0.0068	-0.0044
270	-2.184	-0.100	-0.034	-0.0093	-0.0068	-0.0045
280	-2.175	-0.120	-0.052	-0.0093	-0.0068	-0.0045
288	-2.159	-0.147	-0.069	-0.0021	-0.0060	-0.0060
289	-2.146	-0.158	-0.063	0.0031	-0.0056	-0.0071
290	-2.134	-0.155	-0.048	0.0076	-0.0050	-0.0079
298	-2.114	-0.103	-0.007	0.0125	-0.0034	-0.0104
299	-2.104	-0.079	0.004	0.0129	-0.0024	-0.0111
300	-2.091	-0.055	0.007	0.0125	-0.0015	-0.0112
310	-2.009	-0.000	-0.000	0.0098	-0.0009	0.0054
318	-1.792	-0.605	-0.165	0.0026	-0.0154	0.0226
319	-1.789	-0.636	-0.193	0.0019	-0.0176	0.0194
320	-1.811	-0.649	-0.217	0.0009	-0.0194	0.0163
328	-1.907	-0.648	-0.258	-0.0015	-0.0201	0.0074
329	-1.932	-0.640	-0.269	-0.0031	-0.0192	0.0045
330	-1.939	-0.626	-0.268	-0.0046	-0.0179	0.0017
340	-2.340	0.053	0.209	-0.0440	-0.0105	0.0089
350	-2.598	-0.000	0.338	-0.0257	-0.0013	-0.0645
360	-2.809	-2.507	0.367	-0.0110	0.0017	-0.0334
370	-2.831	-2.579	0.363	-0.0110	0.0017	-0.0332
380	-2.913	-2.841	0.349	-0.0109	0.0017	-0.0324

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
390	-2.936	-2.910	0.346	-0.0109	0.0017	-0.0321
400	-3.011	-2.700	0.296	-0.0057	0.0082	0.0698
410	-3.033	-2.549	0.278	-0.0057	0.0082	0.0700
420	-3.055	-2.397	0.260	-0.0057	0.0082	0.0702
430	-3.069	-2.254	0.244	-0.0047	0.0097	0.0849
440	-3.082	-2.086	0.226	-0.0048	0.0100	0.0976
450	-3.104	-1.874	0.205	-0.0048	0.0100	0.0977
460	-3.187	-1.093	0.125	-0.0048	0.0100	0.0980
470	-3.209	-0.881	0.103	-0.0048	0.0100	0.0980
480	-3.286	-0.000	-0.000	-0.0051	0.0096	0.0519
490	-3.409	0.081	-0.120	-0.0055	0.0033	-0.0147
500	-3.431	0.050	-0.127	-0.0055	0.0033	-0.0147
510	-3.513	-0.067	-0.153	-0.0055	0.0032	-0.0145
520	-3.536	-0.099	-0.160	-0.0055	0.0032	-0.0144
530	-3.549	-0.119	-0.165	-0.0056	0.0011	-0.0073
538	-3.450	-0.113	-0.257	0.0027	0.0127	-0.0123
539	-3.420	-0.109	-0.262	0.0027	0.0115	-0.0135
540	-3.397	-0.084	-0.249	0.0022	0.0091	-0.0155
550	-3.394	-0.079	-0.246	0.0021	0.0089	-0.0157
560	-3.371	-0.045	-0.227	0.0021	0.0089	-0.0158
570	-3.289	0.081	-0.156	0.0021	0.0088	-0.0158
580	-3.266	0.116	-0.137	0.0021	0.0088	-0.0158
590	-3.143	-0.000	-0.033	-0.0006	0.0058	0.0490
600	-3.067	-0.755	0.025	-0.0023	0.0068	0.0973
610	-3.044	-0.965	0.039	-0.0023	0.0068	0.0973
620	-2.962	-1.743	0.094	-0.0023	0.0068	0.0974
630	-2.939	-1.953	0.108	-0.0023	0.0068	0.0973
639	-2.931	-2.136	0.129	-0.0024	0.0089	0.0951
640	-2.944	-2.214	0.152	-0.0026	0.0107	0.0934
650	-3.563	-0.125	-0.160	-0.0096	-0.0058	0.0003
660	-3.586	-0.124	-0.148	-0.0097	-0.0059	0.0004
670	-3.608	-0.123	-0.135	-0.0097	-0.0060	0.0004
680	-3.622	-0.116	-0.117	-0.0138	-0.0131	0.0060
690	-3.688	-0.012	-0.018	-0.0143	-0.0073	0.0077
700	-3.103	-0.329	-0.006	-0.0048	-0.0568	0.0060
710	-2.980	-0.340	0.016	-0.0047	-0.0569	0.0060
720	-2.526	-0.376	0.099	-0.0044	-0.0570	0.0060
730	-2.402	-0.386	0.121	-0.0044	-0.0570	0.0060
740	-0.962	-0.000	0.314	0.0105	-0.0497	0.0060
750	-0.522	0.017	0.397	-0.0008	-0.0354	0.0060
760	-0.445	0.015	0.420	-0.0008	-0.0354	0.0060
770	-0.164	0.010	0.502	-0.0007	-0.0353	0.0060
780	-0.087	0.008	0.525	-0.0007	-0.0353	0.0060
790	0.262	-0.000	0.608	-0.0063	-0.0326	0.0060
800	0.604	-0.138	0.691	-0.0167	-0.0326	0.0060
810	0.674	-0.174	0.713	-0.0167	-0.0326	0.0060
820	0.703	-0.189	0.722	-0.0167	-0.0326	0.0060
830	-3.716	-0.000	-0.000	-0.0145	-0.0016	-0.0052

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
840	-3.797	-0.220	-0.076	-0.0151	0.0148	-0.0246
850	-3.820	-0.273	-0.108	-0.0151	0.0148	-0.0245
860	-3.902	-0.468	-0.226	-0.0151	0.0149	-0.0242
870	-3.925	-0.520	-0.259	-0.0151	0.0149	-0.0241
880	-3.965	-0.585	-0.341	-0.0154	0.0170	-0.0008
890	-4.004	-0.528	-0.431	-0.0156	0.0175	0.0232
900	-4.027	-0.478	-0.469	-0.0157	0.0175	0.0233
910	-4.035	-0.460	-0.482	-0.0157	0.0175	0.0233
920	-4.117	-0.273	-0.622	-0.0157	0.0175	0.0236
930	-4.139	-0.222	-0.659	-0.0157	0.0175	0.0237
940	-4.230	-0.000	-0.777	-0.0162	0.0083	0.0178
950	-4.350	0.060	-0.763	-0.0168	-0.0134	-0.0022
960	-3.892	-0.137	-0.616	-0.0059	-0.0518	-0.0022
970	-3.812	-0.147	-0.600	-0.0059	-0.0519	-0.0022
980	-3.535	-0.178	-0.544	-0.0058	-0.0520	-0.0022
990	-3.455	-0.187	-0.528	-0.0057	-0.0520	-0.0022
1000	-1.887	-0.000	-0.273	0.0072	-0.0675	-0.0022
1010	-1.199	0.024	-0.164	0.0002	-0.0641	-0.0022
1020	-1.100	0.025	-0.148	0.0002	-0.0641	-0.0022
1030	-0.759	0.026	-0.092	0.0002	-0.0641	-0.0022
1040	-0.659	0.026	-0.076	0.0002	-0.0641	-0.0022
1050	0.006	-0.000	0.033	-0.0080	-0.0633	-0.0022
1060	0.669	-0.142	0.143	-0.0163	-0.0633	-0.0022
1070	0.767	-0.168	0.159	-0.0163	-0.0633	-0.0022
1080	0.822	-0.182	0.168	-0.0163	-0.0633	-0.0022
1090	-4.459	-0.000	-0.677	-0.0121	-0.0038	-0.0098
1100	-4.541	-0.098	-0.667	-0.0086	0.0004	-0.0137
1110	-4.557	-0.120	-0.667	-0.0085	0.0004	-0.0137
1118	-2.075	-0.845	0.338	-0.0728	-0.0208	-0.0302
1119	-2.041	-0.929	0.339	-0.0731	-0.0192	-0.0349
1120	-2.018	-0.919	0.318	-0.0729	-0.0172	-0.0406
1130	-2.017	-0.918	0.318	-0.0729	-0.0172	-0.0406
1140	-1.997	-0.839	0.285	-0.0729	-0.0172	-0.0407
1150	-1.925	-0.556	0.165	-0.0729	-0.0172	-0.0408
1160	-1.905	-0.477	0.132	-0.0729	-0.0172	-0.0408
1170	-1.784	-0.000	-0.045	-0.0722	-0.0091	-0.0110
1180	-1.521	-0.888	-0.143	-0.0707	0.0013	0.0302
1190	-1.322	-0.830	-0.075	-0.0696	0.0039	-0.0420
1200	-1.302	-0.749	-0.067	-0.0696	0.0039	-0.0421
1210	-1.233	-0.468	-0.041	-0.0696	0.0039	-0.0422
1220	-1.212	-0.386	-0.033	-0.0696	0.0039	-0.0422
1230	-1.127	-0.000	-0.000	-0.0692	0.0028	-0.0267
1240	-1.046	0.101	0.019	-0.0687	0.0014	0.0002
1250	-1.025	0.101	0.022	-0.0687	0.0014	0.0002
1260	-1.005	0.101	0.025	-0.0687	0.0014	0.0002
1270	-0.601	-0.000	0.014	-0.0664	-0.0008	0.0178
1280	0.000	-0.000	-0.000	-0.0631	0.0000	-0.0253
1290	0.354	-0.000	-0.000	-0.0611	0.0000	0.0322

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1300	0.451	-0.480	0.000	-0.0605	0.0001	0.0465
1310	0.471	-0.570	0.001	-0.0605	0.0001	0.0465
1320	0.492	-0.660	0.001	-0.0605	0.0001	0.0465
1330	0.506	-0.730	0.001	-0.0603	0.0003	0.0415
1340	0.567	-0.788	0.009	-0.0575	0.0022	-0.0287
1350	0.584	-0.741	0.012	-0.0575	0.0022	-0.0288
1360	0.632	-0.610	0.022	-0.0575	0.0022	-0.0289
1370	0.649	-0.563	0.025	-0.0575	0.0022	-0.0289
1380	0.766	-0.000	0.091	-0.0523	0.0088	-0.0187
1390	0.834	-0.047	0.161	-0.0493	0.0104	0.0151
1400	0.848	-0.069	0.177	-0.0490	0.0103	0.0135
1410	0.868	-0.095	0.197	-0.0490	0.0103	0.0135
1420	0.888	-0.121	0.217	-0.0490	0.0103	0.0135
1430	0.964	-0.194	0.302	-0.0486	0.0090	0.0028
1460	1.330	-0.000	0.347	-0.0465	-0.0108	0.0309
1470	1.418	-0.486	0.192	-0.0461	-0.0213	0.0586
1480	1.451	-0.603	0.157	-0.0460	-0.0272	0.0607
1490	1.471	-0.720	0.104	-0.0460	-0.0272	0.0607
1500	1.507	-0.930	0.010	-0.0460	-0.0272	0.0607
1510	1.413	-0.771	-0.026	-0.0460	-0.0273	0.0606
1520	1.360	-0.681	-0.047	-0.0460	-0.0273	0.0606
1530	0.199	-0.000	-0.238	0.0017	-0.0672	-0.0428
1540	-0.057	-0.283	0.112	0.0050	0.0098	0.0324
1550	-0.070	-0.246	0.100	0.0050	0.0095	0.0260
1558	0.101	-0.189	0.059	0.0081	0.0050	0.0369
1559	0.134	-0.164	0.050	0.0082	0.0041	0.0405
1560	0.141	-0.120	0.043	0.0083	0.0034	0.0423
1570	0.112	-0.000	0.037	0.0080	0.0003	0.0293
1580	0.079	0.053	0.042	0.0077	-0.0027	0.0035
1590	0.062	0.053	0.048	0.0077	-0.0031	-0.0012
1595	0.025	0.039	0.060	0.0076	-0.0027	-0.0042
1600	0.010	0.031	0.065	0.0075	-0.0018	-0.0047
1604	0.000	0.023	0.062	0.0074	-0.0006	-0.0044
1605	-0.003	0.015	0.053	0.0068	0.0004	-0.0036
1610	-0.000	0.000	0.020	0.0000	0.0000	-0.0000
1611	-0.000	0.000	-0.000	0.0000	0.0000	-0.0000
1620	-0.000	0.000	-0.000	0.0000	0.0000	-0.0000
1630	-0.089	-0.254	0.104	0.0041	0.0097	0.0176
1640	-0.101	-0.267	0.108	0.0020	0.0101	-0.0011
1650	-0.099	-0.284	0.111	0.0020	0.0101	-0.0012
1651	-1.928	-0.494	-0.143	-0.0108	-0.0042	0.0076
1660	-0.093	-0.340	0.122	0.0020	0.0101	-0.0013
1670	-0.091	-0.357	0.125	0.0020	0.0101	-0.0013
1678	0.178	-0.449	0.003	-0.0293	0.0147	-0.0369
1679	0.200	-0.466	-0.023	-0.0324	0.0148	-0.0349
1680	0.204	-0.493	-0.044	-0.0360	0.0149	-0.0323
1690	0.146	-0.612	-0.142	-0.0646	0.0189	-0.0063
1700	0.129	-0.622	-0.172	-0.0647	0.0189	-0.0063

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1710	0.074	-0.655	-0.273	-0.0647	0.0189	-0.0063
1720	0.057	-0.665	-0.303	-0.0647	0.0189	-0.0063
1728	0.040	-0.681	-0.339	-0.0732	0.0216	-0.0144
1729	0.028	-0.690	-0.332	-0.0769	0.0233	-0.0201
1730	0.009	-0.689	-0.284	-0.0801	0.0252	-0.0272
1740	-0.034	-0.677	-0.178	-0.0848	0.0280	-0.0407
1750	-0.096	-0.664	-0.063	-0.0879	0.0300	-0.0514
1760	-0.210	-0.644	0.118	-0.0895	0.0310	-0.0575
1770	-0.000	-0.000	0.022	-0.0003	0.0001	-0.0001
1780	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1781	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1790	-0.267	-0.807	0.132	-0.0954	0.0340	-0.0676
1800	-0.340	-1.014	0.154	-0.0955	0.0340	-0.0678
1808	-0.362	-0.142	-0.325	0.0124	-0.0550	-0.0901
1809	-0.417	-0.188	-0.313	0.0072	-0.0535	-0.0955
1810	-0.446	-0.282	-0.268	0.0002	-0.0520	-0.0995
1820	-0.446	-0.284	-0.267	0.0001	-0.0520	-0.0995
1830	-0.467	-0.477	-0.166	0.0001	-0.0519	-0.0995
1840	-0.044	-0.325	0.125	0.0050	0.0098	0.0324
1850	-0.414	-1.220	0.177	-0.0955	0.0341	-0.0679
1859	-0.489	-1.465	0.163	-0.0996	0.0390	-0.0808
1860	-0.535	-1.711	0.091	-0.1014	0.0438	-0.0949
1870	-0.602	-2.586	-0.321	-0.1014	0.0567	-0.1108
1880	-0.624	-2.825	-0.443	-0.1014	0.0567	-0.1108
1890	-0.646	-3.065	-0.566	-0.1014	0.0568	-0.1108
1900	-0.668	-3.299	-0.686	-0.1014	0.0568	-0.1108
1910	-0.668	-3.299	-0.686	0.0000	0.0568	-0.1108
1920	-0.690	-3.533	-0.806	0.0000	0.0569	-0.1107
1929	-0.681	-3.739	-0.916	0.0019	0.0668	-0.1031
1930	-0.651	-3.824	-0.956	0.0047	0.0765	-0.0948
1938	-1.402	-3.525	-0.270	0.0329	0.1290	-0.0697
1939	-1.474	-3.437	-0.114	0.0336	0.1303	-0.0702
1940	-1.492	-3.290	0.156	0.0338	0.1312	-0.0710
1950	-1.492	-3.288	0.159	0.0338	0.1312	-0.0711
1960	-1.470	-3.138	0.437	0.0338	0.1312	-0.0711
1970	-1.470	-3.138	0.437	0.0000	0.0000	0.0000
1980	-1.448	-3.138	0.437	0.0000	0.0000	0.0000
1989	-1.433	-3.132	0.438	0.0031	0.0006	0.0020
1990	-1.421	-3.117	0.435	0.0073	0.0003	0.0052
1998	-0.068	-2.545	-0.290	0.0360	-0.0091	0.0911
1999	0.107	-2.428	-0.340	0.0366	-0.0134	0.1065
2000	0.178	-2.197	-0.336	0.0369	-0.0170	0.1204
2010	0.178	-2.194	-0.335	0.0369	-0.0170	0.1204
2020	0.156	-1.939	-0.299	0.0369	-0.0170	0.1205
2030	0.156	-1.939	-0.299	-0.0000	-0.0170	0.1205
2040	0.134	-1.685	-0.263	-0.0000	-0.0170	0.1205
2050	-0.044	-0.325	0.125	0.0050	0.0098	0.0324
2060	-0.024	-0.388	0.144	0.0050	0.0098	0.0325

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
2070	-0.008	-0.437	0.159	0.0050	0.0098	0.0325
2080	-0.008	-0.437	0.159	-0.0000	0.0098	0.0325
2090	0.008	-0.486	0.173	-0.0000	0.0098	0.0325
2099	0.004	-0.520	0.182	-0.0024	0.0094	0.0308
2100	-0.021	-0.541	0.185	-0.0043	0.0090	0.0283
2109	0.395	-1.053	-0.448	-0.0233	-0.0077	-0.0441
2110	0.403	-1.100	-0.449	-0.0233	-0.0080	-0.0450
2120	0.403	-1.101	-0.449	-0.0233	-0.0080	-0.0451
2130	0.387	-1.169	-0.437	-0.0233	-0.0080	-0.0451
2140	0.387	-1.169	-0.437	-0.0000	-0.0000	-0.0000
2150	0.372	-1.169	-0.437	-0.0000	-0.0000	-0.0000
2159	0.362	-1.167	-0.433	-0.0030	-0.0012	-0.0018
2160	0.358	-1.163	-0.424	-0.0053	-0.0026	-0.0035
2169	-0.508	-1.067	0.138	-0.0026	-0.0513	-0.0971
2170	-0.519	-0.972	0.092	-0.0028	-0.0519	-0.0995
2180	-0.518	-0.971	0.092	-0.0028	-0.0519	-0.0995
2190	-0.503	-0.820	0.013	-0.0028	-0.0519	-0.0995
2200	-0.503	-0.820	0.013	0.0000	-0.0519	-0.0995
2201	-2.228	0.003	0.064	-0.0142	-0.0072	-0.0031
2210	-0.487	-0.670	-0.065	0.0000	-0.0519	-0.0995
2220	-0.467	-0.477	-0.166	0.0001	-0.0519	-0.0995
2721	-0.065	-0.260	0.105	0.0050	0.0095	0.0260
3101	-2.329	0.009	0.220	-0.0440	-0.0105	0.0089
3141	-0.152	-0.654	0.029	-0.0895	0.0310	-0.0575
4001	-3.059	-2.249	0.234	-0.0047	0.0097	0.0850
4020	-4.573	-0.141	-0.668	-0.0085	0.0005	-0.0137
4030	-4.845	-0.358	-0.585	0.0106	-0.0156	-0.0023
4038	-4.874	-0.363	-0.535	0.0126	-0.0204	-0.0018
4039	-4.868	-0.354	-0.471	0.0149	-0.0401	0.0000
4040	-4.781	-0.321	-0.415	0.0189	-0.0594	0.0010
4060	-3.396	-0.000	-0.210	0.0054	-0.0746	-0.0010
4080	0.286	-0.000	0.370	0.0149	-0.0536	-0.0065
4098	1.271	0.397	0.597	0.0213	-0.0346	-0.0087
4099	1.317	0.431	0.637	0.0216	-0.0219	-0.0090
4100	1.311	0.429	0.674	0.0210	-0.0097	-0.0107
4120	1.086	-0.000	0.712	0.0197	0.0018	-0.0382
4140	0.543	-2.008	0.585	0.0165	0.0050	0.0061
4160	-0.001	-0.000	-0.000	0.0133	0.0192	0.0137
4180	-1.000	-0.000	-4.158	0.0075	0.0734	0.0122
4198	-1.245	-0.124	-5.891	0.0060	0.0647	-0.0131
4199	-1.307	-0.145	-5.990	0.0058	0.0478	-0.0137
4200	-1.387	-0.145	-6.003	0.0060	0.0300	-0.0136
4220	-1.695	-0.000	-5.799	0.0081	0.0036	-0.0132
4240	-1.222	-0.000	-5.346	-0.0286	-0.0171	-0.0122
4260	-0.000	-0.000	-4.264	0.0076	-0.0051	-0.0100
4280	0.044	-0.000	-3.189	-0.0020	0.0013	-0.0077
4300	-0.003	-0.000	-2.120	0.0005	-0.0000	-0.0054
4320	-0.000	-0.000	-1.057	-0.0001	-0.0000	-0.0031

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
4340	0.000	-0.000	-0.000	0.0000	0.0000	-0.0008
4360	0.000	-0.000	1.047	-0.0000	-0.0000	0.0015
4380	-0.000	-0.000	2.100	0.0000	0.0000	0.0038
4400	-0.001	-0.000	3.159	-0.0000	-0.0000	0.0060
4420	0.016	-0.000	4.225	0.0000	-0.0004	0.0083
4440	0.000	-0.000	5.297	-0.0001	0.0015	0.0106
4460	-0.857	-0.000	6.375	0.0002	0.0194	0.0129
4480	-1.826	-0.000	7.369	0.0287	-0.0231	0.0150
4498	-1.168	0.315	7.559	0.0124	-0.0514	0.0154
4499	-1.045	0.326	7.528	0.0132	-0.0721	0.0151
4500	-0.963	0.309	7.380	0.0140	-0.0922	0.0138
4520	-0.809	0.142	5.853	0.0177	-0.1112	0.0078
4540	-0.523	-0.000	2.763	0.0244	-0.1039	0.0109
4558	-0.110	-0.626	-0.327	0.0340	-0.0513	0.0112
4559	-0.058	-0.619	-0.402	0.0349	-0.0367	0.0099
4560	0.005	-0.559	-0.407	0.0358	-0.0228	0.0095
4580	0.257	-0.000	-0.226	0.0228	-0.0092	0.0096
4600	0.299	-0.000	0.265	-0.0228	0.0082	0.0096
4618	0.048	-0.560	0.446	-0.0359	0.0236	0.0096
4619	-0.018	-0.620	0.440	-0.0351	0.0386	0.0100
4620	-0.071	-0.627	0.360	-0.0342	0.0544	0.0113
4640	-0.484	-0.000	-2.898	-0.0248	0.1085	0.0109
4660	-0.771	0.143	-6.092	-0.0182	0.1138	0.0081
4678	-0.925	0.318	-7.642	-0.0147	0.0925	0.0145
4679	-1.005	0.336	-7.788	-0.0138	0.0704	0.0162
4680	-1.123	0.324	-7.818	-0.0130	0.0477	0.0166
4700	-1.686	-0.000	-7.628	-0.0291	0.0163	0.0171
4720	-0.000	-0.000	-6.634	-0.0001	-0.0210	0.0200
4740	0.261	-0.000	-5.556	0.0000	0.0051	0.0232
4760	-0.003	-0.000	-4.484	0.0000	0.0006	0.0263
4780	-0.000	-0.000	-3.418	-0.0001	-0.0002	0.0294
4800	0.002	-0.000	-2.448	0.0298	0.0001	0.0323
4820	0.000	-0.000	-1.832	-0.0390	0.0000	0.0341
4840	-0.000	-0.000	-0.783	0.0279	-0.0000	0.0372
4860	0.000	-0.000	0.000	-0.0021	-0.0000	0.0396
4880	-0.001	-0.000	0.792	-0.0193	0.0000	0.0419
4900	0.000	-0.000	1.853	0.0046	-0.0002	0.0451
4920	0.022	-0.000	2.921	0.0010	0.0001	0.0482
4940	-0.279	-0.000	3.995	-0.0084	0.0078	0.0513
4950	0.000	-0.000	5.076	0.0328	-0.0307	0.0545
4960	2.856	-0.000	5.700	-0.0209	-0.0545	0.0563
4978	4.480	-1.461	6.095	-0.0534	-0.0208	0.0574
4979	4.457	-1.581	6.066	-0.0661	-0.0047	0.0620
4980	4.336	-1.655	5.937	-0.0788	0.0030	0.0675
4998	3.621	-1.763	5.072	-0.0858	0.0272	0.0712
4999	3.473	-1.724	4.882	-0.0917	0.0349	0.0774
5000	3.392	-1.581	4.728	-0.0961	0.0503	0.0822
5001	-3.548	-0.113	-0.175	-0.0056	0.0011	-0.0073

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 6 (OPE) W+T3+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
5020	3.165	-0.000	3.403	-0.0961	0.0643	0.0469
5040	2.513	-0.000	-0.037	-0.0961	0.0448	-0.0147
5058	2.264	-0.298	-1.013	-0.0961	0.0363	0.0005
5059	2.254	-0.273	-1.003	-0.0970	0.0329	0.0187
5060	2.299	-0.234	-0.844	-0.0980	0.0313	0.0348
5078	2.738	-0.116	0.271	-0.0982	0.0258	0.0377
5079	2.793	-0.072	0.437	-0.0994	0.0241	0.0271
5080	2.791	-0.026	0.473	-0.1005	0.0206	0.0133
5100	2.638	-0.000	0.211	-0.1010	0.0148	-0.0122
5120	1.738	-0.000	-0.078	-0.1041	-0.0019	0.0056
5140	0.843	-0.000	-0.014	-0.1073	0.0005	-0.0102
5150	0.344	-1.022	-0.041	-0.1090	-0.0001	0.0113
5160	0.305	-0.964	-0.040	-0.1091	-0.0003	0.0197
5180	0.289	-0.933	-0.040	-0.1091	-0.0003	0.0198
5200	0.273	-0.902	-0.039	-0.1091	-0.0003	0.0199
5220	-0.000	-0.000	0.000	-0.1101	-0.0028	0.0300
5240	-0.617	-0.000	0.201	-0.1122	-0.0010	-0.0438
5258	-0.794	-1.127	0.169	-0.1129	0.0065	-0.0836
5259	-0.805	-1.213	0.141	-0.1089	0.0143	-0.0987
5260	-0.778	-1.091	0.109	-0.1058	0.0220	-0.1051
5268	-0.357	0.684	-0.063	-0.1094	0.0269	-0.1386
5269	-0.315	0.777	-0.103	-0.1060	0.0257	-0.1451
5270	-0.315	0.572	-0.158	-0.1019	0.0244	-0.1603
5280	-0.351	-0.000	-0.243	-0.1019	0.0242	-0.1666
5300	-0.501	-2.543	-0.591	-0.1019	0.0242	-0.1826
5320	-0.583	-3.987	-0.781	-0.1019	0.0242	-0.1843
5340	-0.600	-4.273	-0.818	-0.1019	0.0242	-0.1843
9301	-4.336	0.043	-0.752	-0.0168	-0.0134	-0.0022

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
10	-0.028	-1.411	-0.284	-0.0000	-0.0217	0.1193
20	-0.028	-1.153	-0.237	-0.0000	-0.0217	0.1193
30	-0.000	-0.000	-0.000	-0.0001	-0.0188	0.0725
40	0.009	0.325	0.109	-0.0002	-0.0143	0.0289
50	0.017	0.403	0.176	-0.0003	-0.0112	0.0030
60	0.042	-0.000	0.291	-0.0005	-0.0026	-0.0376
70	0.076	-0.737	0.259	-0.0008	0.0045	-0.0115
80	0.076	-0.761	0.249	-0.0008	0.0045	-0.0115
90	0.076	-0.786	0.240	-0.0008	0.0045	-0.0114
100	0.135	-0.000	0.013	-0.0013	0.0040	0.0145
110	0.238	-0.000	-0.000	-0.0021	-0.0008	-0.0053
120	0.327	-0.139	0.011	-0.0028	-0.0004	0.0156
130	0.327	-0.106	0.012	-0.0028	-0.0004	0.0156
140	0.328	-0.072	0.013	-0.0028	-0.0005	0.0156
150	0.338	0.008	0.017	-0.0029	-0.0008	0.0045
160	0.345	-0.000	0.022	-0.0030	-0.0011	-0.0097
170	0.395	-0.604	0.043	-0.0033	0.0012	0.0127
180	0.398	-0.574	0.040	-0.0040	0.0016	0.0203
190	0.398	-0.530	0.037	-0.0040	0.0016	0.0204
200	0.398	-0.366	0.024	-0.0040	0.0016	0.0207
210	0.399	-0.321	0.020	-0.0041	0.0016	0.0207
220	0.419	-0.000	0.000	-0.0088	0.0010	0.0155
230	0.438	0.052	-0.015	-0.0129	0.0032	0.0032
240	0.430	0.054	-0.041	-0.0098	0.0048	0.0051
250	0.428	0.055	-0.045	-0.0087	0.0054	0.0059
260	0.416	0.055	-0.061	-0.0086	0.0054	0.0059
270	0.377	0.056	-0.118	-0.0085	0.0055	0.0060
280	0.365	0.056	-0.135	-0.0085	0.0055	0.0060
288	0.346	0.061	-0.154	-0.0046	0.0099	0.0066
289	0.329	0.062	-0.160	-0.0021	0.0121	0.0065
290	0.307	0.062	-0.163	-0.0002	0.0143	0.0062
298	0.231	0.068	-0.171	0.0020	0.0171	0.0048
299	0.205	0.069	-0.163	0.0023	0.0175	0.0044
300	0.192	0.064	-0.139	0.0022	0.0173	0.0043
310	0.175	-0.000	-0.000	0.0010	0.0106	0.0127
318	0.132	-0.550	0.014	-0.0020	-0.0036	0.0221
319	0.127	-0.580	0.010	-0.0022	-0.0032	0.0215
320	0.122	-0.589	0.011	-0.0024	-0.0028	0.0209
328	0.111	-0.578	0.019	-0.0021	-0.0019	0.0193
329	0.120	-0.576	0.022	-0.0015	-0.0016	0.0188
330	0.145	-0.578	0.025	-0.0008	-0.0014	0.0182
340	0.465	0.139	-0.075	-0.0439	0.0050	0.0094
350	0.528	-0.000	-0.089	-0.0257	0.0013	-0.0671
360	0.582	-2.535	-0.182	-0.0112	0.0020	-0.0331
370	0.582	-2.606	-0.187	-0.0112	0.0020	-0.0329
380	0.583	-2.865	-0.202	-0.0111	0.0019	-0.0320

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
390	0.583	-2.934	-0.207	-0.0111	0.0019	-0.0318
400	0.603	-2.719	-0.192	-0.0059	-0.0051	0.0705
410	0.603	-2.567	-0.181	-0.0059	-0.0052	0.0707
420	0.603	-2.414	-0.170	-0.0059	-0.0052	0.0708
430	0.606	-2.269	-0.159	-0.0050	-0.0069	0.0856
440	0.610	-2.099	-0.147	-0.0050	-0.0067	0.0982
450	0.610	-1.887	-0.132	-0.0050	-0.0067	0.0984
460	0.611	-1.100	-0.079	-0.0050	-0.0067	0.0987
470	0.611	-0.887	-0.064	-0.0050	-0.0067	0.0987
480	0.631	-0.000	0.000	-0.0053	-0.0055	0.0523
490	0.666	0.084	0.069	-0.0057	-0.0028	-0.0148
500	0.666	0.052	0.076	-0.0057	-0.0028	-0.0148
510	0.667	-0.065	0.098	-0.0057	-0.0028	-0.0145
520	0.667	-0.096	0.104	-0.0057	-0.0028	-0.0144
530	0.671	-0.117	0.109	-0.0057	-0.0024	-0.0074
538	0.584	-0.109	0.135	0.0025	-0.0099	-0.0123
539	0.564	-0.105	0.132	0.0025	-0.0082	-0.0134
540	0.553	-0.080	0.121	0.0020	-0.0057	-0.0154
550	0.552	-0.074	0.119	0.0019	-0.0054	-0.0157
560	0.552	-0.041	0.108	0.0019	-0.0054	-0.0157
570	0.551	0.085	0.065	0.0019	-0.0054	-0.0157
580	0.550	0.119	0.053	0.0019	-0.0054	-0.0157
590	0.511	-0.000	-0.004	-0.0008	-0.0033	0.0493
600	0.487	-0.759	-0.043	-0.0025	-0.0052	0.0978
610	0.487	-0.971	-0.055	-0.0025	-0.0052	0.0978
620	0.486	-1.752	-0.096	-0.0025	-0.0053	0.0979
630	0.486	-1.963	-0.108	-0.0025	-0.0053	0.0978
639	0.487	-2.147	-0.122	-0.0027	-0.0077	0.0957
640	0.501	-2.226	-0.133	-0.0029	-0.0095	0.0939
650	0.675	-0.123	0.109	-0.0098	0.0015	0.0002
660	0.675	-0.122	0.106	-0.0098	0.0016	0.0003
670	0.675	-0.121	0.103	-0.0099	0.0016	0.0003
680	0.679	-0.115	0.096	-0.0139	0.0052	0.0059
690	0.698	-0.011	0.028	-0.0144	0.0082	0.0076
700	0.521	-0.329	0.062	-0.0048	0.0159	0.0059
710	0.487	-0.339	0.061	-0.0048	0.0159	0.0059
720	0.360	-0.376	0.060	-0.0045	0.0159	0.0059
730	0.326	-0.386	0.060	-0.0044	0.0159	0.0059
740	0.002	-0.000	0.000	0.0105	0.0068	0.0059
750	-0.029	0.017	-0.026	-0.0008	-0.0003	0.0059
760	-0.029	0.015	-0.026	-0.0008	-0.0003	0.0059
770	-0.026	0.010	-0.027	-0.0007	-0.0004	0.0059
780	-0.025	0.008	-0.027	-0.0007	-0.0004	0.0059
790	-0.011	-0.000	-0.053	-0.0063	-0.0017	0.0059
800	0.007	-0.138	-0.079	-0.0167	-0.0017	0.0059
810	0.010	-0.174	-0.080	-0.0167	-0.0017	0.0059
820	0.012	-0.189	-0.080	-0.0167	-0.0017	0.0059
830	0.706	-0.000	0.000	-0.0146	0.0064	-0.0053

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
840	0.731	-0.221	-0.029	-0.0151	-0.0004	-0.0246
850	0.731	-0.274	-0.028	-0.0151	-0.0004	-0.0245
860	0.732	-0.469	-0.025	-0.0151	-0.0004	-0.0243
870	0.732	-0.521	-0.024	-0.0151	-0.0004	-0.0242
880	0.744	-0.586	-0.019	-0.0154	-0.0016	-0.0008
890	0.756	-0.528	-0.008	-0.0157	-0.0023	0.0232
900	0.757	-0.478	-0.003	-0.0157	-0.0023	0.0233
910	0.757	-0.461	-0.002	-0.0157	-0.0023	0.0233
920	0.758	-0.273	0.017	-0.0157	-0.0023	0.0236
930	0.758	-0.222	0.022	-0.0157	-0.0023	0.0237
940	0.757	-0.000	0.033	-0.0162	0.0001	0.0178
950	0.756	0.060	0.001	-0.0168	0.0063	-0.0022
960	0.572	-0.138	0.001	-0.0059	0.0193	-0.0022
970	0.542	-0.147	0.001	-0.0059	0.0193	-0.0022
980	0.439	-0.178	0.001	-0.0058	0.0193	-0.0022
990	0.409	-0.187	0.001	-0.0057	0.0193	-0.0022
1000	0.003	-0.000	0.000	0.0072	0.0085	-0.0022
1010	-0.033	0.024	-0.000	0.0002	-0.0009	-0.0022
1020	-0.032	0.025	-0.000	0.0002	-0.0009	-0.0022
1030	-0.027	0.026	-0.000	0.0002	-0.0009	-0.0022
1040	-0.026	0.026	-0.000	0.0002	-0.0009	-0.0022
1050	-0.001	-0.000	-0.000	-0.0080	-0.0029	-0.0022
1060	0.030	-0.142	-0.000	-0.0163	-0.0029	-0.0022
1070	0.034	-0.168	-0.000	-0.0163	-0.0029	-0.0022
1080	0.037	-0.182	-0.000	-0.0163	-0.0029	-0.0022
1090	0.756	-0.000	-0.032	-0.0121	0.0005	-0.0099
1100	0.756	-0.098	-0.025	-0.0086	-0.0020	-0.0137
1110	0.756	-0.120	-0.022	-0.0086	-0.0020	-0.0137
1118	0.317	-0.767	-0.099	-0.0737	0.0106	-0.0277
1119	0.302	-0.854	-0.096	-0.0741	0.0087	-0.0322
1120	0.294	-0.848	-0.086	-0.0739	0.0064	-0.0376
1130	0.294	-0.847	-0.086	-0.0739	0.0064	-0.0377
1140	0.294	-0.774	-0.074	-0.0739	0.0064	-0.0377
1150	0.294	-0.512	-0.029	-0.0739	0.0064	-0.0378
1160	0.294	-0.439	-0.017	-0.0739	0.0063	-0.0378
1170	0.269	-0.000	0.029	-0.0732	0.0016	-0.0088
1180	0.217	-0.916	0.040	-0.0718	-0.0006	0.0301
1190	0.178	-0.845	0.019	-0.0707	-0.0010	-0.0428
1200	0.178	-0.762	0.017	-0.0707	-0.0010	-0.0428
1210	0.178	-0.476	0.011	-0.0707	-0.0010	-0.0430
1220	0.177	-0.393	0.009	-0.0707	-0.0010	-0.0430
1230	0.161	-0.000	0.000	-0.0703	-0.0007	-0.0273
1240	0.145	0.105	-0.005	-0.0698	-0.0003	-0.0001
1250	0.145	0.105	-0.005	-0.0698	-0.0003	-0.0001
1260	0.145	0.105	-0.006	-0.0698	-0.0003	-0.0000
1270	0.068	-0.000	-0.001	-0.0677	0.0002	0.0180
1280	-0.045	-0.000	0.000	-0.0644	-0.0001	-0.0253
1290	-0.112	-0.000	0.000	-0.0625	0.0000	0.0323

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1300	-0.130	-0.481	0.001	-0.0620	0.0001	0.0466
1310	-0.131	-0.571	0.001	-0.0620	0.0001	0.0466
1320	-0.131	-0.661	0.001	-0.0620	0.0001	0.0466
1330	-0.132	-0.732	0.001	-0.0617	0.0001	0.0416
1340	-0.136	-0.790	0.001	-0.0591	0.0000	-0.0287
1350	-0.136	-0.743	0.001	-0.0591	0.0000	-0.0287
1360	-0.136	-0.612	0.001	-0.0590	0.0000	-0.0288
1370	-0.136	-0.565	0.002	-0.0590	0.0000	-0.0289
1380	-0.142	-0.000	-0.000	-0.0540	-0.0003	-0.0192
1390	-0.146	-0.043	-0.004	-0.0511	-0.0006	0.0145
1400	-0.148	-0.064	-0.004	-0.0508	-0.0006	0.0129
1410	-0.148	-0.089	-0.006	-0.0508	-0.0006	0.0129
1420	-0.148	-0.114	-0.007	-0.0508	-0.0006	0.0129
1430	-0.163	-0.182	-0.013	-0.0504	-0.0007	0.0023
1460	-0.234	-0.000	-0.044	-0.0484	-0.0008	0.0323
1470	-0.252	-0.504	-0.047	-0.0480	0.0009	0.0608
1480	-0.239	-0.629	-0.032	-0.0478	0.0026	0.0633
1490	-0.239	-0.750	-0.027	-0.0478	0.0026	0.0633
1500	-0.239	-0.970	-0.018	-0.0478	0.0026	0.0633
1510	-0.230	-0.804	-0.018	-0.0478	0.0026	0.0632
1520	-0.225	-0.712	-0.018	-0.0478	0.0026	0.0632
1530	-0.001	-0.000	0.000	0.0011	0.0114	-0.0316
1540	-0.230	-0.211	0.076	0.0019	0.0108	0.0297
1550	-0.229	-0.176	0.061	0.0019	0.0107	0.0246
1558	-0.059	-0.179	0.037	0.0057	0.0080	0.0377
1559	-0.021	-0.164	0.028	0.0060	0.0075	0.0414
1560	-0.003	-0.123	0.019	0.0061	0.0070	0.0433
1570	-0.001	-0.000	0.000	0.0064	0.0047	0.0299
1580	0.002	0.054	-0.010	0.0067	0.0012	0.0035
1590	0.004	0.054	-0.011	0.0068	0.0003	-0.0013
1595	0.011	0.039	-0.010	0.0069	-0.0008	-0.0045
1600	0.013	0.031	-0.008	0.0070	-0.0019	-0.0049
1604	0.013	0.023	-0.005	0.0070	-0.0030	-0.0046
1605	0.010	0.015	-0.003	0.0065	-0.0036	-0.0037
1610	0.000	0.000	-0.000	0.0000	-0.0000	-0.0000
1611	0.000	0.000	0.000	0.0000	-0.0000	-0.0000
1620	0.000	0.000	0.000	0.0000	-0.0000	-0.0000
1630	-0.247	-0.175	0.063	0.0010	0.0109	0.0172
1640	-0.260	-0.174	0.063	-0.0013	0.0114	0.0007
1650	-0.261	-0.174	0.061	-0.0013	0.0114	0.0006
1651	0.382	-0.603	0.040	-0.0033	0.0012	0.0127
1660	-0.264	-0.174	0.054	-0.0013	0.0114	0.0005
1670	-0.265	-0.174	0.052	-0.0013	0.0114	0.0005
1678	0.001	-0.168	-0.100	-0.0317	0.0168	-0.0417
1679	0.031	-0.180	-0.128	-0.0346	0.0170	-0.0411
1680	0.044	-0.208	-0.151	-0.0380	0.0172	-0.0399
1690	0.048	-0.395	-0.262	-0.0650	0.0210	-0.0201
1700	0.048	-0.427	-0.296	-0.0650	0.0210	-0.0201

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1710	0.048	-0.534	-0.408	-0.0650	0.0210	-0.0201
1720	0.048	-0.566	-0.442	-0.0650	0.0210	-0.0201
1728	0.050	-0.604	-0.480	-0.0730	0.0232	-0.0250
1729	0.042	-0.623	-0.475	-0.0765	0.0246	-0.0290
1730	0.020	-0.632	-0.427	-0.0795	0.0261	-0.0342
1740	-0.029	-0.633	-0.322	-0.0839	0.0284	-0.0443
1750	-0.094	-0.634	-0.209	-0.0869	0.0300	-0.0524
1760	-0.207	-0.635	-0.030	-0.0884	0.0307	-0.0571
1770	-0.000	-0.000	-0.000	-0.0003	0.0001	-0.0001
1780	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1781	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1790	-0.264	-0.797	-0.034	-0.0943	0.0338	-0.0667
1800	-0.337	-1.001	-0.034	-0.0944	0.0338	-0.0668
1808	0.079	-0.140	0.008	0.0124	0.0071	-0.0749
1809	0.086	-0.180	0.006	0.0072	0.0069	-0.0800
1810	0.089	-0.260	0.000	0.0002	0.0068	-0.0835
1820	0.089	-0.261	0.000	0.0001	0.0068	-0.0836
1830	0.089	-0.423	-0.013	0.0001	0.0068	-0.0836
1840	-0.230	-0.249	0.089	0.0019	0.0108	0.0297
1850	-0.410	-1.205	-0.035	-0.0944	0.0338	-0.0669
1859	-0.476	-1.446	-0.068	-0.0984	0.0385	-0.0793
1860	-0.503	-1.688	-0.147	-0.1001	0.0432	-0.0926
1870	-0.482	-2.537	-0.554	-0.1001	0.0564	-0.1069
1880	-0.482	-2.768	-0.676	-0.1001	0.0564	-0.1069
1890	-0.482	-2.999	-0.798	-0.1001	0.0564	-0.1069
1900	-0.482	-3.225	-0.917	-0.1001	0.0565	-0.1069
1910	-0.482	-3.225	-0.917	0.0000	0.0565	-0.1069
1920	-0.481	-3.451	-1.037	0.0000	0.0566	-0.1068
1929	-0.456	-3.642	-1.153	0.0031	0.0673	-0.0984
1930	-0.425	-3.706	-1.204	0.0075	0.0777	-0.0892
1938	-1.452	-2.977	-0.588	0.0429	0.1356	-0.0618
1939	-1.552	-2.867	-0.427	0.0437	0.1372	-0.0623
1940	-1.598	-2.723	-0.144	0.0440	0.1383	-0.0632
1950	-1.598	-2.722	-0.141	0.0440	0.1383	-0.0632
1960	-1.598	-2.588	0.151	0.0440	0.1383	-0.0632
1970	-1.598	-2.588	0.151	0.0000	0.0000	-0.0000
1980	-1.599	-2.588	0.151	0.0000	0.0000	0.0000
1989	-1.603	-2.590	0.151	0.0017	0.0008	0.0015
1990	-1.601	-2.594	0.145	0.0040	0.0006	0.0043
1998	-0.302	-2.483	-0.409	0.0241	-0.0129	0.0890
1999	-0.119	-2.394	-0.441	0.0245	-0.0175	0.1048
2000	-0.029	-2.176	-0.423	0.0247	-0.0216	0.1191
2010	-0.029	-2.173	-0.422	0.0247	-0.0216	0.1192
2020	-0.028	-1.921	-0.376	0.0247	-0.0216	0.1192
2030	-0.028	-1.921	-0.376	-0.0000	-0.0216	0.1192
2040	-0.028	-1.669	-0.331	-0.0000	-0.0216	0.1193
2050	-0.230	-0.249	0.089	0.0019	0.0108	0.0297
2060	-0.230	-0.306	0.110	0.0019	0.0107	0.0297

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
2070	-0.230	-0.351	0.126	0.0019	0.0107	0.0297
2080	-0.230	-0.351	0.126	-0.0000	0.0107	0.0297
2090	-0.230	-0.396	0.143	-0.0000	0.0107	0.0297
2099	-0.242	-0.423	0.152	-0.0018	0.0101	0.0264
2100	-0.267	-0.432	0.153	-0.0033	0.0093	0.0223
2109	1.093	-0.529	-0.341	-0.0151	-0.0144	-0.0777
2110	1.124	-0.604	-0.333	-0.0151	-0.0146	-0.0786
2120	1.124	-0.605	-0.333	-0.0151	-0.0146	-0.0787
2130	1.124	-0.724	-0.311	-0.0151	-0.0146	-0.0787
2140	1.124	-0.724	-0.311	-0.0000	-0.0000	-0.0000
2150	1.124	-0.724	-0.311	-0.0000	-0.0000	-0.0000
2159	1.125	-0.725	-0.310	-0.0028	-0.0005	-0.0008
2160	1.125	-0.728	-0.307	-0.0050	-0.0010	-0.0015
2169	0.118	-0.916	-0.052	0.0016	0.0056	-0.0806
2170	0.089	-0.839	-0.047	0.0015	0.0067	-0.0835
2180	0.089	-0.838	-0.046	0.0015	0.0067	-0.0836
2190	0.089	-0.712	-0.036	0.0015	0.0068	-0.0836
2200	0.089	-0.712	-0.036	0.0000	0.0068	-0.0836
2201	0.435	0.053	-0.028	-0.0129	0.0032	0.0032
2210	0.089	-0.585	-0.026	0.0000	0.0068	-0.0836
2220	0.089	-0.423	-0.013	0.0001	0.0068	-0.0836
2721	-0.229	-0.189	0.067	0.0019	0.0107	0.0247
3101	0.460	0.095	-0.075	-0.0439	0.0050	0.0094
3141	-0.150	-0.635	-0.118	-0.0884	0.0307	-0.0571
4001	0.600	-2.264	-0.159	-0.0050	-0.0069	0.0856
4020	0.756	-0.141	-0.019	-0.0085	-0.0020	-0.0137
4030	0.756	-0.358	0.019	0.0106	0.0036	-0.0023
4038	0.756	-0.363	0.006	0.0126	0.0057	-0.0018
4039	0.747	-0.354	-0.012	0.0149	0.0145	0.0000
4040	0.712	-0.321	-0.026	0.0189	0.0226	0.0010
4060	0.233	-0.000	-0.027	0.0054	0.0205	-0.0010
4080	-0.028	-0.000	-0.029	0.0149	-0.0018	-0.0065
4098	-0.005	0.397	-0.030	0.0213	-0.0008	-0.0087
4099	-0.003	0.431	-0.029	0.0216	-0.0010	-0.0090
4100	-0.002	0.429	-0.027	0.0210	-0.0012	-0.0107
4120	-0.001	-0.000	-0.000	0.0197	-0.0011	-0.0382
4140	-0.001	-2.008	0.018	0.0165	0.0002	0.0061
4160	0.000	-0.000	0.000	0.0133	0.0003	0.0137
4180	0.001	-0.000	0.000	0.0075	-0.0002	0.0122
4198	0.002	-0.124	0.004	0.0060	-0.0001	-0.0131
4199	0.003	-0.145	0.004	0.0058	0.0000	-0.0137
4200	0.003	-0.145	0.004	0.0060	0.0001	-0.0136
4220	0.000	-0.000	0.003	0.0081	0.0001	-0.0132
4240	-0.000	-0.000	0.001	-0.0286	-0.0000	-0.0122
4260	0.000	-0.000	0.000	0.0076	0.0000	-0.0100
4280	-0.000	-0.000	0.000	-0.0020	-0.0000	-0.0077
4300	0.000	-0.000	0.000	0.0005	0.0000	-0.0054
4320	-0.000	-0.000	0.000	-0.0001	-0.0000	-0.0031

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
4340	0.000	-0.000	0.000	0.0000	0.0000	-0.0008
4360	0.000	-0.000	-0.000	-0.0000	-0.0000	0.0015
4380	-0.000	-0.000	-0.000	0.0000	0.0000	0.0038
4400	0.000	-0.000	-0.000	-0.0000	-0.0000	0.0060
4420	-0.000	-0.000	-0.000	0.0000	0.0000	0.0083
4440	0.000	-0.000	-0.000	-0.0001	-0.0000	0.0106
4460	-0.000	-0.000	-0.000	0.0002	0.0000	0.0129
4480	0.000	-0.000	-0.001	0.0287	-0.0001	0.0150
4498	0.002	0.315	-0.002	0.0124	-0.0001	0.0154
4499	0.002	0.326	-0.002	0.0132	-0.0000	0.0151
4500	0.002	0.309	-0.003	0.0140	0.0000	0.0138
4520	0.001	0.142	-0.002	0.0177	0.0001	0.0078
4540	0.000	-0.000	-0.000	0.0244	0.0001	0.0109
4558	-0.002	-0.626	0.002	0.0340	-0.0000	0.0112
4559	-0.002	-0.619	0.002	0.0349	-0.0001	0.0099
4560	-0.002	-0.559	0.002	0.0358	-0.0001	0.0095
4580	-0.000	-0.000	0.001	0.0228	-0.0001	0.0096
4600	-0.000	-0.000	-0.001	-0.0228	0.0001	0.0096
4618	-0.002	-0.560	-0.002	-0.0359	0.0001	0.0096
4619	-0.002	-0.620	-0.002	-0.0351	0.0001	0.0099
4620	-0.002	-0.627	-0.002	-0.0342	0.0000	0.0113
4640	0.000	-0.000	0.000	-0.0247	-0.0001	0.0109
4660	0.001	0.143	0.002	-0.0182	-0.0001	0.0081
4678	0.002	0.317	0.003	-0.0147	-0.0000	0.0145
4679	0.002	0.335	0.002	-0.0137	0.0000	0.0161
4680	0.002	0.323	0.002	-0.0129	0.0001	0.0165
4700	0.000	-0.000	0.001	-0.0291	0.0001	0.0170
4720	-0.000	-0.000	0.000	-0.0001	-0.0000	0.0195
4740	0.000	-0.000	0.000	0.0000	0.0000	0.0222
4760	-0.000	-0.000	0.000	0.0000	-0.0000	0.0250
4780	0.000	-0.000	0.000	-0.0001	0.0000	0.0277
4800	-0.000	-0.000	0.000	0.0298	-0.0000	0.0302
4820	0.000	-0.000	0.000	-0.0390	-0.0000	0.0318
4840	-0.000	-0.000	0.000	0.0279	0.0000	0.0346
4860	0.000	-0.000	-0.000	-0.0021	-0.0000	0.0366
4880	-0.000	-0.000	-0.000	-0.0193	0.0000	0.0387
4900	0.000	-0.000	-0.000	0.0046	-0.0001	0.0414
4920	-0.000	-0.000	-0.000	0.0010	0.0003	0.0441
4940	0.000	-0.000	-0.000	-0.0085	-0.0011	0.0469
4950	-0.000	-0.000	-0.001	0.0332	0.0042	0.0496
4960	0.001	-0.000	-0.002	-0.0221	-0.0126	0.0512
4978	0.792	-1.202	-0.003	-0.0288	-0.0240	0.0522
4979	0.796	-1.253	-0.024	-0.0263	-0.0229	0.0526
4980	0.714	-1.273	-0.072	-0.0244	-0.0222	0.0535
4998	0.151	-1.272	-0.319	-0.0235	-0.0202	0.0554
4999	0.044	-1.227	-0.348	-0.0237	-0.0196	0.0585
5000	-0.002	-1.115	-0.331	-0.0241	-0.0180	0.0603
5001	0.668	-0.111	0.109	-0.0057	-0.0024	-0.0074

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 8 (SUS) W+P1

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
5020	-0.001	-0.000	-0.000	-0.0241	-0.0114	0.0292
5040	0.000	-0.000	-0.000	-0.0241	0.0091	0.0013
5058	0.001	0.011	-0.258	-0.0242	0.0114	0.0011
5059	0.002	0.013	-0.260	-0.0243	0.0110	0.0003
5060	0.001	0.012	-0.223	-0.0246	0.0107	-0.0007
5078	-0.006	0.012	0.061	-0.0256	0.0099	0.0000
5079	-0.004	0.014	0.103	-0.0271	0.0093	0.0023
5080	-0.002	0.020	0.108	-0.0280	0.0082	0.0033
5100	-0.001	-0.000	0.000	-0.0313	0.0059	-0.0122
5120	-0.000	-0.000	-0.000	-0.0505	-0.0016	0.0055
5140	-0.000	-0.000	0.000	-0.0698	0.0004	-0.0099
5150	-0.000	-1.010	-0.005	-0.0805	-0.0001	0.0114
5160	-0.000	-0.952	-0.004	-0.0814	-0.0001	0.0196
5180	-0.000	-0.921	-0.004	-0.0814	-0.0001	0.0197
5200	-0.000	-0.890	-0.004	-0.0814	-0.0001	0.0198
5220	0.000	-0.000	-0.000	-0.0873	-0.0001	0.0291
5240	0.001	-0.000	-0.000	-0.1004	0.0001	-0.0409
5258	0.002	-1.056	-0.002	-0.1042	0.0001	-0.0787
5259	0.002	-1.139	-0.002	-0.1009	-0.0000	-0.0930
5260	0.002	-1.026	-0.001	-0.0991	-0.0001	-0.0988
5268	0.000	0.651	-0.001	-0.1039	-0.0001	-0.1315
5269	-0.000	0.740	-0.000	-0.1008	-0.0001	-0.1377
5270	-0.000	0.545	-0.000	-0.0967	-0.0001	-0.1527
5280	0.000	-0.000	0.000	-0.0967	-0.0001	-0.1590
5300	0.001	-2.433	0.001	-0.0967	-0.0001	-0.1750
5320	0.001	-3.817	0.002	-0.0967	-0.0001	-0.1767
5340	0.001	-4.091	0.002	-0.0967	-0.0001	-0.1767
9301	0.750	0.043	0.001	-0.0168	0.0063	-0.0022

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
10	-0.010	-1.412	-0.274	-0.0000	-0.0208	0.1194
20	-0.010	-1.154	-0.228	-0.0000	-0.0209	0.1194
30	-0.000	-0.000	-0.000	-0.0003	-0.0181	0.0725
40	0.000	0.325	0.105	-0.0004	-0.0138	0.0289
50	0.000	0.403	0.169	-0.0005	-0.0107	0.0030
60	0.001	-0.000	0.278	-0.0009	-0.0024	-0.0377
70	0.001	-0.738	0.243	-0.0015	0.0045	-0.0116
80	0.001	-0.763	0.233	-0.0015	0.0045	-0.0116
90	0.001	-0.788	0.224	-0.0015	0.0045	-0.0115
100	0.002	-0.000	0.000	-0.0025	0.0039	0.0147
110	0.007	-0.000	-0.000	-0.0041	-0.0012	-0.0061
120	0.012	-0.170	0.012	-0.0054	0.0007	0.0169
130	0.012	-0.134	0.010	-0.0054	0.0007	0.0169
140	0.012	-0.097	0.009	-0.0054	0.0007	0.0169
150	0.013	-0.005	0.004	-0.0056	0.0008	0.0068
160	0.013	-0.000	-0.000	-0.0057	0.0009	-0.0067
170	0.018	-0.539	-0.018	-0.0064	-0.0004	0.0103
180	0.018	-0.514	-0.018	-0.0068	-0.0002	0.0180
190	0.018	-0.475	-0.017	-0.0068	-0.0002	0.0181
200	0.018	-0.329	-0.015	-0.0068	-0.0002	0.0184
210	0.018	-0.289	-0.014	-0.0068	-0.0002	0.0185
220	0.020	-0.000	0.000	-0.0096	-0.0024	0.0124
230	0.023	0.007	0.033	-0.0120	-0.0020	-0.0022
240	0.028	0.008	0.009	-0.0088	-0.0011	-0.0019
250	0.028	0.008	0.006	-0.0075	-0.0008	-0.0018
260	0.032	0.008	-0.009	-0.0075	-0.0008	-0.0018
270	0.044	0.008	-0.058	-0.0074	-0.0007	-0.0018
280	0.048	0.008	-0.072	-0.0074	-0.0007	-0.0018
288	0.053	0.010	-0.088	-0.0033	0.0017	-0.0016
289	0.053	0.010	-0.091	-0.0008	0.0030	-0.0014
290	0.049	0.011	-0.092	0.0012	0.0044	-0.0013
298	0.022	0.024	-0.094	0.0034	0.0070	-0.0011
299	0.011	0.029	-0.091	0.0035	0.0081	-0.0010
300	0.005	0.033	-0.079	0.0032	0.0089	-0.0004
310	-0.001	-0.000	-0.000	0.0016	0.0058	0.0108
318	-0.017	-0.527	-0.088	-0.0027	-0.0105	0.0189
319	-0.025	-0.550	-0.104	-0.0031	-0.0113	0.0172
320	-0.042	-0.555	-0.109	-0.0035	-0.0119	0.0156
328	-0.098	-0.537	-0.107	-0.0041	-0.0115	0.0110
329	-0.108	-0.532	-0.103	-0.0041	-0.0107	0.0095
330	-0.102	-0.530	-0.097	-0.0041	-0.0099	0.0081
340	0.027	0.055	0.057	-0.0424	-0.0013	0.0092
350	0.036	-0.000	-0.000	-0.0247	0.0033	-0.0646
360	0.047	-2.508	-0.079	-0.0106	0.0010	-0.0334
370	0.047	-2.580	-0.081	-0.0106	0.0010	-0.0332
380	0.047	-2.841	-0.089	-0.0105	0.0010	-0.0323

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
390	0.047	-2.911	-0.091	-0.0105	0.0010	-0.0321
400	0.051	-2.701	-0.085	-0.0054	-0.0024	0.0699
410	0.051	-2.549	-0.079	-0.0054	-0.0024	0.0701
420	0.051	-2.398	-0.074	-0.0054	-0.0024	0.0703
430	0.052	-2.254	-0.069	-0.0045	-0.0031	0.0850
440	0.052	-2.086	-0.064	-0.0046	-0.0030	0.0976
450	0.052	-1.875	-0.057	-0.0046	-0.0030	0.0977
460	0.053	-1.093	-0.033	-0.0046	-0.0030	0.0980
470	0.053	-0.881	-0.027	-0.0046	-0.0030	0.0981
480	0.057	-0.000	-0.000	-0.0049	-0.0020	0.0519
490	0.066	0.082	0.023	-0.0055	-0.0010	-0.0147
500	0.066	0.050	0.025	-0.0055	-0.0010	-0.0147
510	0.067	-0.067	0.034	-0.0055	-0.0010	-0.0145
520	0.067	-0.098	0.036	-0.0055	-0.0010	-0.0144
530	0.068	-0.118	0.038	-0.0055	-0.0011	-0.0073
538	0.031	-0.114	0.047	0.0027	-0.0040	-0.0123
539	0.023	-0.109	0.046	0.0027	-0.0032	-0.0135
540	0.019	-0.084	0.042	0.0022	-0.0019	-0.0156
550	0.019	-0.079	0.041	0.0021	-0.0018	-0.0158
560	0.019	-0.045	0.037	0.0021	-0.0018	-0.0158
570	0.018	0.082	0.023	0.0021	-0.0018	-0.0158
580	0.018	0.116	0.019	0.0021	-0.0018	-0.0158
590	0.004	-0.000	-0.000	-0.0005	-0.0013	0.0490
600	-0.004	-0.756	-0.018	-0.0021	-0.0025	0.0974
610	-0.004	-0.967	-0.023	-0.0021	-0.0025	0.0975
620	-0.005	-1.745	-0.043	-0.0022	-0.0025	0.0975
630	-0.005	-1.955	-0.049	-0.0022	-0.0025	0.0975
639	-0.004	-2.138	-0.055	-0.0023	-0.0036	0.0952
640	0.003	-2.216	-0.060	-0.0025	-0.0044	0.0934
650	0.069	-0.124	0.039	-0.0096	0.0001	0.0003
660	0.069	-0.124	0.038	-0.0096	0.0002	0.0003
670	0.069	-0.123	0.038	-0.0097	0.0002	0.0004
680	0.070	-0.116	0.037	-0.0138	0.0012	0.0060
690	0.076	-0.012	0.012	-0.0142	0.0034	0.0077
700	0.050	-0.329	0.024	-0.0048	0.0016	0.0060
710	0.046	-0.340	0.024	-0.0047	0.0016	0.0060
720	0.033	-0.376	0.024	-0.0044	0.0016	0.0060
730	0.030	-0.386	0.024	-0.0044	0.0016	0.0060
740	0.000	-0.000	0.002	0.0105	0.0006	0.0060
750	-0.002	0.017	-0.006	-0.0008	-0.0000	0.0060
760	-0.002	0.015	-0.007	-0.0008	-0.0001	0.0060
770	-0.002	0.010	-0.007	-0.0007	-0.0001	0.0060
780	-0.002	0.008	-0.007	-0.0007	-0.0001	0.0060
790	-0.000	-0.000	-0.016	-0.0063	-0.0002	0.0060
800	0.002	-0.138	-0.025	-0.0167	-0.0002	0.0060
810	0.002	-0.174	-0.025	-0.0167	-0.0002	0.0060
820	0.002	-0.189	-0.025	-0.0167	-0.0002	0.0060
830	0.078	-0.000	0.000	-0.0145	0.0029	-0.0052

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
840	0.086	-0.220	-0.017	-0.0150	0.0004	-0.0246
850	0.086	-0.273	-0.018	-0.0151	0.0004	-0.0245
860	0.086	-0.468	-0.021	-0.0151	0.0004	-0.0242
870	0.086	-0.521	-0.021	-0.0151	0.0004	-0.0241
880	0.090	-0.585	-0.021	-0.0153	-0.0005	-0.0008
890	0.094	-0.528	-0.016	-0.0156	-0.0013	0.0232
900	0.094	-0.478	-0.014	-0.0156	-0.0013	0.0233
910	0.094	-0.460	-0.013	-0.0156	-0.0013	0.0233
920	0.095	-0.273	-0.002	-0.0156	-0.0013	0.0236
930	0.095	-0.222	0.000	-0.0156	-0.0013	0.0237
940	0.102	-0.000	0.015	-0.0161	-0.0020	0.0178
950	0.112	0.060	0.035	-0.0168	-0.0004	-0.0022
960	0.093	-0.137	0.023	-0.0059	0.0029	-0.0022
970	0.089	-0.147	0.023	-0.0059	0.0029	-0.0022
980	0.073	-0.178	0.023	-0.0058	0.0029	-0.0022
990	0.069	-0.187	0.023	-0.0057	0.0029	-0.0022
1000	0.001	-0.000	0.001	0.0072	0.0015	-0.0022
1010	-0.006	0.024	-0.008	0.0002	-0.0001	-0.0022
1020	-0.006	0.025	-0.008	0.0002	-0.0001	-0.0022
1030	-0.005	0.026	-0.008	0.0002	-0.0001	-0.0022
1040	-0.005	0.026	-0.008	0.0002	-0.0001	-0.0022
1050	-0.001	-0.000	-0.017	-0.0080	-0.0005	-0.0022
1060	0.004	-0.142	-0.026	-0.0163	-0.0005	-0.0022
1070	0.005	-0.168	-0.026	-0.0163	-0.0005	-0.0022
1080	0.005	-0.182	-0.026	-0.0163	-0.0005	-0.0022
1090	0.121	-0.000	0.040	-0.0121	-0.0008	-0.0098
1100	0.128	-0.098	0.048	-0.0086	-0.0012	-0.0137
1110	0.128	-0.120	0.050	-0.0085	-0.0013	-0.0137
1118	0.067	-0.823	0.048	-0.0716	-0.0030	-0.0294
1119	0.070	-0.906	0.046	-0.0719	-0.0026	-0.0340
1120	0.071	-0.896	0.042	-0.0717	-0.0022	-0.0396
1130	0.071	-0.895	0.042	-0.0717	-0.0022	-0.0397
1140	0.071	-0.819	0.038	-0.0717	-0.0022	-0.0397
1150	0.071	-0.542	0.023	-0.0717	-0.0022	-0.0398
1160	0.071	-0.465	0.018	-0.0717	-0.0022	-0.0399
1170	0.062	-0.000	-0.001	-0.0711	-0.0009	-0.0103
1180	0.046	-0.897	-0.012	-0.0698	0.0001	0.0302
1190	0.033	-0.835	-0.006	-0.0688	0.0003	-0.0423
1200	0.033	-0.753	-0.006	-0.0688	0.0003	-0.0423
1210	0.033	-0.471	-0.004	-0.0688	0.0003	-0.0424
1220	0.033	-0.388	-0.003	-0.0688	0.0003	-0.0425
1230	0.027	-0.000	-0.000	-0.0684	0.0002	-0.0269
1240	0.023	0.103	0.002	-0.0679	0.0001	0.0001
1250	0.023	0.102	0.002	-0.0679	0.0001	0.0001
1260	0.023	0.102	0.002	-0.0679	0.0001	0.0001
1270	0.000	-0.000	0.000	-0.0659	-0.0001	0.0178
1280	-0.034	-0.000	-0.000	-0.0629	0.0000	-0.0253
1290	-0.055	-0.000	-0.000	-0.0611	-0.0000	0.0323

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1300	-0.061	-0.481	-0.000	-0.0606	-0.0000	0.0466
1310	-0.061	-0.571	-0.000	-0.0606	-0.0000	0.0465
1320	-0.061	-0.661	-0.000	-0.0606	-0.0000	0.0465
1330	-0.061	-0.731	-0.001	-0.0604	-0.0000	0.0416
1340	-0.062	-0.789	-0.001	-0.0580	-0.0000	-0.0287
1350	-0.062	-0.743	-0.001	-0.0580	-0.0000	-0.0287
1360	-0.062	-0.611	-0.001	-0.0580	-0.0000	-0.0288
1370	-0.062	-0.565	-0.001	-0.0580	-0.0000	-0.0289
1380	-0.062	-0.000	0.000	-0.0533	0.0002	-0.0191
1390	-0.063	-0.043	0.002	-0.0506	0.0002	0.0146
1400	-0.064	-0.065	0.003	-0.0503	0.0002	0.0130
1410	-0.064	-0.090	0.003	-0.0503	0.0002	0.0130
1420	-0.064	-0.115	0.003	-0.0503	0.0002	0.0129
1430	-0.069	-0.184	0.005	-0.0500	0.0001	0.0023
1460	-0.093	-0.000	-0.010	-0.0481	-0.0011	0.0321
1470	-0.099	-0.501	-0.020	-0.0477	-0.0005	0.0605
1480	-0.085	-0.625	-0.008	-0.0475	0.0004	0.0630
1490	-0.085	-0.747	-0.008	-0.0475	0.0004	0.0630
1500	-0.085	-0.965	-0.007	-0.0475	0.0004	0.0629
1510	-0.084	-0.800	-0.006	-0.0475	0.0004	0.0629
1520	-0.083	-0.708	-0.006	-0.0475	0.0004	0.0629
1530	-0.000	-0.000	-0.000	0.0012	0.0030	-0.0328
1540	-0.226	-0.215	0.076	0.0025	0.0104	0.0301
1550	-0.226	-0.180	0.062	0.0025	0.0103	0.0249
1558	-0.055	-0.180	0.034	0.0063	0.0074	0.0378
1559	-0.017	-0.164	0.025	0.0065	0.0068	0.0415
1560	0.000	-0.123	0.016	0.0066	0.0062	0.0434
1570	0.001	-0.000	0.001	0.0068	0.0038	0.0300
1580	0.002	0.054	-0.007	0.0070	0.0006	0.0035
1590	0.003	0.054	-0.007	0.0070	-0.0001	-0.0013
1595	0.005	0.040	-0.005	0.0071	-0.0007	-0.0044
1600	0.006	0.031	-0.004	0.0071	-0.0011	-0.0049
1604	0.006	0.023	-0.002	0.0071	-0.0015	-0.0046
1605	0.004	0.015	-0.001	0.0066	-0.0016	-0.0037
1610	0.000	0.000	-0.000	0.0000	-0.0000	-0.0000
1611	0.000	0.000	-0.000	0.0000	-0.0000	-0.0000
1620	0.000	0.000	-0.000	0.0000	-0.0000	-0.0000
1630	-0.244	-0.180	0.064	0.0016	0.0105	0.0174
1640	-0.257	-0.180	0.065	-0.0006	0.0110	0.0007
1650	-0.258	-0.180	0.064	-0.0006	0.0110	0.0006
1651	0.008	-0.539	-0.024	-0.0064	-0.0004	0.0103
1660	-0.261	-0.180	0.060	-0.0007	0.0110	0.0005
1670	-0.262	-0.180	0.059	-0.0007	0.0110	0.0005
1678	0.005	-0.178	-0.087	-0.0313	0.0165	-0.0417
1679	0.035	-0.190	-0.115	-0.0342	0.0166	-0.0411
1680	0.047	-0.218	-0.137	-0.0376	0.0168	-0.0398
1690	0.049	-0.403	-0.246	-0.0648	0.0207	-0.0197
1700	0.049	-0.435	-0.280	-0.0648	0.0207	-0.0196

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
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DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
1710	0.049	-0.539	-0.390	-0.0649	0.0207	-0.0196
1720	0.049	-0.570	-0.423	-0.0649	0.0207	-0.0196
1728	0.050	-0.607	-0.461	-0.0729	0.0230	-0.0246
1729	0.042	-0.626	-0.456	-0.0765	0.0244	-0.0287
1730	0.020	-0.634	-0.408	-0.0795	0.0260	-0.0340
1740	-0.029	-0.635	-0.303	-0.0839	0.0284	-0.0443
1750	-0.094	-0.635	-0.189	-0.0870	0.0300	-0.0525
1760	-0.208	-0.636	-0.010	-0.0885	0.0308	-0.0573
1770	-0.000	-0.000	-0.000	-0.0003	0.0001	-0.0001
1780	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1781	-0.000	-0.000	0.000	-0.0000	0.0000	-0.0000
1790	-0.265	-0.798	-0.012	-0.0944	0.0339	-0.0669
1800	-0.338	-1.002	-0.012	-0.0945	0.0339	-0.0671
1808	0.011	-0.140	0.003	0.0124	0.0005	-0.0766
1809	0.011	-0.181	0.003	0.0072	0.0005	-0.0816
1810	0.012	-0.263	0.002	0.0002	0.0006	-0.0853
1820	0.012	-0.264	0.002	0.0001	0.0006	-0.0853
1830	0.012	-0.429	0.001	0.0001	0.0006	-0.0853
1840	-0.226	-0.254	0.089	0.0025	0.0104	0.0302
1850	-0.411	-1.206	-0.012	-0.0945	0.0339	-0.0672
1859	-0.479	-1.448	-0.043	-0.0985	0.0387	-0.0796
1860	-0.508	-1.690	-0.121	-0.1002	0.0434	-0.0930
1870	-0.501	-2.543	-0.529	-0.1002	0.0565	-0.1075
1880	-0.501	-2.776	-0.651	-0.1002	0.0566	-0.1075
1890	-0.501	-3.008	-0.774	-0.1002	0.0566	-0.1075
1900	-0.501	-3.235	-0.893	-0.1002	0.0567	-0.1074
1910	-0.501	-3.235	-0.893	0.0000	0.0567	-0.1074
1920	-0.501	-3.462	-1.013	0.0000	0.0567	-0.1074
1929	-0.478	-3.656	-1.128	0.0030	0.0673	-0.0991
1930	-0.448	-3.723	-1.178	0.0071	0.0777	-0.0900
1938	-1.444	-3.056	-0.548	0.0416	0.1350	-0.0628
1939	-1.541	-2.950	-0.387	0.0424	0.1366	-0.0633
1940	-1.583	-2.806	-0.105	0.0426	0.1376	-0.0641
1950	-1.583	-2.804	-0.101	0.0426	0.1376	-0.0642
1960	-1.583	-2.668	0.189	0.0426	0.1376	-0.0642
1970	-1.583	-2.668	0.189	0.0000	0.0000	-0.0000
1980	-1.583	-2.668	0.189	0.0000	0.0000	0.0000
1989	-1.585	-2.669	0.189	0.0019	0.0008	0.0015
1990	-1.581	-2.670	0.184	0.0044	0.0006	0.0044
1998	-0.278	-2.491	-0.389	0.0256	-0.0122	0.0892
1999	-0.097	-2.398	-0.424	0.0261	-0.0168	0.1050
2000	-0.010	-2.178	-0.407	0.0263	-0.0208	0.1192
2010	-0.010	-2.175	-0.407	0.0263	-0.0208	0.1193
2020	-0.010	-1.923	-0.363	0.0263	-0.0208	0.1193
2030	-0.010	-1.923	-0.363	-0.0000	-0.0208	0.1193
2040	-0.010	-1.671	-0.319	-0.0000	-0.0208	0.1194
2050	-0.226	-0.254	0.089	0.0025	0.0104	0.0302
2060	-0.226	-0.312	0.109	0.0025	0.0104	0.0302

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
2070	-0.226	-0.358	0.125	0.0025	0.0104	0.0302
2080	-0.226	-0.358	0.125	-0.0000	0.0104	0.0302
2090	-0.226	-0.404	0.140	-0.0000	0.0104	0.0302
2099	-0.238	-0.431	0.149	-0.0018	0.0097	0.0270
2100	-0.263	-0.441	0.150	-0.0033	0.0089	0.0231
2109	1.000	-0.556	-0.328	-0.0147	-0.0142	-0.0742
2110	1.030	-0.628	-0.320	-0.0147	-0.0145	-0.0751
2120	1.030	-0.629	-0.320	-0.0147	-0.0145	-0.0752
2130	1.030	-0.742	-0.298	-0.0147	-0.0145	-0.0752
2140	1.030	-0.742	-0.298	-0.0000	-0.0000	-0.0000
2150	1.030	-0.742	-0.298	-0.0000	-0.0000	-0.0000
2159	1.030	-0.743	-0.297	-0.0029	-0.0006	-0.0009
2160	1.030	-0.746	-0.294	-0.0051	-0.0012	-0.0017
2169	0.039	-0.933	-0.001	0.0004	-0.0004	-0.0824
2170	0.012	-0.854	-0.001	0.0003	0.0005	-0.0853
2180	0.012	-0.852	-0.001	0.0003	0.0006	-0.0853
2190	0.012	-0.724	-0.001	0.0003	0.0006	-0.0853
2200	0.012	-0.724	-0.001	0.0000	0.0006	-0.0853
2201	0.025	0.007	0.021	-0.0120	-0.0020	-0.0022
2210	0.012	-0.595	0.000	0.0000	0.0006	-0.0853
2220	0.012	-0.429	0.001	0.0001	0.0006	-0.0853
2721	-0.226	-0.193	0.068	0.0025	0.0103	0.0250
3101	0.028	0.013	0.057	-0.0424	-0.0013	0.0092
3141	-0.151	-0.635	-0.099	-0.0885	0.0308	-0.0573
4001	0.049	-2.250	-0.069	-0.0045	-0.0031	0.0850
4020	0.128	-0.141	0.052	-0.0085	-0.0013	-0.0137
4030	0.179	-0.358	0.062	0.0106	0.0017	-0.0023
4038	0.184	-0.363	0.056	0.0126	0.0025	-0.0018
4039	0.184	-0.354	0.048	0.0149	0.0055	0.0000
4040	0.173	-0.321	0.039	0.0189	0.0083	0.0010
4060	0.001	-0.000	0.001	0.0054	0.0072	-0.0010
4080	-0.154	-0.000	-0.107	0.0149	0.0035	-0.0065
4098	-0.243	0.397	-0.150	0.0213	0.0028	-0.0087
4099	-0.244	0.431	-0.154	0.0216	-0.0004	-0.0090
4100	-0.239	0.429	-0.151	0.0210	-0.0037	-0.0107
4120	-0.198	-0.000	-0.032	0.0197	-0.0059	-0.0382
4140	-0.098	-2.008	0.141	0.0165	-0.0005	0.0061
4160	0.001	-0.000	-0.000	0.0133	0.0061	0.0137
4180	0.185	-0.000	0.254	0.0075	-0.0268	0.0122
4198	0.231	-0.124	0.974	0.0060	-0.0281	-0.0131
4199	0.252	-0.145	1.019	0.0058	-0.0199	-0.0137
4200	0.282	-0.145	1.028	0.0060	-0.0110	-0.0136
4220	0.365	-0.000	0.991	0.0081	0.0011	-0.0132
4240	0.115	-0.000	0.907	-0.0286	0.0063	-0.0122
4260	-0.000	-0.000	0.712	0.0076	-0.0008	-0.0100
4280	0.001	-0.000	0.524	-0.0020	0.0002	-0.0077
4300	-0.000	-0.000	0.343	0.0005	-0.0000	-0.0054
4320	0.000	-0.000	0.168	-0.0001	0.0000	-0.0031

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
4340	-0.000	-0.000	0.000	0.0000	-0.0000	-0.0008
4360	-0.000	-0.000	-0.159	-0.0000	0.0000	0.0015
4380	0.000	-0.000	-0.325	0.0000	-0.0000	0.0038
4400	-0.000	-0.000	-0.497	-0.0000	0.0000	0.0060
4420	0.001	-0.000	-0.676	0.0000	-0.0002	0.0083
4440	-0.000	-0.000	-0.861	-0.0001	0.0006	0.0106
4460	0.035	-0.000	-1.052	0.0002	-0.0032	0.0129
4480	0.262	-0.000	-1.234	0.0287	0.0038	0.0150
4498	0.129	0.315	-1.269	0.0124	0.0122	0.0154
4499	0.098	0.326	-1.259	0.0132	0.0197	0.0151
4500	0.077	0.309	-1.217	0.0140	0.0269	0.0138
4520	0.049	0.142	-0.770	0.0177	0.0319	0.0078
4540	-0.005	-0.000	-0.017	0.0244	0.0175	0.0109
4558	-0.082	-0.626	0.096	0.0340	-0.0048	0.0112
4559	-0.082	-0.619	0.085	0.0349	-0.0050	0.0099
4560	-0.074	-0.559	0.077	0.0358	-0.0047	0.0095
4580	-0.003	-0.000	0.043	0.0228	-0.0032	0.0096
4600	-0.003	-0.000	-0.048	-0.0228	0.0032	0.0096
4618	-0.074	-0.560	-0.082	-0.0359	0.0047	0.0096
4619	-0.082	-0.620	-0.090	-0.0351	0.0050	0.0100
4620	-0.082	-0.627	-0.101	-0.0342	0.0049	0.0113
4640	-0.005	-0.000	0.017	-0.0248	-0.0180	0.0109
4660	0.049	0.143	0.789	-0.0183	-0.0327	0.0081
4678	0.077	0.318	1.247	-0.0148	-0.0275	0.0146
4679	0.099	0.337	1.291	-0.0138	-0.0201	0.0163
4680	0.130	0.325	1.301	-0.0130	-0.0124	0.0167
4700	0.262	-0.000	1.266	-0.0292	-0.0036	0.0174
4720	0.000	-0.000	1.084	-0.0001	0.0034	0.0207
4740	-0.010	-0.000	0.893	0.0000	-0.0009	0.0244
4760	0.002	-0.000	0.709	0.0000	0.0002	0.0280
4780	-0.000	-0.000	0.531	-0.0001	-0.0000	0.0317
4800	0.000	-0.000	0.374	0.0298	0.0000	0.0350
4820	-0.000	-0.000	0.277	-0.0390	-0.0000	0.0371
4840	-0.000	-0.000	0.116	0.0279	0.0000	0.0408
4860	0.000	-0.000	-0.000	-0.0021	-0.0000	0.0435
4880	-0.000	-0.000	-0.125	-0.0193	0.0000	0.0463
4900	0.000	-0.000	-0.298	0.0045	-0.0000	0.0499
4920	-0.001	-0.000	-0.478	0.0010	0.0001	0.0536
4940	0.003	-0.000	-0.664	-0.0087	-0.0004	0.0572
4950	-0.000	-0.000	-0.858	0.0339	0.0014	0.0609
4960	0.024	-0.000	-0.972	-0.0243	-0.0055	0.0630
4978	0.459	-1.217	-1.045	-0.0233	-0.0182	0.0643
4979	0.447	-1.251	-1.062	-0.0151	-0.0219	0.0634
4980	0.346	-1.257	-1.084	-0.0079	-0.0235	0.0626
4998	-0.304	-1.237	-1.142	-0.0044	-0.0290	0.0626
4999	-0.420	-1.184	-1.126	-0.0033	-0.0306	0.0621
5000	-0.465	-1.067	-1.067	-0.0027	-0.0334	0.0604
5001	0.067	-0.113	0.038	-0.0055	-0.0011	-0.0073

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
 Job:: R-52-2-P (ID-28711)
 Licensed To:: Edit name in <system>\company.txt

DISPLACEMENTS REPORT: Nodal Movements
 CASE 9 (SUS) W+P2

Node	DX mm.	DY mm.	DZ mm.	RX deg.	RY deg.	RZ deg.
5020	-0.422	-0.000	-0.365	-0.0049	-0.0267	0.0262
5040	-0.302	-0.000	0.048	-0.0112	0.0059	0.0060
5058	-0.256	0.101	-0.122	-0.0137	0.0077	0.0003
5059	-0.256	0.093	-0.125	-0.0141	0.0073	-0.0077
5060	-0.277	0.082	-0.104	-0.0149	0.0070	-0.0152
5078	-0.465	0.060	0.074	-0.0165	0.0067	-0.0151
5079	-0.486	0.049	0.101	-0.0182	0.0062	-0.0077
5080	-0.486	0.040	0.105	-0.0190	0.0054	-0.0008
5100	-0.458	-0.000	0.033	-0.0226	0.0041	-0.0123
5120	-0.297	-0.000	-0.006	-0.0439	-0.0008	0.0055
5140	-0.139	-0.000	0.000	-0.0651	0.0002	-0.0099
5150	-0.054	-1.009	-0.004	-0.0770	-0.0000	0.0114
5160	-0.047	-0.950	-0.003	-0.0779	-0.0001	0.0196
5180	-0.047	-0.920	-0.003	-0.0779	-0.0001	0.0197
5200	-0.047	-0.889	-0.003	-0.0780	-0.0001	0.0198
5220	0.000	-0.000	0.000	-0.0845	-0.0002	0.0290
5240	0.113	-0.000	-0.043	-0.0989	0.0028	-0.0405
5258	0.147	-1.048	-0.086	-0.1031	0.0009	-0.0781
5259	0.149	-1.129	-0.083	-0.0999	-0.0028	-0.0923
5260	0.142	-1.018	-0.076	-0.0983	-0.0065	-0.0980
5268	0.009	0.647	-0.044	-0.1032	-0.0084	-0.1306
5269	-0.005	0.736	-0.034	-0.1001	-0.0073	-0.1368
5270	-0.006	0.541	-0.020	-0.0961	-0.0061	-0.1518
5280	0.000	-0.000	0.000	-0.0961	-0.0060	-0.1580
5300	0.029	-2.419	0.086	-0.0961	-0.0060	-0.1740
5320	0.044	-3.796	0.133	-0.0961	-0.0060	-0.1758
5340	0.044	-4.068	0.143	-0.0961	-0.0060	-0.1758
9301	0.112	0.043	0.035	-0.0168	-0.0004	-0.0022

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 9

RESTRAINT SUMMARY

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

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CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45
Job:: R-52-2-P (ID-28711)
Licensed To:: Edit name in <system>\company.txt

LISTING OF STATIC LOAD CASES FOR THIS ANALYSIS

- 1 (HYD) WW+HP
- 2 (OPE) W+D1+T1+P1
- 3 (Alt-SUS) W+P1
- 4 (OPE) W+T2+P2
- 5 (Alt-SUS) W+P2
- 6 (OPE) W+T3+P1
- 7 (Alt-SUS) W+P1
- 8 (SUS) W+P1
- 9 (SUS) W+P2
- 10 (EXP) L10=L2-L8
- 11 (EXP) L11=L4-L9
- 12 (EXP) L12=L6-L8
- 13 (EXP) L13=L10-L12
- 14 (EXP) L14=L11-L12

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

LOAD CASE DEFINITION KEY

CASE 1 (HYD) WW+HP

CASE 2 (OPE) W+D1+T1+P1

CASE 4 (OPE) W+T2+P2

CASE 6 (OPE) W+T3+P1

CASE 8 (SUS) W+P1

CASE 9 (SUS) W+P2

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
30		TYPE=Rigid +Y; Rigid GUI; Rigid LIM;Tag=SL-001					
	1(HYD)	-13546	-4146	-420	0	0	0
	2(OPE)	-17277	0	-1370	0	0	0
	4(OPE)	-16529	-4778	-1724	0	0	0
	6(OPE)	15260	-4820	-390	0	0	0
	8(SUS)	-13437	-4819	-476	0	0	0
	9(SUS)	-10677	-4818	-466	0	0	0
	MAX	-17277/L2	-4820/L6	-1724/L4			
60		TYPE=Rigid +Y;Tag=SL-004					
	1(HYD)	4	-40	10	0	0	0
	2(OPE)	366	-1229	78	0	0	0
	4(OPE)	70	-247	29	0	0	0
	6(OPE)	-5	-26	6	0	0	0
	8(SUS)	2	-35	11	0	0	0
	9(SUS)	0	-36	11	0	0	0
	MAX	366/L2	-1229/L2	78/L2			
100		TYPE=Rigid +Y;Tag=SL-005 (4" PIPE)					
	1(HYD)	543	-1690	50	0	0	0
	2(OPE)	681	-2275	36	0	0	0
	4(OPE)	591	-1970	63	0	0	0
	6(OPE)	-580	-1982	47	0	0	0
	8(SUS)	636	-1990	62	0	0	0
	9(SUS)	423	-1985	60	0	0	0
	MAX	681/L2	-2275/L2	63/L4			
110		TYPE=Rigid +Y; Rigid GUI;Tag=SL- 006 (4" PIPE)					
	1(HYD)	417	-1316	-16	0	0	0
	2(OPE)	510	-1523	103	0	0	0
	4(OPE)	480	-1597	140	0	0	0
	6(OPE)	-543	-1620	-16	0	0	0

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	8(SUS)	508	-1586	-26	0	0	0
	9(SUS)	458	-1604	-18	0	0	0
	MAX	-543/L6	-1620/L6	140/L4			
160		TYPE=Rigid +Y;Tag=SL-007					
	1(HYD)	799	-2670	56	0	0	0
	2(OPE)	884	-2994	4	0	0	0
	4(OPE)	901	-2984	5	0	0	0
	6(OPE)	-914	-2769	-68	0	0	0
	8(SUS)	848	-3015	53	0	0	0
	9(SUS)	789	-2890	-19	0	0	0
	MAX	-914/L6	-3015/L8	-68/L6			
220		TYPE=Rigid +Y; Rigid GUI;Tag=PS- 010 (4" PIPE)					
	1(HYD)	1480	-4125	417	0	0	0
	2(OPE)	2033	-5472	1267	0	0	0
	4(OPE)	1993	-5419	1148	0	0	0
	6(OPE)	-1315	-3977	68	0	0	0
	8(SUS)	1538	-4436	388	0	0	0
	9(SUS)	1531	-4393	340	0	0	0
	MAX	2033/L2	-5472/L2	1267/L2			
310		TYPE=Rigid +Y; Rigid GUI;Tag=PS- 010 (3" BY PASS LINE)					
	1(HYD)	213	-708	-410	0	0	0
	2(OPE)	0	0	-1303	0	0	0
	4(OPE)	0	0	-1186	0	0	0
	6(OPE)	-508	-1519	-71	0	0	0
	8(SUS)	303	-923	-383	0	0	0
	9(SUS)	-181	-1006	-332	0	0	0
	MAX	-508/L6	-1519/L6	-1303/L2			
350		TYPE=Rigid +Y;Tag=SL-009					
	1(HYD)	1417	-4352	-294	0	0	0
	2(OPE)	1447	-4822	-16	0	0	0
	4(OPE)	1452	-4819	-27	0	0	0
	6(OPE)	-1517	-4752	197	0	0	0
	8(SUS)	1464	-4677	-246	0	0	0
	9(SUS)	1433	-4745	-13	0	0	0
	MAX	-1517/L6	-4822/L2	-294/L1			

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
480		TYPE=Rigid +Y; Rigid GUI;Tag=SL-010					
	1(HYD)	2325	-7089	148	0	0	0
	2(OPE)	2200	-7331	1395	0	0	0
	4(OPE)	2212	-7365	1559	0	0	0
	6(OPE)	-2409	-7499	-138	0	0	0
	8(SUS)	2375	-7512	5	0	0	0
	9(SUS)	2510	-7496	-37	0	0	0
	MAX	2510/L9	-7512/L8	1559/L4			
590		TYPE=Rigid +Y;Tag=SL-010 (COMMON SLEEPER FOR FUTURE)					
	1(HYD)	1591	-4923	10	0	0	0
	2(OPE)	1851	-5383	56	0	0	0
	4(OPE)	1609	-5343	48	0	0	0
	6(OPE)	-1714	-5205	-18	0	0	0
	8(SUS)	1634	-5209	-13	0	0	0
	9(SUS)	719	-5209	-25	0	0	0
	MAX	1851/L2	-5383/L2	56/L2			
740		TYPE=Rigid +Y;Tag=SL-016					
	1(HYD)	715	-2983	-339	0	0	0
	2(OPE)	942	-3176	-113	0	0	0
	4(OPE)	1013	-3180	-125	0	0	0
	6(OPE)	-824	-3197	269	0	0	0
	8(SUS)	316	-3198	4	0	0	0
	9(SUS)	23	-3197	419	0	0	0
	MAX	1013/L4	-3198/L8	419/L9			
790		TYPE=Rigid +Y;Tag=SL-017					
	1(HYD)	-275	-2152	-541	0	0	0
	2(OPE)	702	-2286	-174	0	0	0
	4(OPE)	638	-2285	-178	0	0	0
	6(OPE)	256	-2281	593	0	0	0
	8(SUS)	-128	-2281	-600	0	0	0
	9(SUS)	-11	-2281	-648	0	0	0
	MAX	702/L2	-2286/L2	-648/L9			
830		TYPE=Rigid +Y; Rigid GUI;Tag=SL-011					
	1(HYD)	1923	-5978	982	0	0	0

CAESAR II Ver.13.00.00.0890, (Build 220811) Date: MAR 26, 2025 Time: 8:45

Job:: R-52-2-P (ID-28711)

Licensed To:: Edit name in <system>\company.txt

RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	2(OPE)	1909	-6361	64	0	0	0
	4(OPE)	1913	-6362	-180	0	0	0
	6(OPE)	-2025	-6359	-1094	0	0	0
	8(SUS)	2000	-6351	827	0	0	0
	9(SUS)	1947	-6358	349	0	0	0
	MAX	-2025/L6	-6362/L4	-1094/L6			
940		TYPE=Rigid +Y;Tag=SL-012					
	1(HYD)	1198	-3736	81	0	0	0
	2(OPE)	1372	-3982	-4	0	0	0
	4(OPE)	1196	-3982	-33	0	0	0
	6(OPE)	-1258	-3982	-231	0	0	0
	8(SUS)	1214	-3983	53	0	0	0
	9(SUS)	1166	-3982	174	0	0	0
	MAX	1372/L2	-3983/L8	-231/L6			
1000		TYPE=Rigid +Y;Tag=SL-014					
	1(HYD)	426	-1535	22	0	0	0
	2(OPE)	538	-1726	-10	0	0	0
	4(OPE)	517	-1726	-88	0	0	0
	6(OPE)	-517	-1729	-75	0	0	0
	8(SUS)	472	-1729	20	0	0	0
	9(SUS)	94	-1729	238	0	0	0
	MAX	538/L2	-1729/L8	238/L9			
1050		TYPE=Rigid +Y;Tag=SL-015					
	1(HYD)	-126	-1477	-96	0	0	0
	2(OPE)	510	-1591	-11	0	0	0
	4(OPE)	486	-1590	-145	0	0	0
	6(OPE)	72	-1590	411	0	0	0
	8(SUS)	-194	-1590	-47	0	0	0
	9(SUS)	-34	-1590	-482	0	0	0
	MAX	510/L2	-1591/L2	-482/L9			
1090		TYPE=Rigid +Y;Tag=SL-014					
	1(HYD)	256	-805	8	0	0	0
	2(OPE)	311	-905	-11	0	0	0
	4(OPE)	271	-904	-22	0	0	0
	6(OPE)	-284	-903	-43	0	0	0
	8(SUS)	275	-902	-12	0	0	0
	9(SUS)	265	-903	87	0	0	0
	MAX	311/L2	-905/L2	87/L9			

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RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
1170		TYPE=Rigid +Y;Tag=PS-010 (3" PIPE)					
	1(HYD)	615	-1933	113	0	0	0
	2(OPE)	541	-1800	-22	0	0	0
	4(OPE)	573	-1865	-21	0	0	0
	6(OPE)	-686	-2084	-17	0	0	0
	8(SUS)	693	-2060	76	0	0	0
	9(SUS)	622	-2077	-13	0	0	0
	MAX	693/L8	-2084/L6	113/L1			
1230		TYPE=Rigid +Y; Rigid GUI;Tag=SL- 018 (3" PIPE)					
	1(HYD)	603	-1926	28	0	0	0
	2(OPE)	629	-2119	-1	0	0	0
	4(OPE)	655	-2104	21	0	0	0
	6(OPE)	-626	-2050	-26	0	0	0
	8(SUS)	647	-2056	7	0	0	0
	9(SUS)	618	-2051	-3	0	0	0
	MAX	655/L4	-2119/L2	28/L1			
1270		TYPE=Rigid +Y;Tag=SL-006 (3" PIPE)					
	1(HYD)	311	-903	-10	0	0	0
	2(OPE)	333	-997	12	0	0	0
	4(OPE)	315	-1000	7	0	0	0
	6(OPE)	-340	-1017	8	0	0	0
	8(SUS)	321	-1015	-2	0	0	0
	9(SUS)	11	-1016	1	0	0	0
	MAX	-340/L6	-1017/L6	12/L2			
1280		TYPE=Rigid +Y;Tag=SL-005 (3" PIPE)					
	1(HYD)	219	-691	1	0	0	0
	2(OPE)	248	-778	17	0	0	0
	4(OPE)	258	-794	10	0	0	0
	6(OPE)	20	-791	-1	0	0	0
	8(SUS)	-220	-791	0	0	0	0
	9(SUS)	-234	-791	-0	0	0	0
	MAX	258/L4	-794/L4	17/L2			
1290		TYPE=Rigid +Y; Rigid GUI;Tag=SL- 019					
	1(HYD)	-282	-1013	1	0	0	0

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RESTRAINT SUMMARY REPORT: Loads On Restraints

Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	2(OPE)	371	-1130	-95	0	0	0
	4(OPE)	330	-1085	-29	0	0	0
	6(OPE)	321	-1082	-3	0	0	0
	8(SUS)	-288	-1083	0	0	0	0
	9(SUS)	-326	-1083	-0	0	0	0
	MAX	371/L2	-1130/L2	-95/L2			
1380		TYPE=Rigid +Y;Tag=SL-020					
	1(HYD)	-242	-917	-1	0	0	0
	2(OPE)	150	-812	-222	0	0	0
	4(OPE)	151	-986	-217	0	0	0
	6(OPE)	276	-1004	33	0	0	0
	8(SUS)	-277	-1001	-0	0	0	0
	9(SUS)	-284	-1001	2	0	0	0
	MAX	-284/L9	-1004/L6	-222/L2			
1460		TYPE=Rigid +Y;Tag=SL-003					
	1(HYD)	-381	-1476	-93	0	0	0
	2(OPE)	-494	-1834	-207	0	0	0
	4(OPE)	-423	-1566	-173	0	0	0
	6(OPE)	417	-1588	109	0	0	0
	8(SUS)	-427	-1585	-80	0	0	0
	9(SUS)	-453	-1585	-49	0	0	0
	MAX	-494/L2	-1834/L2	-207/L2			
1530		TYPE=Rigid Y; Rigid GUI w/gap; Rigid LIM w/gap;Tag=SL-002					
	1(HYD)	-127	-962	29	0	0	0
	2(OPE)	-1340	633	775	0	0	0
	4(OPE)	-1225	-1139	193	0	0	0
	6(OPE)	170	-1014	-204	0	0	0
	8(SUS)	-116	-1031	10	0	0	0
	9(SUS)	-83	-1030	-23	0	0	0
	MAX	-1340/L2	-1139/L4	775/L2			
1570		TYPE=Rigid +Y;Tag=SL-001					
	1(HYD)	-552	-2258	-1	0	0	0
	2(OPE)	0	0	0	0	0	0
	4(OPE)	-1118	-3572	-304	0	0	0
	6(OPE)	632	-2343	209	0	0	0
	8(SUS)	-119	-2405	74	0	0	0
	9(SUS)	154	-2403	117	0	0	0
	MAX	-1118/L4	-3572/L4	-304/L4			

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Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
1611		TYPE=Displ. Reaction;					
	1(HYD)	833	220	75	125	-239	-27
	2(OPE)	1482	-3239	-167	-1229	-781	1679
	4(OPE)	1095	533	428	194	-80	-52
	6(OPE)	-297	215	-125	131	67	-27
	8(SUS)	421	227	9	132	-123	-28
	9(SUS)	151	227	-33	133	-47	-28
	MAX	1482/L2	-3239/L2	428/L4	-1229/L2	-781/L2	1679/L2
1620		TYPE=Rigid ANC;					
	1(HYD)	833	220	75	125	-239	-27
	2(OPE)	1482	-3239	-167	-1229	-781	1679
	4(OPE)	1095	533	428	194	-80	-52
	6(OPE)	-297	215	-125	131	67	-27
	8(SUS)	421	227	9	132	-123	-28
	9(SUS)	151	227	-33	133	-47	-28
	MAX	1482/L2	-3239/L2	428/L4	-1229/L2	-781/L2	1679/L2
1780		TYPE=Rigid ANC;					
	1(HYD)	-638	-3249	363	-3328	963	-936
	2(OPE)	-2458	-7587	1092	-8643	2684	-3011
	4(OPE)	-787	-2588	1648	-2854	1237	-853
	6(OPE)	-747	-3751	332	-3859	1125	-1118
	8(SUS)	-722	-3681	408	-3799	1108	-1109
	9(SUS)	-723	-3685	399	-3803	1110	-1113
	MAX	-2458/L2	-7587/L2	1648/L4	-8643/L2	2684/L2	-3011/L2
1781		TYPE=Displ. Reaction;					
	1(HYD)	-638	-3249	363	-3328	963	-936
	2(OPE)	-2458	-7587	1092	-8643	2684	-3011
	4(OPE)	-787	-2588	1648	-2854	1237	-853
	6(OPE)	-747	-3751	332	-3859	1125	-1118
	8(SUS)	-722	-3681	408	-3799	1108	-1109
	9(SUS)	-723	-3685	399	-3803	1110	-1113
	MAX	-2458/L2	-7587/L2	1648/L4	-8643/L2	2684/L2	-3011/L2
1830		TYPE=Rigid ANC;					
	1(HYD)	41	467	65	165	17	14
	2(OPE)	15	-1079	-281	-602	324	-100
	4(OPE)	182	569	211	195	241	-58
	6(OPE)	18	482	66	178	-39	28
	8(SUS)	43	493	70	180	15	15
	9(SUS)	40	492	70	180	9	16
	MAX	182/L4	-1079/L2	-281/L2	-602/L2	324/L2	-100/L2

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
1900		TYPE=Rigid X; Rigid RZ; Rigid RY; Rigid Z; Rigid Y;					
	1(HYD)	398	824	-371	0	-632	-388
	2(OPE)	991	4395	-1206	0	-1574	-1282
	4(OPE)	993	843	-1561	0	-1577	-1279
	6(OPE)	429	1008	-351	0	-681	-379
	8(SUS)	462	998	-421	0	-733	-431
	9(SUS)	458	1000	-412	0	-728	-426
	MAX	993/L4	4395/L2	-1561/L4		-1577/L4	-1282/L2
1910		TYPE=Bi-linear RX2;					
	1(HYD)	0	0	0	334	0	0
	2(OPE)	0	0	0	-3797	0	0
	4(OPE)	0	0	0	2403	0	0
	6(OPE)	0	0	0	209	0	0
	8(SUS)	0	0	0	337	0	0
	9(SUS)	0	0	0	320	0	0
	MAX				-3797/L2		
1960		TYPE=Rigid X; Rigid Z; Rigid Y;					
	1(HYD)	398	-498	-371	0	0	0
	2(OPE)	991	2821	-1206	0	0	0
	4(OPE)	993	-731	-1561	0	0	0
	6(OPE)	429	-566	-351	0	0	0
	8(SUS)	462	-575	-421	0	0	0
	9(SUS)	458	-574	-412	0	0	0
	MAX	993/L4	2821/L2	-1561/L4			
1970		TYPE=Bi-linear RY2; Bi-linear RX2; Bi-linear RZ2;					
	1(HYD)	0	0	0	128	134	-29
	2(OPE)	0	0	0	-4285	247	-1501
	4(OPE)	0	0	0	-2205	100	-758
	6(OPE)	0	0	0	332	153	19
	8(SUS)	0	0	0	185	159	-26
	9(SUS)	0	0	0	204	159	-21
	MAX				-4285/L2	247/L2	-1501/L2
2020		TYPE=Rigid X; Rigid RZ; Rigid RY; Rigid Z; Rigid Y;					
	1(HYD)	398	-1854	-371	0	187	-591
	2(OPE)	991	1205	-1206	0	550	-372
	4(OPE)	993	-2347	-1561	0	698	-1113

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 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	6(OPE)	429	-2182	-351	0	193	-646
	8(SUS)	462	-2192	-421	0	213	-673
	9(SUS)	458	-2190	-412	0	210	-670
	MAX	993/L4	-2347/L4	-1561/L4		698/L4	-1113/L4
2030		TYPE=Bi-linear RX2;					
	1(HYD)	0	0	0	-1	0	0
	2(OPE)	0	0	0	81	0	0
	4(OPE)	0	0	0	61	0	0
	6(OPE)	0	0	0	-9	0	0
	8(SUS)	0	0	0	-3	0	0
	9(SUS)	0	0	0	-5	0	0
	MAX				81/L2		
2050		TYPE=Rigid ANC;					
	1(HYD)	41	-847	65	0	-5	116
	2(OPE)	15	-2479	-281	0	-62	499
	4(OPE)	182	-831	211	0	-38	-195
	6(OPE)	18	-918	66	0	5	186
	8(SUS)	43	-907	70	0	-5	129
	9(SUS)	40	-908	70	0	-4	134
	MAX	182/L4	-2479/L2	-281/L2	0/L9	-62/L2	499/L2
2070		TYPE=Rigid X; Rigid RZ; Rigid RY; Rigid Z; Rigid Y;					
	1(HYD)	-41	664	-65	0	18	32
	2(OPE)	-15	2290	281	0	6	-29
	4(OPE)	-182	642	-211	0	79	339
	6(OPE)	-18	729	-66	0	8	-24
	8(SUS)	-43	718	-70	0	19	30
	9(SUS)	-40	719	-70	0	18	25
	MAX	-182/L4	2290/L2	281/L2		79/L4	339/L4
2080		TYPE=Bi-linear RX2;					
	1(HYD)	0	0	0	-44	0	0
	2(OPE)	0	0	0	-1508	0	0
	4(OPE)	0	0	0	296	0	0
	6(OPE)	0	0	0	-62	0	0
	8(SUS)	0	0	0	-49	0	0
	9(SUS)	0	0	0	-48	0	0
	MAX				-1508/L2		
2130		TYPE=Rigid X; Rigid Z; Rigid Y;					
	1(HYD)	-41	198	-65	0	0	0

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 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	2(OPE)	-15	1788	281	0	0	0
	4(OPE)	-182	140	-211	0	0	0
	6(OPE)	-18	227	-66	0	0	0
	8(SUS)	-43	216	-70	0	0	0
	9(SUS)	-40	217	-70	0	0	0
	MAX	-182/L4	1788/L2	281/L2			
2140		TYPE=Bi-linear RY2; Bi-linear RX2; Bi-linear RZ2;					
	1(HYD)	0	0	0	-69	-26	-40
	2(OPE)	0	0	0	-1517	248	96
	4(OPE)	0	0	0	-249	37	137
	6(OPE)	0	0	0	-84	-50	-80
	8(SUS)	0	0	0	-77	-29	-47
	9(SUS)	0	0	0	-79	-31	-50
	MAX				-1517/L2	248/L2	137/L4
2190		TYPE=Rigid X; Rigid RZ; Rigid RY; Rigid Z; Rigid Y;					
	1(HYD)	-41	-284	-65	0	-30	58
	2(OPE)	-15	1268	281	0	-268	-133
	4(OPE)	-182	-380	-211	0	-283	150
	6(OPE)	-18	-293	-66	0	26	47
	8(SUS)	-43	-304	-70	0	-29	63
	9(SUS)	-40	-303	-70	0	-23	61
	MAX	-182/L4	1268/L2	281/L2		-283/L4	150/L4
2200		TYPE=Bi-linear RX2;					
	1(HYD)	0	0	0	165	0	0
	2(OPE)	0	0	0	-602	0	0
	4(OPE)	0	0	0	195	0	0
	6(OPE)	0	0	0	178	0	0
	8(SUS)	0	0	0	180	0	0
	9(SUS)	0	0	0	180	0	0
	MAX				-602/L2		
4060		TYPE=Rigid +Y;Tag=RA\SL-001					
	1(HYD)	304	-1031	3	0	0	0
	2(OPE)	347	-1032	-74	0	0	0
	4(OPE)	331	-1032	-125	0	0	0
	6(OPE)	-293	-1032	-18	0	0	0
	8(SUS)	319	-1032	-36	0	0	0
	9(SUS)	126	-1032	97	0	0	0
	MAX	347/L2	-1032/L8	-125/L4			

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 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
4080		TYPE=Rigid +Y;Tag=RA\SL-002					
	1(HYD)	-73	-393	-76	0	0	0
	2(OPE)	93	-394	-76	0	0	0
	4(OPE)	43	-394	-109	0	0	0
	6(OPE)	66	-394	86	0	0	0
	8(SUS)	-75	-394	-75	0	0	0
	9(SUS)	-105	-394	-73	0	0	0
	MAX	-105/L9	-394/L6	-109/L4			
4120		TYPE=Rigid +Y;Tag=RA\SL-003					
	1(HYD)	-298	-1179	-104	0	0	0
	2(OPE)	26	-1179	-395	0	0	0
	4(OPE)	-8	-1179	-322	0	0	0
	6(OPE)	283	-1179	185	0	0	0
	8(SUS)	-231	-1179	-10	0	0	0
	9(SUS)	-344	-1179	-56	0	0	0
	MAX	-344/L9	-1179/L1	-395/L2			
4160		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-005					
	1(HYD)	172	-1465	-26	0	0	0
	2(OPE)	801	-1466	-892	0	0	0
	4(OPE)	439	-1466	117	0	0	0
	6(OPE)	-190	-1466	-0	0	0	0
	8(SUS)	7	-1466	2	0	0	0
	9(SUS)	169	-1466	-34	0	0	0
	MAX	801/L2	-1466/L6	-892/L2			
4180		TYPE=Rigid +Y;Tag=RA\SL-007					
	1(HYD)	96	-1082	274	0	0	0
	2(OPE)	257	-1082	256	0	0	0
	4(OPE)	115	-1082	295	0	0	0
	6(OPE)	-77	-1082	-322	0	0	0
	8(SUS)	239	-1082	1	0	0	0
	9(SUS)	178	-1082	244	0	0	0
	MAX	257/L2	-1082/L4	-322/L6			
4220		TYPE=Rigid +Y;Tag=RA\SL-009					
	1(HYD)	25	-253	76	0	0	0
	2(OPE)	78	-253	0	0	0	0
	4(OPE)	29	-253	69	0	0	0
	6(OPE)	-22	-253	-74	0	0	0

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 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	8(SUS)	3	-253	74	0	0	0
	9(SUS)	29	-253	77	0	0	0
	MAX	78/L2	-253/L6	77/L9			
4240		TYPE=Rigid +Y;Tag=RA\SL-010					
	1(HYD)	61	-1180	368	0	0	0
	2(OPE)	-32	-1180	268	0	0	0
	4(OPE)	128	-1180	324	0	0	0
	6(OPE)	-81	-1180	-352	0	0	0
	8(SUS)	-1	-1180	205	0	0	0
	9(SUS)	48	-1180	376	0	0	0
	MAX	128/L4	-1180/L4	376/L9			
4260		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-012					
	1(HYD)	-6	-1474	465	0	0	0
	2(OPE)	-15	-1474	64	0	0	0
	4(OPE)	399	-1474	569	0	0	0
	6(OPE)	-7	-1474	-450	0	0	0
	8(SUS)	0	-1474	48	0	0	0
	9(SUS)	-7	-1474	478	0	0	0
	MAX	399/L4	-1474/L2	569/L4			
4280		TYPE=Rigid +Y;Tag=RA\SL-014					
	1(HYD)	0	-1414	443	0	0	0
	2(OPE)	4	-1415	14	0	0	0
	4(OPE)	-69	-1415	409	0	0	0
	6(OPE)	6	-1415	-430	0	0	0
	8(SUS)	-0	-1415	11	0	0	0
	9(SUS)	1	-1415	453	0	0	0
	MAX	-69/L4	-1415/L2	453/L9			
4300		TYPE=Rigid +Y;Tag=RA\SL-016					
	1(HYD)	-0	-1430	448	0	0	0
	2(OPE)	-1	-1430	3	0	0	0
	4(OPE)	-42	-1430	417	0	0	0
	6(OPE)	-1	-1430	-435	0	0	0
	8(SUS)	0	-1430	3	0	0	0
	9(SUS)	-0	-1430	458	0	0	0
	MAX	-42/L4	-1430/L2	458/L9			
4320		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-018					

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	1(HYD)	0	-1426	449	0	0	0
	2(OPE)	0	-1426	1	0	0	0
	4(OPE)	-21	-1426	433	0	0	0
	6(OPE)	-0	-1426	-434	0	0	0
	8(SUS)	-0	-1426	1	0	0	0
	9(SUS)	0	-1426	460	0	0	0
	MAX	-21/L4	-1426/L2	460/L9			
4340		TYPE=Rigid LIM; Rigid +Y;Tag=RA\SL-020					
	1(HYD)	-0	-1427	651	0	0	0
	2(OPE)	-0	-1427	0	0	0	0
	4(OPE)	24	-1427	1483	0	0	0
	6(OPE)	0	-1427	-687	0	0	0
	8(SUS)	0	-1427	0	0	0	0
	9(SUS)	-0	-1427	625	0	0	0
	MAX	24/L4	-1427/L2	1483/L4			
4360		TYPE=Rigid +Y;Tag=RA\SL-022					
	1(HYD)	-0	-1427	-440	0	0	0
	2(OPE)	0	-1427	-0	0	0	0
	4(OPE)	5	-1427	-428	0	0	0
	6(OPE)	0	-1427	437	0	0	0
	8(SUS)	0	-1427	-0	0	0	0
	9(SUS)	-0	-1427	-453	0	0	0
	MAX	5/L4	-1427/L2	-453/L9			
4380		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-024					
	1(HYD)	0	-1427	-440	0	0	0
	2(OPE)	-0	-1427	-0	0	0	0
	4(OPE)	-16	-1427	-428	0	0	0
	6(OPE)	-0	-1427	437	0	0	0
	8(SUS)	-0	-1427	-0	0	0	0
	9(SUS)	0	-1427	-453	0	0	0
	MAX	-16/L4	-1427/L4	-453/L9			
4400		TYPE=Rigid +Y;Tag=RA\SL-026					
	1(HYD)	-0	-1427	-438	0	0	0
	2(OPE)	0	-1427	-1	0	0	0
	4(OPE)	-24	-1427	-428	0	0	0
	6(OPE)	-0	-1427	437	0	0	0
	8(SUS)	0	-1427	-1	0	0	0
	9(SUS)	-0	-1427	-450	0	0	0

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	MAX	-24/L4	-1427/L2	-450/L9			
4420		TYPE=Rigid +Y;Tag=RA\SL-028					
	1(HYD)	1	-1427	-437	0	0	0
	2(OPE)	-0	-1427	-3	0	0	0
	4(OPE)	-40	-1427	-428	0	0	0
	6(OPE)	2	-1427	437	0	0	0
	8(SUS)	-0	-1427	-3	0	0	0
	9(SUS)	1	-1427	-446	0	0	0
	MAX	-40/L4	-1427/L4	-446/L9			
4440		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-030					
	1(HYD)	-6	-1427	-435	0	0	0
	2(OPE)	0	-1427	-15	0	0	0
	4(OPE)	334	-1427	-428	0	0	0
	6(OPE)	7	-1427	436	0	0	0
	8(SUS)	0	-1427	-15	0	0	0
	9(SUS)	-4	-1427	-444	0	0	0
	MAX	334/L4	-1427/L2	-444/L9			
4460		TYPE=Rigid +Y;Tag=RA\SL-032					
	1(HYD)	25	-1428	-434	0	0	0
	2(OPE)	-0	-1428	-64	0	0	0
	4(OPE)	77	-1428	-431	0	0	0
	6(OPE)	-58	-1428	433	0	0	0
	8(SUS)	-0	-1428	-64	0	0	0
	9(SUS)	15	-1428	-442	0	0	0
	MAX	77/L4	-1428/L4	-442/L9			
4480		TYPE=Rigid +Y;Tag=RA\SL-034					
	1(HYD)	73	-1079	-365	0	0	0
	2(OPE)	1	-1079	-258	0	0	0
	4(OPE)	-37	-1079	-309	0	0	0
	6(OPE)	-79	-1079	319	0	0	0
	8(SUS)	1	-1079	-258	0	0	0
	9(SUS)	68	-1079	-318	0	0	0
	MAX	-79/L6	-1079/L2	-365/L1			
4540		TYPE=Rigid +Y;Tag=RA\SL-036					
	1(HYD)	6	-781	-213	0	0	0
	2(OPE)	0	-781	-0	0	0	0
	4(OPE)	-147	-781	-164	0	0	0

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	6(OPE)	-43	-781	229	0	0	0
	8(SUS)	0	-781	-0	0	0	0
	9(SUS)	-61	-781	-217	0	0	0
	MAX	-147/L4	-781/L2	229/L6			
4580		TYPE=Rigid +Y;Tag=RA\SL-037					
	1(HYD)	-100	-770	225	0	0	0
	2(OPE)	-1	-770	124	0	0	0
	4(OPE)	-212	-770	43	0	0	0
	6(OPE)	180	-770	-157	0	0	0
	8(SUS)	-1	-770	124	0	0	0
	9(SUS)	-18	-770	227	0	0	0
	MAX	-212/L4	-770/L2	227/L9			
4600		TYPE=Rigid +Y;Tag=RA\SL-038					
	1(HYD)	-95	-770	-226	0	0	0
	2(OPE)	-1	-770	-124	0	0	0
	4(OPE)	-218	-770	17	0	0	0
	6(OPE)	167	-770	147	0	0	0
	8(SUS)	-1	-770	-124	0	0	0
	9(SUS)	-17	-770	-229	0	0	0
	MAX	-218/L4	-770/L2	-229/L9			
4640		TYPE=Rigid +Y;Tag=RA\SL-039					
	1(HYD)	9	-786	218	0	0	0
	2(OPE)	0	-785	0	0	0	0
	4(OPE)	-141	-797	155	0	0	0
	6(OPE)	-39	-785	-231	0	0	0
	8(SUS)	0	-785	0	0	0	0
	9(SUS)	-64	-785	223	0	0	0
	MAX	-141/L4	-797/L4	-231/L6			
4700		TYPE=Rigid +Y;Tag=RA\SL-040					
	1(HYD)	64	-1075	324	0	0	0
	2(OPE)	1	-1076	258	0	0	0
	4(OPE)	-63	-1062	281	0	0	0
	6(OPE)	-71	-1076	-320	0	0	0
	8(SUS)	1	-1076	258	0	0	0
	9(SUS)	69	-1075	332	0	0	0
	MAX	-71/L6	-1076/L2	332/L9			
4720		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-042					

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	1(HYD)	30	-1429	443	0	0	0
	2(OPE)	-0	-1429	64	0	0	0
	4(OPE)	458	-1432	430	0	0	0
	6(OPE)	-83	-1429	-437	0	0	0
	8(SUS)	-0	-1429	64	0	0	0
	9(SUS)	16	-1429	453	0	0	0
	MAX	458/L4	-1432/L4	453/L9			
4740		TYPE=Rigid +Y;Tag=RA\SL-044					
	1(HYD)	-9	-1427	444	0	0	0
	2(OPE)	0	-1427	15	0	0	0
	4(OPE)	10	-1427	427	0	0	0
	6(OPE)	21	-1427	-437	0	0	0
	8(SUS)	0	-1427	15	0	0	0
	9(SUS)	-5	-1427	457	0	0	0
	MAX	21/L6	-1427/L2	457/L9			
4760		TYPE=Rigid +Y;Tag=RA\SL-046					
	1(HYD)	1	-1427	447	0	0	0
	2(OPE)	-0	-1427	3	0	0	0
	4(OPE)	6	-1427	428	0	0	0
	6(OPE)	-0	-1427	-438	0	0	0
	8(SUS)	-0	-1427	3	0	0	0
	9(SUS)	1	-1427	462	0	0	0
	MAX	6/L4	-1427/L4	462/L9			
4780		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-048					
	1(HYD)	-0	-1430	451	0	0	0
	2(OPE)	0	-1430	1	0	0	0
	4(OPE)	5	-1430	429	0	0	0
	6(OPE)	-1	-1430	-439	0	0	0
	8(SUS)	0	-1430	1	0	0	0
	9(SUS)	-0	-1430	469	0	0	0
	MAX	5/L4	-1430/L2	469/L9			
4800		TYPE=Rigid +Y;Tag=RA\SL-050					
	1(HYD)	0	-960	304	0	0	0
	2(OPE)	-0	-960	0	0	0	0
	4(OPE)	-1	-960	288	0	0	0
	6(OPE)	0	-960	-295	0	0	0
	8(SUS)	-0	-960	0	0	0	0
	9(SUS)	0	-960	319	0	0	0
	MAX	-1/L4	-960/L4	319/L9			

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
4820		TYPE=Rigid +Y;Tag=RA\SL-051					
	1(HYD)	-0	-1157	367	0	0	0
	2(OPE)	0	-1158	0	0	0	0
	4(OPE)	-2	-1158	348	0	0	0
	6(OPE)	0	-1158	-355	0	0	0
	8(SUS)	0	-1158	0	0	0	0
	9(SUS)	-0	-1158	384	0	0	0
	MAX	-2/L4	-1158/L4	384/L9			
4840		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-053					
	1(HYD)	-0	-1349	428	0	0	0
	2(OPE)	-0	-1349	0	0	0	0
	4(OPE)	9	-1349	405	0	0	0
	6(OPE)	-0	-1349	-414	0	0	0
	8(SUS)	-0	-1349	0	0	0	0
	9(SUS)	-0	-1349	448	0	0	0
	MAX	9/L4	-1349/L6	448/L9			
4860		TYPE=Rigid LIM; Rigid +Y;Tag=RA\SL-055					
	1(HYD)	0	-922	-761	0	0	0
	2(OPE)	0	-922	-0	0	0	0
	4(OPE)	-19	-923	-1698	0	0	0
	6(OPE)	0	-922	833	0	0	0
	8(SUS)	0	-922	-0	0	0	0
	9(SUS)	0	-922	-879	0	0	0
	MAX	-19/L4	-923/L4	-1698/L4			
4880		TYPE=Rigid +Y;Tag=R8\SL-057					
	1(HYD)	-0	-1290	-403	0	0	0
	2(OPE)	-0	-1290	-0	0	0	0
	4(OPE)	-21	-1289	-381	0	0	0
	6(OPE)	-0	-1290	389	0	0	0
	8(SUS)	-0	-1290	-0	0	0	0
	9(SUS)	-0	-1290	-409	0	0	0
	MAX	-21/L4	-1290/L6	-409/L9			
4900		TYPE=Rigid GUI; Rigid +Y;Tag=RA\SL-059					
	1(HYD)	0	-1463	-457	0	0	0
	2(OPE)	0	-1463	-1	0	0	0

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	4(OPE)	46	-1466	-453	0	0	0
	6(OPE)	1	-1463	442	0	0	0
	8(SUS)	0	-1463	-1	0	0	0
	9(SUS)	0	-1463	-465	0	0	0
	MAX	46/L4	-1466/L4	-465/L9			
4920		TYPE=Rigid +Y;Tag=RA/SL-061					
	1(HYD)	-1	-1403	-438	0	0	0
	2(OPE)	-2	-1404	-5	0	0	0
	4(OPE)	71	-1394	-396	0	0	0
	6(OPE)	3	-1404	424	0	0	0
	8(SUS)	-2	-1404	-5	0	0	0
	9(SUS)	-1	-1404	-446	0	0	0
	MAX	71/L4	-1404/L6	-446/L9			
4940		TYPE=Rigid +Y;Tag=RA/SL-063					
	1(HYD)	5	-1486	-464	0	0	0
	2(OPE)	7	-1484	-23	0	0	0
	4(OPE)	121	-1522	-421	0	0	0
	6(OPE)	-35	-1483	501	0	0	0
	8(SUS)	7	-1484	-23	0	0	0
	9(SUS)	2	-1485	-472	0	0	0
	MAX	121/L4	-1522/L4	501/L6			
4950		TYPE=Rigid GUI; Rigid +Y;Tag=RA/SL-065					
	1(HYD)	-18	-1125	-349	0	0	0
	2(OPE)	-50	-1139	-98	0	0	0
	4(OPE)	-837	-860	-280	0	0	0
	6(OPE)	83	-1144	346	0	0	0
	8(SUS)	-50	-1139	-98	0	0	0
	9(SUS)	-19	-1131	-356	0	0	0
	MAX	-837/L4	-1144/L6	-356/L9			
4960		TYPE=Rigid +Y;Tag=RA/SL-066					
	1(HYD)	-34	-868	-268	0	0	0
	2(OPE)	112	-774	-176	0	0	0
	4(OPE)	-162	-1436	-391	0	0	0
	6(OPE)	86	-642	172	0	0	0
	8(SUS)	112	-774	-176	0	0	0
	9(SUS)	6	-836	-260	0	0	0
	MAX	-162/L4	-1436/L4	-391/L4			

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

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5020		TYPE=Rigid +Y;Tag=RA/PS-001					
	1(HYD)	-268	-1152	-262	0	0	0
	2(OPE)	-220	-1224	-49	0	0	0
	4(OPE)	-160	-895	-261	0	0	0
	6(OPE)	256	-1360	276	0	0	0
	8(SUS)	-220	-1224	-49	0	0	0
	9(SUS)	-242	-1177	-209	0	0	0
	MAX	-268/L1	-1360/L6	276/L6			
5040		TYPE=Rigid +Y;Tag=RA/PS-002					
	1(HYD)	-99	-335	11	0	0	0
	2(OPE)	67	-489	-22	0	0	0
	4(OPE)	0	0	0	0	0	0
	6(OPE)	215	-776	-3	0	0	0
	8(SUS)	67	-489	-22	0	0	0
	9(SUS)	-101	-365	16	0	0	0
	MAX	215/L6	-776/L6	-22/L2			
5100		TYPE=Rigid +Y;Tag=RA/SL-68					
	1(HYD)	-388	-1310	29	0	0	0
	2(OPE)	-190	-1165	44	0	0	0
	4(OPE)	-300	-1586	-302	0	0	0
	6(OPE)	267	-870	21	0	0	0
	8(SUS)	-190	-1165	44	0	0	0
	9(SUS)	-382	-1281	27	0	0	0
	MAX	-388/L1	-1586/L4	-302/L4			
5120		TYPE=Rigid +Y;Tag=RA/SL-70					
	1(HYD)	-327	-1195	-4	0	0	0
	2(OPE)	-49	-1195	-14	0	0	0
	4(OPE)	-290	-1173	-108	0	0	0
	6(OPE)	350	-1194	-16	0	0	0
	8(SUS)	-49	-1195	-14	0	0	0
	9(SUS)	-348	-1195	-7	0	0	0
	MAX	350/L6	-1195/L9	-108/L4			
5140		TYPE=Rigid +Y;Tag=RA/SL-72					
	1(HYD)	-369	-1283	1	0	0	0
	2(OPE)	-12	-1283	4	0	0	0
	4(OPE)	-381	-1297	8	0	0	0
	6(OPE)	377	-1286	-6	0	0	0
	8(SUS)	-12	-1283	4	0	0	0
	9(SUS)	-372	-1283	1	0	0	0

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RESTRAINT SUMMARY REPORT: Loads On Restraints
 Various Load Cases

Node	Load Case	FX N.	FY N.	FZ N.	MX N.m.	MY N.m.	MZ N.m.
	MAX	-381/L4	-1297/L4	8/L4			
5220		TYPE=Rigid GUI; Rigid LIM; Rigid +Y;Tag=MANIFOLD SUPPORT (NOT IN ODC SCOPE)					
	1(HYD)	1122	-993	3	0	0	0
	2(OPE)	133	-990	-1	0	0	0
	4(OPE)	1470	-942	35	0	0	0
	6(OPE)	-1194	-977	22	0	0	0
	8(SUS)	133	-990	-1	0	0	0
	9(SUS)	1146	-992	6	0	0	0
	MAX	1470/L4	-993/L1	35/L4			
5240		TYPE=Rigid +Y;Tag=MANIFOLD SUPPORT (NOT IN ODC SCOPE)					
	1(HYD)	247	-837	-125	0	0	0
	2(OPE)	200	-846	-2	0	0	0
	4(OPE)	284	-961	-15	0	0	0
	6(OPE)	-262	-878	86	0	0	0
	8(SUS)	200	-846	-2	0	0	0
	9(SUS)	229	-842	-88	0	0	0
	MAX	284/L4	-961/L4	-125/L1			
5280		TYPE=Rigid +Y;Tag=MANIFOLD SUPPORT (NOT IN ODC SCOPE)					
	1(HYD)	129	-674	121	0	0	0
	2(OPE)	2	-669	2	0	0	0
	4(OPE)	169	-594	47	0	0	0
	6(OPE)	-147	-647	-102	0	0	0
	8(SUS)	2	-669	2	0	0	0
	9(SUS)	81	-671	82	0	0	0
	MAX	169/L4	-674/L1	121/L1			

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 10

LINE LIST

Service	Line number	From	To	PEFS Number	Unit	Fluid	Steam Out	Phase	AT WORKING CONDITIONS										DESIGN CONDITIONS					Material	Insulation Type	Insulation Thickness [mm]	Tracing		Test Medium	Test Pressure	External Coating	PWHT	Toxicity	Construction Class	Stress Critical Line	Process Notes	Piping Notes	Remarks
									Pressure kPa(g)	Temp. °C	Density			Viscosity	Flow Rate		Continuous/Intermittent	Nominal Pipe size inch	Piping Class	Pressure Max kPa(g)	Pressure Min kPa(g)	Temp Max °C	Temp Min °C				Temperature °C [degC]	Type										
											Gas kg/m³	Liquid kg/m³	Mixed kg/m³		Gas cP	Gas M³/sm/hr																						
P	8811801	WELL NO. U_61_1_P Tubing	Sonolog Coupling	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	G	5000	265	50	-	-	0.013	0.032	-	187	I	2	91441C4	11600	FV	325	5	CSS	PP	50	-	Water	22980	PCS-3	No	Yes(A)	Class-1b	Yes	6		AFD_OOC
P	8811802	WELL NO. U_61_1_P Casing	4"-P8811604-91441C4	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	G	5000	265	50	-	-	0.013	0.032	-	332.42	I	1.5	91441C4	11600	FV	325	5	CSS	PP	50	-	Water	22980	PCS-3	No	Yes(A)	Class-1b	No	6		AFD_OOC
P	8811804	WELL NO. U_61_1_P	4"-P8811605-31441T3	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	M	1719	151	21.3	983	983	0.013	0.04	11.7	0.5	C	4	91441C4	11600	FV	325	5	CSS	PP	50	-	Water	22980	PCS-2	No	Yes(A)	Class-1b	Yes	6		AFD_OOC
P	8811805	11/2"-P8811602 Casing vent	88-VMC-20001	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	M	3456	244	17.2	803	17.43	0.02	0.0263	0.00519	12	C	2	91441C4	11600	FV	325	5	CSS	PP	50	-	Water	22980	PCS-3	No	Yes(A)	Class-1b	Yes	6,10		AFD_OOC
P	8811806	4"-P8811604-91441C4	FL-4"-MB810001	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	M	1719	151	21.3	983	983	0.013	0.04	11.7	0.5	C	4	31441T3	4140	FV	260	5	CSS	PP	50		Water	7725	PCS-2	No	Yes(A)	Class-1b	Yes	6		AFD_OOC
P	8811807	88-VMC-20001	4"-P8811604-91441C4	AML-88-PDFEED-PX-2365-00001-0001	88	HC	N	G	1043	199	5.7	-	-	0.02	0.0263	-	15	C	3	91441C4	11600	FV	325	5	CSS	PP	50		Water	22980	PCS-3	No	Yes(A)	Class-1b	Yes	6,10		AFD_OOC

		
PETROLEUM DEVELOPMENT OMAN LLC	AL TASNIM ENTERPRISES LLC	TKAVIN ENGINEERING CONSULTANCY LLC

ANNEXURE – 11

API 6A FLANGE LOAD CHECK

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API 6A flange load check

For API 6A flanges, actual load due to various piping loads shall be compared with the allowable load as specified in API TR 6AF

Flange Node No.	API 6A flange Size / Rated Pressure	Bore Pressure Kpa (Psi)	Allowable Flange load		Actual Flange load		Result
			Tension Force (N)	Bending Moment (Nm)	Tension Force (N)	Bending Moment (Nm)	
3170	4 1/16" & 3000 psi	10000 (1450)	355857	21693	7734	8077	Pass
2850	2 1/16" & 5000 psi	10000 (1450)	177928	12202	3452	1071	Pass

Actual loads are within the allowable loads, hence API 6A flanges are SAFE in condition

